Contents

1 On Manifolds with Boundary

In this section we discuss how known theorems/lemmas fail for manifolds with non-emtpy boundary and how they can be generalized.

1.1 Regular values and transversality

It is known that for the regular value theorem, and more generally statements about transversality, to still hold for manifolds with boundary one needs to impose additional conditions on the boundary behaviour of a smooth function $f\colon M \to N$. Here is another lemma, that fits into the same category. We call $q \in N$ a **boundary value** of f if $q \in f(\partial M)$.

Lemma 1.1. *Let M be a manifold with boundary and $f\colon M \to \mathbb{R}$ a smooth function. Furthermore, let $a \in \mathbb{R}$ be a regular value of f, that is not a boundary value. Then $M_a := f^{-1}(-\infty, a]$ is a submanifold with boundary*

$$\partial M_a = f^{-1}(a) \cup (M_a \cap \partial M).$$

Proof sketch. For $p \in f^{-1}(a)$ we have $p \in \operatorname{int}(M)$ by assumption. Since $\operatorname{int}(M)$ is a boundaryless manifold (and an open subset of M) the local submersion theorem holds, i.e. f looks like the orthogonal projection $\mathbb{R}^n \to \mathbb{R}$ around p. Then the proof goes essentially the same as in the boundaryless case. □

Counterexample 1.2. To see that Lemma 1.1 fails without the assumption that a is not a boundary value consider $M := S^1 \times [0,1]$ and $f\colon M \to \mathbb{R}, (z,t) \mapsto t$. Then f is everywhere regular but $f^{-1}(-\infty, 0]$ does not satisfy the statements in Lemma 1.1.

Let M, N be manifolds with boundary and $f\colon M \to N$ smooth. We define the **boundary map** ∂f of f by

$$\partial f := f|_{\partial M}.$$

We now generalize three theorems from [GP74] about transversal maps by allowing the codomain to have boundary as well.

Theorem 1.3 (Transversality Theorem)**.** *Let M, N be manifolds with boundary, $f\colon M \to N$ smooth and $X \subseteq N$ a submanifold without boundary. If*

$$f \pitchfork X \text{ and } \partial f \pitchfork X$$

then $X' := f^{-1}(X)$ is a submanifold with boundary

$$\partial X' = X' \cap \partial M$$

and

$$\operatorname{codim}(X') = \operatorname{codim}(X).$$

Proof. The case $\partial N = \emptyset$ is shown in [GP74] (p. 60). If $\partial N \neq \emptyset$ consider the inclusion

$$\iota\colon N \to \mathrm{D}(N),$$

where $\mathrm{D}(N)$ denotes the double of N and define $F := \iota \circ f$. Then $Y := \iota(X)$ is a submanifold of $\mathrm{D}(N)$ and $X' = F^{-1}(Y)$. Furthermore, since ι is an embedding (and $\dim(N) = \dim(\mathrm{D}(N))$) it follows that

$$F \pitchfork Y \text{ and } \partial F \pitchfork Y,$$

such that the statement follows from the boundaryless case. □

We can also make a statement about the tangent space of X' in the above theorem. To this end let $f\colon M \to N$ and $X \subseteq N, X' \subseteq M$ be as above. Then we can also characterize transversality $f \pitchfork X$ as follows: Choose a metric on N and for $x \in X', y = f(x)$ define the **vertical differential**

$$D^V_x f\colon T_xM \to N_yX$$

as the composition

$$T_xM \xrightarrow{d_xf} T_yN = T_yX \oplus N_yX \xrightarrow{\text{proj.}} N_yX.$$

Then $f \pitchfork X$ if and only if $D_x^V f$ is surjective for all $x \in X'$. If this is the case we moreover have (at least if $\partial M = \emptyset$, otherwise we also require $\partial f \pitchfork X$)

$$\operatorname{codim}(\ker D_x^V f) = \dim(\operatorname{im}(D_x^V f)) = \operatorname{codim}(X) = \operatorname{codim}(X')$$

and thus (since $T_x X' \subseteq \ker D_x^V f$) we get

$$T_x X' = \ker D_x^V f.$$

This generalizes the formula for the tangent space of the preimage of a regular value. It should also be noted, that the Riemannian metric on N was only chosen, to define a complement of $T_y X$ in a consistent manner, such that the vertical differential $D_x^V f$ depends smoothly on x.

Theorem 1.4 (Parametric Transversality Theorem)**.** *Let M, N be manifolds with boundary and $X \subseteq N$ a submanifold without boundary. Furthermore, let S be a manifold without boundary and $F\colon M \times S \to N$ a smooth map with*

$$F \pitchfork X \text{ and } \partial F \pitchfork X.$$

Then for almost every $s \in S$ the map $f_s\colon M \to N$ defined by $f_s(p) := F(p, s)$ satisfies

$$f_s \pitchfork X \text{ and } \partial f_s \pitchfork X.$$

Proof. By Theorem 1.3 we know that

$$X' := F^{-1}(X)$$

is a submanifold (with boundary) of $M \times S$. Consider the projection $\pi\colon M \times S \to S$. We can then show, that for a regular value $s \in S$ of the map $\pi|_{X'}$ the maps f_s and ∂f_s are transversal to X, which is just some linear algebra. The proof is exactly the same as in the case $\partial N = \emptyset$, see [GP74] (p. 68). Then the statement follows from Sard‘s Theorem. □

Theorem 1.5 (Transversality Homotopy Theorem)**.** *Let M, N be manifolds with boundary, $f\colon M \to N$ smooth and $X \subseteq N$ a submanifold without boundary. Then f is smoothly homotopic to a map $g\colon M \to N$ with*

$$g \pitchfork X \text{ and } \partial g \pitchfork X.$$

Proof. First we may assume that $\operatorname{im}(f) \subseteq \operatorname{int}(N)$, since f is always smoothly homotopic to such a map (the inclusion $\operatorname{int}(N) \hookrightarrow N$ is a smooth homotopy equivalence). Define

$$Y := X \cap \operatorname{int}(N)$$

and

$$f'\colon M \to \operatorname{int}(N),\ p \mapsto f(p).$$

Using the Transversality Homotopy Theorem for the case $\partial N = \emptyset$ (see [GP74] p. 70) we get that f' is smoothly homotopic to a map $g'\colon M \to \operatorname{int}(N)$, such that

$$g' \pitchfork Y \text{ and } \partial g' \pitchfork Y.$$

Now define $g := \iota \circ f'$, where $\iota\colon \operatorname{int}(N) \hookrightarrow N$ is the inclusion. Since g does not intersect X at boundary points, i.e. $\operatorname{im}(g) \cap X \subseteq \operatorname{int}(N)$, the transversality properties of g' directly imply

$$g \pitchfork X \text{ and } \partial g \pitchfork X.$$

□

1.2 Morse Theory

For a smooth function $f\colon M \to \mathbb{R}$ and a regular value $a \in \mathbb{R}$ that is not a boundary value we define the **sublevel set** of a to be

$$M_a := f^{-1}(-\infty, a].$$

Lemma 1.1 guarantees, that M_a is a (sub-)manifold with boundary.

Lemma 1.6. *Let M be a compact manifold with boundary and $f\colon M \to \mathbb{R}$ a Morse function. Furthermore let $a, b \in \mathbb{R}$ be regular values of f, such that $[a, b]$ does not contain any critical values nor any boundary values. Then*

$$M_a \cong M_b.$$

Proof. Let $K := f^{-1}[a, b]$ and let $U := \{p \in \mathrm{int}(M) \mid p \text{ is a regular point of } f\}$. Then $U \subseteq M$ is open and $K \subseteq U$. Choose a cutoff function $\phi\colon M \to \mathbb{R}$ with

- $\phi|_K = 1$
- $\mathrm{supp}(\phi) \subseteq U$
- $0 \leq \phi \leq 1$

Now choose a pseudo gradient[1] X of f and define the smooth vector field

$$Y := \phi \frac{-1}{X \cdot f} X,$$

which we define to be 0 outside of U. Since Y vanishes in a neighbourhood around ∂M (or more generally since Y is tangent to ∂M, see [Lee11] Theorem 9.34) we can consider the flow of Y:

$$\theta\colon M \times \mathbb{R} \to M.$$

Since M is compact (and Y tangent to ∂M) each integral curve is defined for all time. Let $\theta_t := \theta(-, t)\colon M \to M$. We claim that

$$\theta_{b-a}(M_b) \subseteq M_a, \tag{1}$$
$$\theta_{a-b}(M_a) \subseteq M_b. \tag{2}$$

Let $p \in f^{-1}[a, b]$ and let $\gamma_p\colon \mathbb{R} \to M$ be the integral curve of Y starting at p. Consider

$$h\colon \mathbb{R} \to \mathbb{R},\ t \mapsto f(\gamma_p(t)).$$

Then for $t \in \mathbb{R}$ with $h(t) \in [a, b]$ we have $\phi(\gamma_p(t)) = 1$ and thus

$$h'(t) = \dot{\gamma_p}(t) \cdot f = Y_{\gamma_p(t)} \cdot f = -1.$$

So h is a smooth function with $c := h(0) = f(p) \in [a, b]$ and $h'(t) = -1$ as long as $h(t) \in [a, b]$. We claim that $h(c - a) = a$. To see this let

$$s := \sup\{t \in [0, c - a] \mid h([0, t]) \subseteq [a, b]\}.$$

Assume $s < c - a$. For continuity reasons we get that $h([0, s]) \subseteq [a, b]$ and thus $h'(t) = -1$ for $t \in [0, s]$. Then $h(t) = c - t$ for $t \in [0, s]$. But then $h(s) \in (a, b]$ and h is strictly decreasing around $t = s$ such that there exists $\varepsilon > 0$ with $h([0, s + \varepsilon]) \subseteq [a, b]$, contradicting the definition of s. So $s = c - a$ and we get that $h(t) = c - t$ for $t \in [0, c - a]$. Thus $h(c - a) = a$.

As $h'(t) \leq 0$ for all $t \in \mathbb{R}$ we conclude

$$f(\theta_{b-a}(p)) = h(b - a) \leq h(c - a) = a,$$

thus $\theta_{b-a}(p) \in M_a$. For $p \in M_b \setminus f^{-1}[a, b] = \mathrm{int}M_a$ we also have

$$\theta_{b-a}(p) \in M_a$$

as f decreases along gradient flow lines of Y. This shows (1). We argue similiarly for (2). Let $p \in M_a$ and let $\gamma_p\colon \mathbb{R} \to M$ be the integral curve of Y starting at p. Consider

$$h\colon \mathbb{R} \to \mathbb{R},\ t \mapsto f(\gamma_p(-t)).$$

Then $h'(t) = \phi(\gamma_p(-t)) \leq 1$. Assume that $h(b - a) = f(\theta_{a-b}(p)) > b$. Then by the mean value theorem there exists some $\xi \in [0, b - a]$ with

$$h'(\xi) = \frac{h(b - a) - h(0)}{b - a} \geq \frac{h(b - a) - a}{b - a} > 1,$$

which is a contradiction. So $f(\theta_{a-b}(p)) \leq b$, showing (2). Thus θ_{b-a} restricts to a diffeomorphism $M_b \xrightarrow{\cong} M_a$ with inverse θ_{a-b}. □

[1] Here we agree to the convention that $X \cdot f < 0$ outisde of critical points.

Counterexample 1.7. Example 1.2 also shows, that we need to impose conditions on the boundary values in Lemma 1.6. To see that we also need that condition for Lemma 1.9 we may similiarly define f as the height function on the disjoint union of two cylinders which are slightly shifted in height. Similiar examples show, that Theorem 1.10 fails without the assumption on boundary values.

Remark 1.8. We can weaken the assumptions of Lemma 1.6. First of all f does not need to be a Morse function. We can then still construct a vector field X, such that $X \cdot f < 0$ outisde of critical points. Secondly, we may replace the assumption that M is compact by instead requiring $f^{-1}[a, b]$ to be compact, which is for example the case if f is proper. We then only have to modify the proof in the definition of U. Instead we define U to be a *compact* neighbourhood of $K := f^{-1}[a, b]$, i.e. $K \subseteq \mathrm{int}U$, which does not contain any critical points nor boundary points. For example we may define U as a union of finitely many (here we need compactness of $f^{-1}[a, b]$) compact sets K_i not containing neither critical points nor boundary points such that $\mathrm{int}K_i$ still cover K.

Similiarly we may proof the following, for which we can also weaken the assumptions as described in the remark above.

Lemma 1.9. *Let M be a compact manifold with boundary and $f\colon M \to \mathbb{R}$ a Morse function. Furthermore let $a, b \in \mathbb{R}$ be regular values of f, such that $[a, b]$ does not contain any critical values nor any boundary values. Then M_a is a (not necessarily smooth) deformation retract of M_b.*

Proof. We consider the flow θ as in Lemma 1.6 and define a retraction

$$r\colon M_b \to M_a,\ p \mapsto \begin{cases} \theta(f(p) - a, p), & f(p) \geq a \\ p, & f(p) \leq a \end{cases}$$

As we have shown above $\theta(f(p) - a, p) \in f^{-1}(a)$. It follows that r is continous and well-defined. To see that r is a deformation retract consider

$$H\colon M_b \times [0, 1] \to M_b, (p, t) \mapsto \begin{cases} \theta(t(f(p) - a), p), & f(p) \geq a \\ p, & f(p) \leq a \end{cases}$$

Then $H(-, 0) = id_{M_b}$ and $H(-, 1) = r$. $\square$

We would now also like to generalize the result, that the topology of M changes by attaching a λ-cell when crossing a critical point c_0 of index λ to compact manifolds with boundary. This statement can now be shown using the two lemmas above by requiring $f^{-1}[a, b]$ to only contain the critical point c_0 and no boundary points. The proof now works exactly the same as for the boundaryless case, see [Mil63]. Let us state the theorem for manifolds with boundary:

Theorem 1.10. *Let M be a compact manifold with boundary and $f\colon M \to \mathbb{R}$ a Morse function. Let $c_0 \in M$ be a critical point of f of index λ and $c := f(c_0)$ as well as $\varepsilon > 0$, such that $f^{-1}[c-\varepsilon, c+\varepsilon]$ contains no boundary points and no critical points besides c. Then for ε sufficiently small $M_{c+\varepsilon}$ has the homotopy type of $M_{c-\varepsilon}$ with a λ-cell attached. More precisely, there exists an attaching map*

$$\phi\colon S^{\lambda-1} = \partial D^\lambda \to \partial M_{c-\varepsilon}$$

and a homotopy equivalence

$$r\colon M_{c+\varepsilon} \to M_{c-\varepsilon} \cup_\phi D^\lambda$$

with homotopy inverse

$$\iota\colon M_{c-\varepsilon} \cup_\phi D^\lambda \to M_{c+\varepsilon},$$

such that $r \circ \iota = id_{M_{c-\varepsilon} \cup_\phi D^\lambda}$.

Proof Sketch. Following [Mil63] we first construct $F\colon M \to \mathbb{R}$, such that

$$F^{-1}(-\infty, c + \varepsilon] = f^{-1}(-\infty, c + \varepsilon] = M_{c+\varepsilon}.$$

Furthermore $F^{-1}[c - \varepsilon, c + \varepsilon]$ does not contain any critical points or boundary points, such that $F^{-1}(-\infty, c - \varepsilon]$ is a deformation retract of $F^{-1}(-\infty, c + \varepsilon]$ by Lemma 1.9. We then show that $M_{c-\varepsilon} \cup_\phi D^\lambda$ is a deformation retract of $F^{-1}(-\infty, c-\varepsilon]$ (or more precisely the image of $M_{c-\varepsilon} \cup_\phi D^\lambda$ under an embedding is a deformation retract). This then shows that $M_{c-\varepsilon} \cup_\phi D^\lambda$ is a deformation retract of $M_{c+\varepsilon}$. $\square$

However this does not yield a cell decomposition of M as we do not know how the topology changes when crossing boundary points. Instead we can use the fact that any compact manifold without boundary admits a cell-decomposition directly, to proof the case when M has non-empty boundary.

Theorem 1.11 (Cell-decompositions of Manifolds)**.** *Let M be compact manifold with or without boundary. Then M has the homotopy type of a CW-complex.*

Proof. The case when M has no boundary is shown in [Mil63]. If $\partial M \neq \emptyset$ we can argue as follows: consider the double $D(M)$ of M. Then $D(M)$ is a compact manifold without boundary. Now we construct a Morse function on $D(M)$. Choose a Morse function $f\colon M \to \mathbb{R}$ with $f \leq 0$ and $f(p) = 0$ if and only if $p \in \partial M$, such that f has no critical points on the boundary. Such Morse functions always exist, see [Vin24]. We can then glue f and $-f$ together to get a Morse function F on $D(M)$, again see [Vin24]. Then $M \cong F^{-1}(-\infty, 0]$ and 0 is a regular value of F, allowing us to refer back to the boundaryless case, which tells us that every sublevel set M_a for a regular value $a \in \mathbb{R}$ has the homotopy type of a CW-complex. $\square$

Remark 1.12. We may also formulate Theorem 1.10 and 1.11 for handle bodies instead of CW-complexes. The proof is essentially the same but instead of constructing a deformation retract $F^{-1}(-\infty, c-\varepsilon] \to M_{c-\varepsilon} \cup_\phi D^\lambda$ one constructs a homeomorphism $F^{-1}(-\infty, c-\varepsilon] \to M_{c-\varepsilon} \cup_\phi D^\lambda \times D^{n-\lambda}$.

1.3 Integral curves and flows

For manifolds with boundary we essentially define integral curves in the same way as for boundaryless manifolds, but allow the domain to be any non-trivial interval (i.e. not a single point) containing 0. In particular this interval may only be half-open or closed. In general, the existence of integral curves on manifolds with boundary fails as illustrated by the following example:

Counterexample 1.13. Consider the lower halfspace $M := \mathbb{R}^2_{y \leq 0}$ and the vector field $X := \frac{\partial}{\partial x} + 2x\frac{\partial}{\partial y}$. Then for $p = (0, 0)$ the integral curve is only defined for $t = 0$, since on $\mathbb{R}^2$ it would be given by $\gamma(t) = (t, t^2)$ but $t^2 > 0$ for $t \neq 0$.

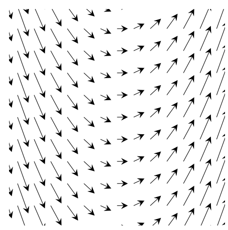

Figure 1: The vector field X. The integral curves are parabolas.

We do however still get a uniqueness result *if* the integral curve exists. Furthermore, if the vector field is sufficiently well-behaved we can guarantee the existence.

Theorem 1.14 (Uniqueness and existence of Integral curves)**.** *Let M be a manifold with boundary, X a smooth vector field on M and $p \in M$. If there exists an integral curve starting at p, then there exists a unique maximal integral curve $\gamma\colon I \to M$ starting at p, where one of the four following cases holds for $a \leq 0 \leq b$, $a < b$:*

(i) $I = (a, b)$

(ii) $I = [a, b)$ *and* $\gamma(a) \in \partial M$

(iii) $I = (a, b]$ *and* $\gamma(b) \in \partial M$

(iv) $I = [a, b]$ *and* $\gamma(a), \gamma(b) \in \partial M$

Furthermore, if X is either transversal to the boundary or everywhere tangent, the unique maximal integral curve starting at p always exists.

Proof. Let's begin by showing uniqueness. The proof is essentially the same as in the boundaryless case (see for example [Lee11]): We show that two integral curves $\gamma\colon I \to M$ and $\gamma'\colon I' \to M$ starting at p agree on their common domain. This then allows us to define the maximal integral curve on some interval I and every interval containing 0 is of the form $(i) - (iv)$ [2]. The only difference is that I and I' may not be open intervalls. So we have to show that uniqueness still applies. We claim: Let $U \subseteq \mathbb{R}^n$ be open, $F\colon U \to \mathbb{R}^n$ a smooth vector field, I an interval of the form $(i) - (iv)$ and $\gamma, \gamma'\colon I \to U$ integral curves both starting at $p \in U$. Then $\gamma = \gamma'$. If 0 is an interior point of I we can apply the usual uniqueness theorem for γ, γ' restricted to $\operatorname{int}(I)$. By continuity they must then also agree at the endpoint. So it remains to show uniqueness if 0 is a boundary point of I (say for example $I = [0, b)$). But by the existence theorem for integral curves we can find an integral curve $\eta\colon (-\varepsilon, \varepsilon) \to U$ starting at p and define

$$\beta\colon (-\varepsilon, b) \to U, t \mapsto \begin{cases} \eta(t), t \leq 0 \\ \gamma(t), t \geq 0 \end{cases}$$

Since $\eta(0) = \gamma(0)$ and $\dot\eta(0) = F(p) = \dot\gamma(0)$ we get that β is at least C^1. Similiarly we define

$$\beta'\colon (-\varepsilon, b) \to U, t \mapsto \begin{cases} \eta(t), t \leq 0 \\ \gamma'(t), t \geq 0 \end{cases}$$

By the uniqueness theorem we have

$$\beta = \beta'$$

and our claim follows. The case $I = (a, 0]$ goes analogous and for $I = [0, b]$ or $I = [a, 0]$ it is either trivial (if $a = b = 0$) or we consider $I \setminus \{b\}$ resp. $I \setminus \{a\}$ and apply the above argument. Equality at $t = a, t = b$ respectively then again follows by continuity.

To see that $\gamma(t) \in \partial M$ for $t \in \{a, b\}$ a boundary point of I we note that for $p = \gamma(t) \in \operatorname{int}(M)$ we can define the integral curve starting at p on some open interval $(-\varepsilon, \varepsilon)$ by the existence theorem for integral curves (Picard-Lindelöff). But then t is not a boundary point of I since I is maximal.

Next we show the existence if X is transversal to ∂M. For $p \in \operatorname{int}(M)$ this follows directly from the Picard-Lindelöff Theorem as in the boundaryless case. If $p \in \partial M$, we may argue similiarly: Consider a chart $h\colon \mathbb{R}^n_{x_n \geq 0} \supseteq U \to M$ around $p = h(0)$ and let Y be the vector field X in this chart, i.e. $D_q h(Y_q) = X_{h(q)}$. Then by definition of smoothness we can extend Y to a smooth vector field Y' on some open neighbourhood $U' \subseteq \mathbb{R}^n$ around $0 = h^{-1}(p)$. Applying Picard-Lindelöff we get the existence of an integral curve $\gamma\colon (-\varepsilon, \varepsilon) \to U'$ of Y' starting at 0. Write

$$Y' = \sum_i F_i \frac{\partial}{\partial x_i}$$

and

$$\gamma = (\gamma_1, \ldots, \gamma_n).$$

Assume that X is inwards pointing along the boundary. After restricting to a smaller domain we may assume $F_n \geq 0$. Then

$$\gamma_n'(t) = F_n(\gamma(t)) \geq 0,$$

such that γ_n is increasing. In particular $\gamma_n(t) \geq 0$ for $t \in [0, \varepsilon)$, hence $\gamma(t) \in U$ for $t \in [0, \varepsilon)$. Then

$$\eta := h \circ \gamma\colon [0, \varepsilon) \to M$$

is the desired integral curve.

For the case that X is everywhere tangent to ∂M see [Lee11] (Theorem 9.34).

□

On compact manifolds we can say more about the maximal Interval I:

[2] If we put together integral curves defined on open intervals that agree on their common domain, it is clear that this results in a smooth curve, since around each t this curve is given by one of the smooth curves which we put together. If we put together integral curves γ_1, γ_2 defined on something like $I_1 = (a, b]$ and $I_2 = [b, c)$ we have to be more careful around the boundary point b. But the same argument as we applied to β in the proof of Lemma 1.14 shows that putting γ_1 and γ_2 together results in a C^1-curve. We can then apply the usual uniqueness theorem to see that this curve is already given by a C^∞-curve.

Lemma 1.15 (Existence of Integral curves on compact manifolds)**.** *If M is compact in Lemma 1.14 then the maximal interval I has one of the following forms:*

(i) $I = (-\infty, \infty)$

(ii) $I = [a, \infty)$ *and* $\gamma(a) \in \partial M$

(iii) $I = (-\infty, b]$ *and* $\gamma(b) \in \partial M$

(iv) $I = [a, b]$ *and* $\gamma(a), \gamma(b) \in \partial M$

Proof. Let $\gamma\colon I \to M$ be the unique maximal integral curve guaranteed by Lemma 1.14. We will show that any open boundary point of I must be $\pm\infty$. For simplicity let us only consider $I = [a, b)$. The other cases are completely analogous. By compactness we can find a sequence $0 < t_n \nearrow b$ such that $\gamma(t_n)$ coverges to a point $q \in M$.
Case 1: $q \in \mathrm{int}(M)$**.** By Picard-Lindelöff we can find an open neighbourhood $U \subseteq M$ around q and $\varepsilon > 0$, such that for every point $x \in U$ the intgeral curve starting at x is defined for $t \in (-\varepsilon, \varepsilon)$. For some fixed m sufficiently large we have $q_0 := \gamma(t_m) \in U$. Assume that $b - t_m < \frac{\varepsilon}{2}$. But then we can extend γ to $[a, b + \frac{\varepsilon}{2})$ as follows. Consider

$$\gamma'\colon [a - t_m, b - t_m) \to M,\ t \mapsto \gamma(t + t_m).$$

Then γ' is a integral curve starting at q_0. Let $\alpha\colon (-\varepsilon, \varepsilon) \to M$ be another integral curve starting at q_0. Then α and γ' agree on their common domain by the arguments in Lemma 1.14 and thus we can put them together to define an integral curve

$$\beta\colon J \to M$$

for $J := [a - t_m, b - t_m) \cup (-\varepsilon, \varepsilon) \supseteq [a - t_m, \varepsilon)$. But then we can define

$$\gamma''\colon [a, \varepsilon + t_m) \to M,\ t \mapsto \beta(t - t_m),$$

which is an integral curve starting at $\gamma(0) = p$. However

$$b + \frac{\varepsilon}{2} < t_m + \varepsilon$$

and so $[a, b + \frac{\varepsilon}{2}) \subseteq [a, \varepsilon + t_m)$, contradicting maximality of γ.
Case 2: $q \in \partial M$**.** We argue similiarly as above. In a local chart around q we can again apply Picard-Lindelöff (smoothness of X guarantees by definition the existence of a smooth extension of X to an open neighbourhood in $\mathbb{R}^n$). We then again take $q_0 := \gamma(t_m)$ sufficiently close to q, choose an integral curve α starting at q, which we can use to define an integral curve α' starting at q_0 defined on some open interval $(-\varepsilon, \varepsilon)$ by translation in time as above. Also we define γ' as above. The same argument as above then shows that we can extend γ' (in the local chart) to an integral curve β on $J := [a - t_m, b - t_m) \cup (-\varepsilon, \varepsilon) \supseteq [a - t_m, b - t_m]$. In particular we have $\beta([a - t_m, b - t_m]) \subseteq M$. Translating in time again yields an integral curve $\gamma''\colon [a, b] \to M$ contradicting maximality. ☐

As an application we can show:

Lemma 1.16. *Let I be some proper interval containing* 0 *and $\gamma\colon I \to M$ any smooth regular curve, i.e. $\dot\gamma(t) \neq 0$ for all $t \in I$. Then there exists a unique unit speed reparametrization of γ, that is an orientation preserving diffeomorphism $\phi\colon J \to I$ with $\phi(0) = 0$, such that $\gamma \circ \phi$ is a unit speed curve.*

Proof. We have

$$(\gamma \circ \phi)^{\cdot}(t) = \phi'(t)\dot\gamma(\phi(t)),$$

so we need to find ϕ, such that (note that we also need $\phi' > 0$ to preserve orientation)

$$\phi'(t) = \frac{1}{|\dot\gamma(\phi(t))|}$$

This is a flow equation for the smooth vector field

$$X(p) := \frac{1}{|\dot\gamma(p)|}$$

on I. Since X is transversal to the boundary of I (if I even has boundary), Theorem 1.14 yields the existence of a unique maximal integral curve $\phi\colon J \to I$ of X with $\phi(0) = 0$. We now only need to show that ϕ is surjective, since $\phi' > 0$ already shows that it is an embedding.

To this end, consider any compact subinterval $I' := [a, b] \subseteq I$ $(a < b)$ containing 0 and let $\psi\colon J' \to I'$ be the unique maximal integral curve of $X|_{I'}$ starting at 0. By Lemma 1.15 ψ has to be defined on a compact interval $J' = [c, d]$, since $X|_{I'}$ is bounded below by a positive constant, so if J' would be infinite in the positive/negative direction, ψ would also tend to positive/negative ininity, but by definition it is conatined in the compact interval I'. Moreover Lemma 1.15 then yields

$$\psi(c) = a, \quad \psi(d) = b.$$

Since ϕ is the unique maximal integral curve of X we get that $J' \subseteq J$ and $\phi|_{J'} = \psi$. In particular $I' \subseteq \mathrm{im}(\phi)$. This completes the proof. $\square$

Remark 1.17 (A note on flows for manifolds with boundary). In general the concept of a flow is not well behaved if M has boundary, as integral curves may not even exist (see Counterexample 1.13). In [Lee11] the author shows, that if X is tangent to ∂M we still get a well-behaved flow, i.e. it is defined on some open subset $\mathcal{O} \subseteq M \times \mathbb{R}$. I think that if we instead assume, that X is transversal to ∂M, we still get a "good" flow domain in the sense that we can define the flow on some subset $\mathcal{D} \subseteq M \times \mathbb{R}$, which is a codimension 0 submanifold with boundary and $I_p := \{t \in \mathbb{R} \mid (p, t) \in \mathcal{D}\}$ is an interval around 0 for each $p \in M$.

The statement in the above remark should follow from the following lemma, which allows us to define submanifold charts for $\mathcal{D}$.

Lemma 1.18. *Let M be a manifold with boundary and X a vector field of M, which is inward pointing along ∂M. Let*

$$M_\partial := \{p \in M \mid \mathrm{im}\gamma_p \cap \partial M \neq \emptyset\}$$

be all points in M whose trajectory contains boundary points (this is at most one for each $p \in M$). For $p \in M_\partial$ the integral curve γ_p starting at p is defined on some interval $[-\tau(p), a)$ for some $\tau(p) \geq 0$ and $a > -\tau(p)$. Then the following statements hold true

(i) M_∂ is an open subset of M

(ii) $\tau\colon M_\partial \to \mathbb{R}_{\geq 0}$, $p \mapsto \tau(p)$ is smooth

Proof sketch. The idea is to consider the maximal flow

$$\theta\colon \mathcal{D} \to M,\ (p, t) \mapsto \gamma_p(t)$$

on some open subset $\mathcal{D} \subseteq \partial M \times [0, \infty)$ and show that this is an embedding. Then

$$M_\partial = \mathrm{im}\theta$$

is open and

$$\tau = \pi \circ \theta^{-1}\colon M_\partial \to [0, \infty)$$

where $\pi\colon \partial M \times [0, \infty) \to [0, \infty)$ is the canonical projection map. $\square$

Lemma 1.19. *Injective Flowouts are already embeddings*

2 Cartan Calculus

2.1 Lie derivative and Cartan's magic formula

In this section we discuss two differential operators on the graded algebra $\Omega^\bullet(M)$ of differential forms on a manifold M, namely the Lie derivative $\mathcal{L}_X$ and the interior product ι_X, where X is a smooth vector field on M. The main goal of this section is to show well-definedness of $\mathcal{L}_X$ and proof the following theorem, describing the relationships between $\mathcal{L}_X, \iota_X$ and the exterior derivative d.

Theorem 2.1. *Let M be a manifold and $X, Y \in \mathscr{X}(M)$. Then the maps*

$$\begin{aligned} d\colon \Omega^r(M) &\to \Omega^{r+1}(M), \\ \mathcal{L}_X\colon \Omega^r(M) &\to \Omega^r(M), \\ \iota_X\colon \Omega^r(M) &\to \Omega^{r-1}(M) \end{aligned}$$

satisfy the following Leibniz rules for $\alpha, \beta \in \Omega^\bullet(M)$:

$$\begin{aligned} d(\alpha \wedge \beta) &= d\alpha \wedge \beta + (-1)^{|\alpha|}\alpha \wedge d\beta, \\ \mathcal{L}_X(\alpha \wedge \beta) &= \mathcal{L}_X\alpha \wedge \beta + \alpha \wedge \mathcal{L}_X\beta, \\ \iota_X(\alpha \wedge \beta) &= \iota_X\alpha \wedge \beta + (-1)^{|\alpha|}\alpha \wedge \iota_X\beta, \end{aligned}$$

as well as the following identities:

(i) $d^2 = 0$,

(ii) $d\mathcal{L}_X = \mathcal{L}_X d$,

(iii) $\mathcal{L}_X = d\iota_X + \iota_X d$,

(iv) $\mathcal{L}_X\mathcal{L}_Y - \mathcal{L}_Y\mathcal{L}_X = \mathcal{L}_{[X,Y]}$,

(v) $\mathcal{L}_X\iota_Y - \iota_Y\mathcal{L}_X = \iota_{[X,Y]}$,

(vi) $\iota_X\iota_Y + \iota_Y\iota_X = 0$.

Firstly let us restate the definition of the Lie derivative. For a smooth manifold M without boundary, $\alpha \in \Omega^r(M)$ and $X \in \mathscr{X}(M)$ we define

$$(\mathcal{L}_X\alpha)_p := \left.\frac{d}{dt}\right|_{t=0} (\phi_t^*\alpha)_p,$$

where $\phi_t = \phi(t, -)$ denotes the flow of X.

Lemma 2.2. *$(\mathcal{L}_X\alpha)_p$ exists for all $p \in M$ and defines a smooth r-form on M.*

Before showing the lemma, let us introduce the notion of a smooth family of r-forms.

Definition 2.3 (Smooth family of r-forms)**.** Let β_t be a family of r-forms on M indexed over some open Interval I. We call β_t smooth if in some (then all) local coordinates $x^1, \ldots, x^n$ we can write

$$(\beta_t)_p = \sum \beta^{i_1,\ldots,i_r}(t,p) d_p x_{i_1} \wedge \ldots \wedge d_p x_{i_r}$$

for smooth functions $\beta^{i_1,\ldots,i_r}$. If I is closed we call β_t smooth if it can be extended to a smooth family β'_t indexed over some open open interval $I' \supseteq I$.

Then $t \mapsto (\beta_t)_p$ is a smooth curve in the vector space $\Lambda^r(T_p^*M)$, such that

$$\left(\left.\frac{d}{dt}\right|_{t=t_0} \beta_t\right)_p := \left.\frac{d}{dt}\right|_{t=t_0} (\beta_t)_p$$

is a well-defined r-form. Furthermore, in local coordinates we have

$$\left.\frac{d}{dt}\right|_{t=t_0} \beta_t = \sum \frac{\partial \beta^{i_1,\ldots,i_r}}{\partial t}(t_0, -) dx_{i_1} \wedge \ldots \wedge dx_{i_r},$$

such that this r-form is also smooth and moreover depends smoothly on t_0. So

$$\dot{\beta}_t := \frac{d}{dt}\beta_t$$

defines a smooth family of r-forms by

$$\left(\frac{d}{dt}\beta_t\right)_{t_0} := \frac{d}{dt}\bigg|_{t=t_0} \beta_t.$$

Note that if $\partial M = \emptyset$ we can also directly allow for closed intervals I in the definition above. However we may then always extend the family β_t to an open interval (in fact to all of $\mathbb{R}$), as β_t can equivalently be considered a smooth r-form β on $M \times I \subseteq M \times \mathbb{R}$. Under this correspondence we have the following interesting relationship (without proof):

$$\dot{\beta}_t = \mathcal{L}_{\frac{\partial}{\partial t}}\beta.$$

Proof of Lemma 2.2. Let $x_1, \ldots, x_n$ be local coordinates on some open neighbourhood $U \subseteq M$ of p. Choose a smaller neighbourhood $U' \subseteq U$ of p and $\varepsilon > 0$, such that the flow

$$\phi\colon (-\varepsilon, \varepsilon) \times U' \to U$$

of X is defined. Write

$$\alpha = \sum \alpha^{i_1,\ldots,i_r} dx_{i_1} \wedge \ldots \wedge dx_{i_r}.$$

Then:

$$(\phi_t^*\alpha)_p = \sum \alpha^{i_1,\ldots,i_r}(\phi_t(p))\, d_p(x_{i_1} \circ \phi_t) \wedge \ldots \wedge d_p(x_{i_r} \circ \phi_t)$$

Expanding the differentials

$$d_p(x_{i_k} \circ \phi_t) = \sum_{j=1}^{n} \frac{\partial(x_{i_k} \circ \phi_t)}{\partial x_j}(p) d_p x_j$$

we see that $\beta_t := \phi_t^*\alpha$ is a smooth family of r-forms on U' indexed over $t \in (-\varepsilon, \varepsilon)$. Thus

$$(\mathcal{L}_X\alpha)_p = \left(\frac{d}{dt}\bigg|_{t=0} \beta_t\right)_p$$

exists and depends smoothly on p.

$\square$

Proposition 2.4 (Basic properties of the Lie derivative)**.** *Let $\alpha \in \Omega^r(M), \beta \in \Omega^s(M)$, $\lambda \in \mathbb{R}$ and $X \in \mathscr{X}(M)$. Then the Lie derivative satisfies the following properties:*

(i) Locality: $(\mathcal{L}_X\alpha)_U = \mathcal{L}_{X|_U}\alpha|_U$ for $U \subseteq M$ open

(ii) Linearity in α: $\mathcal{L}_X(\lambda\alpha + \beta) = \lambda\mathcal{L}_X\alpha + \mathcal{L}_X\beta$

(iii) Leibniz rule: $\mathcal{L}_X(\alpha \wedge \beta) = \mathcal{L}_X\alpha \wedge \beta + \alpha \wedge \mathcal{L}_X\beta$

(iv) Compatibility with d: $\mathcal{L}_X(d\alpha) = d(\mathcal{L}_X\alpha)$

Proof. (i) follows from the fact that locally both sides are given by

$$\frac{d}{dt}\bigg|_{t=0} \beta_t$$

for the smooth family β_t defined in the proof of Lemma 2.2. Moreover, (ii) follows immediately from the definition. To see (iii) write

$$\mathcal{L}_X(\alpha \wedge \beta)_p = \frac{d}{dt}\bigg|_{t=0} (\phi_t^*\alpha)_p \wedge (\phi_t^*\beta)_p.$$

We claim: For a vector space V and smooth curves $\alpha_t \in \Lambda^r(V^*), \beta_t \in \Lambda^s(V^*)$ the following holds:

$$\left.\frac{d}{dt}\right|_{t=0}(\alpha_t \wedge \beta_t) = \left(\left.\frac{d}{dt}\right|_{t=0}\alpha_t\right) \wedge \beta_0 + \alpha_0 \wedge \left(\left.\frac{d}{dt}\right|_{t=0}\beta_t\right)$$

W.l.o.g. let

$$\alpha_t = a(t)\alpha',\ \beta_t = b(t)\beta'$$

for some $\alpha' \in \Lambda^r(V^*), \beta' \in \Lambda^s(V^*)$ and smooth functions a, b. Then the general case follows from linearity. We compute:

$$\begin{aligned}
\left.\frac{d}{dt}\right|_{t=0}(\alpha_t \wedge \beta_t) &= \left.\frac{d}{dt}\right|_{t=0} a(t)b(t)\alpha' \wedge \beta' \\
&= \left(\frac{da}{dt}(0)b(0) + a(0)\frac{db}{dt}(0)\right)\alpha' \wedge \beta' \\
&= \frac{da}{dt}(0)\alpha' \wedge b(0)\beta' + a(0)\alpha' \wedge \frac{db}{dt}(0)\beta' \\
&= \left(\left.\frac{d}{dt}\right|_{t=0}\alpha_t\right) \wedge \beta_0 + \alpha_0 \wedge \left(\left.\frac{d}{dt}\right|_{t=0}\beta_t\right).
\end{aligned}$$

This shows the claim and thus (iii). To see (iv) let β_t be again the smooth family of r-forms defined in the proof of Lemma 2.2 and write in local coordinates $x_1, \ldots, x_n$:

$$\beta_t = \sum \beta^{i_1,\ldots,i_r}(t,-)dx_{i_1} \wedge \ldots \wedge dx_{i_r}.$$

By linearity we may assume that all functions $\beta^{i_1,\ldots,i_r}$ except for one vanish, i.e.

$$\beta_t = f(t,-)dx_{j_1} \wedge \ldots \wedge dx_{j_r}$$

for some smooth function f and $1 \leq j_1 < \cdots < j_r \leq n$. Then

$$d\beta_t = \sum_{k=1}^{n} \frac{\partial f(t,-)}{\partial x_k} dx^k \wedge dx_{j_1} \wedge \ldots \wedge dx_{j_r}.$$

Thus

$$\begin{aligned}
\mathcal{L}_X(d\alpha) &= \left.\frac{d}{dt}\right|_{t=0} \phi_t^* d\alpha = \left.\frac{d}{dt}\right|_{t=0} d\beta_t \\
&= \sum_{k=1}^{n} \frac{\partial^2 f}{\partial t \partial x_k}(0,-)dx_k \wedge dx_{j_1} \wedge \ldots \wedge dx_{j_r} \\
&= d\left(\frac{\partial f}{\partial t}(0,-)dx_{j_1} \wedge \ldots \wedge dx_{j_r}\right) \\
&= d(\left.\frac{d}{dt}\right|_{t=0}\beta_t) = d(\mathcal{L}_X\alpha)
\end{aligned}$$

□

For convenience, let us also restate the definition of ι_X

Definition 2.5. Let $X \in \mathscr{X}(M)$ and $\alpha \in \Omega^r(M)$. For $r > 0$ define

$$\iota_X\alpha := \alpha(X,-) \in \Omega^{r-1}(M)$$

For $r = 0$ let $\iota_X\alpha := 0$.

Clearly ι_X is linear and only depends on the local behaviour of X and α. Furthermore, it is easily seen that

$$\iota_X(\alpha \wedge \beta) = \iota_X\alpha \wedge \beta + (-1)^{|\alpha|}\alpha \wedge \iota_X\beta.$$

Thus far the Lie derivative has only been defined on manfiolds without boundary. To extend the definition of $\mathcal{L}_X\alpha$ to manifolds with boundary, first choose extensions X' and α' of X and α, respectively, to the double of M and then define

$$\mathcal{L}_X\alpha := (\mathcal{L}_{X'}\alpha')|_M$$

Of course a priori it is not clear that this definition does not depend on the choice of X' and α'. However, the following proposition, which is one of the formulas of Theorem 2.1, shows that it is in fact independent.

Proposition 2.6 (Cartan's magic formula)**.** *Let $X \in \mathscr{X}(M)$. Then the following identity holds:*

$$\mathcal{L}_X = d\iota_X + \iota_X d$$

Proof. Let $\mathcal{D}_X := d\iota_X + \iota_X d$. From the discussion above it is easy to verify:

$$\mathcal{D}_X(\alpha \wedge \beta) = \mathcal{D}_X\alpha \wedge \beta + \alpha \wedge \mathcal{D}_X\beta.$$

Moreover, $\mathcal{D}_X$ clearly satisfies locality, linearity and commutes with d. Thus we can establish

$$\mathcal{D}_X = \mathcal{L}_X$$

by showing equality on $\Omega^0(M)$. So let $f \in \Omega^0(M)$, then

$$\mathcal{D}_X f = \iota_X df = X \cdot f$$

and

$$\mathcal{L}_X f = \left.\frac{d}{dt}\right|_{t=0} (f \circ \phi_t) = df(X) = X \cdot f$$

This shows the proposition. ☐

Now let us proof the remaining formulas of Theorem 2.1, which are shown similiarly.

Proof of Theorem 2.1. Equation (i) is only mentioned for the sake of completeness and will not be shown here. See for example [Ste23a] for a proof. Equations (ii) and (iii) have already been shown above.

To see (iv) note that again both sides of the equation are differential operators satisfying the same Leibniz rule and commute with d, such that it suffices to show equality on $\Omega^0(M)$. But this follows immediately from the definition of the Lie bracket.

We argue similiarly for (v), but in this equation the differential operators on each side do not commute with d. However they still satisfy the same Leibniz rule, so we only need to show equality on $\Omega^0(M)$ as well as on the exact one-forms. Clearly both sides vanish on $\Omega^0(M)$ (by definition of the interior product) and for $\alpha = df \in \Omega^1(M)$ we compute:

$$\begin{aligned}(\mathcal{L}_X\iota_Y - \iota_Y\mathcal{L}_X)(df) &= \mathcal{L}_X(Y \cdot f) - (\iota_Y d\mathcal{L}_X)(f) \\ &= XYf - YXf \\ &= [X,Y] \cdot f = \iota_{[X,Y]}(df).\end{aligned}$$

Finally, (vi) is just a way to rewrite

$$\alpha(X, Y, -) = -\alpha(Y, X, -).$$

☐

As an applictaion of Theorem 2.1 we can show the following identity:

Corollary 2.7. *For $\alpha \in \Omega^1(M)$ and $X, Y \in \mathscr{X}(M)$ the following identity holds:*

$$d\alpha(X, Y) = X \cdot \alpha(Y) - Y \cdot \alpha(X) - \alpha([X, Y])$$

Proof. By Theorem 2.1 (iii) we have:

$$(\iota_Y \mathcal{L}_X)(\alpha) = Y \cdot \alpha(X) + d\alpha(X, Y).$$

Moreover, applying (v) yields:

$$(\iota_Y \mathcal{L}_X)(\alpha) = X \cdot \alpha(Y) - \alpha([X, Y]).$$

The corollary follows. $\square$

Integrating time dependent forms

Let β_t be a smooth time dependent family of differential forms on a manifold M defined on some interval I. At the beginning of the section we described what it means to take the derivative of β_t with respect to time. Often one is also interested in the *integral* with respect to time, which is defined similiarly, i.e. pointwise. For $a, b \in I$ and $p \in M$ we set

$$\left(\int_a^b \beta_t dt \right)_p := \int_a^b (\beta_t)_p dt,$$

where the right side is just the integral of a smooth curve $[a, b] \to \Lambda^r(T_p^* M)$ (see section 11.2 for a definition of such an integral). Furthermore, in local coordinates we have

$$\int_a^b \beta_t dt = \sum \int_a^b \beta^{i_1, \ldots, i_r}(t, -) dt \; dx_{i_1} \wedge \ldots \wedge dx_{i_r},$$

so $\int_a^b \beta_t dt$ is a smooth r-form on M.

2.2 Computation rules for time-dependent forms

The goal of this section is to establish a few coordinate free rules for computing expressions involving the derivative or integral of time dependent differential forms in practice. We begin by discussing some elementary common cases and explain how to deal with more complex ones later on.

The Lie derivative $\mathcal{L}_X \alpha$ was defined as

$$\left. \frac{d}{dt} \right|_{t=0} \phi_t^* \alpha,$$

where ϕ_t is the flow of X. We might also want to differentiate this expression at some other time $t = t_0$. The following proposition shows how this can be computed in terms of $\mathcal{L}_X \alpha$. In fact, we show a more general statement, where ϕ_t is the flow of a time dependent vector field.

Proposition 2.8. *Let M be a manifold without boundary, X_t a time dependent vector field on M and $\alpha \in \Omega^r(M)$. Denote by ϕ_t the flow of X_t and consider the open set $M_{t_0} = \{p \in M \mid \phi_t(p)$ exists for $t = t_0\}$. Then*

$$\left. \frac{d}{dt} \right|_{t=t_0} (\phi_t^* \alpha)_p = \phi_{t_0}^* (\mathcal{L}_{X_{t_0}} \alpha)_p$$

for all points $p \in M_{t_0}$.

Proof. We begin by introducing some notation. Say X_t is defined for time $t = t_1$. We then write $\phi_{t_1,t}$ for the solution of the following nonautonomous ODE

$$\frac{d}{dt} \phi_{t_1,t} = X_{t+t_1}(\phi_{t_1,t}),$$

i.e. the flow of the reparametrised time-dependent vector field X_{t+t_1}. Then we have $\phi_{0,t} = \phi_t$ and

$$\phi_{t+t_0} = \phi_{t_0,t} \circ \phi_{0,t_0}$$

wherever this equation makes sense. Then we can compute:

$$\begin{aligned}
\frac{d}{dt}\bigg|_{t=t_0}(\phi_t^*\alpha)_p &= \frac{d}{dt}\bigg|_{t=0}(\phi_{t+t_0}^*\alpha)_p \\
&= \frac{d}{dt}\bigg|_{t=0}(\phi_{0,t_0}^*\phi_{t_0,t}^*\alpha)_p \\
&= \phi_{0,t_0}^*\frac{d}{dt}\bigg|_{t=0}(\phi_{t_0,t}^*\alpha)_p
\end{aligned}$$

We claim that

$$\frac{d}{dt}\bigg|_{t=0}(\phi_{t_0,t}^*\alpha)_p = (\mathcal{L}_{X_{t_0}}\alpha)_p$$

To see this, we can explicitly express both sides in local coordinates and then use (ii) from Lemma 2.9 and (i) from Lemma 2.10 to reduce this to the case, where α is a 0-form, which follows immediately. Essentially, this is again using the trick that two differential operators are equal if they satisfy the same Leibniz rule, commute with d and agree on 0-forms. □

Lemma 2.9. *Let α_t, β_t be smooth time-dependent families of differential forms on a manifold M and $f\colon M \to N$ a smooth map. Then the following holds:*

(i) $\frac{d}{dt}f^*\alpha_t = f^*\frac{d}{dt}\alpha_t$

(ii) $\frac{d}{dt}(\alpha_t \wedge \beta_t) = \frac{d}{dt}\alpha_t \wedge \beta_t + \alpha_t \wedge \frac{d}{dt}\beta_t$

Proof. For (i) note that for a given point $p \in M$ the pullback along f is given by a linear map

$$f^*\colon \Lambda^r(T_{f(p)}^*N) \to \Lambda^r(T_p^*M),$$

so the statement follows directly from the chain rule. (ii) follows from the ordinary Leibniz rule for smooth functions and can be seen by computing the derivative in local coordinates. We have in fact already seen this computation in the proof of Proposition 2.4. □

This next Lemma shows, that the time derivative of time-dependent differential forms commutes with the differential operators from Theorem 2.1.

Lemma 2.10. *Let α_t be a time-dependent family of differential forms and X a vector field on a manifold M. Then the following holds:*

(i) $\frac{d}{dt}d\alpha_t = d\frac{d}{dt}\alpha_t$

(ii) $\frac{d}{dt}\mathcal{L}_X\alpha_t = \mathcal{L}_X\frac{d}{dt}\alpha_t$

(iii) $\frac{d}{dt}\iota_X\alpha_t = \iota_X\frac{d}{dt}\alpha_t$

Proof. (i) follows from Schwarz's Theorem by computing the expression in local coordinates. This computation has essentially already been carried out in the proof of Proposition 2.4. (ii) follows from (i) and (iii) using Cartan's magic formula. For (iii) note that for $p \in M$ the map

$$\iota_{X_p}\colon \Lambda^r(T_p^*M) \to \Lambda^{r-1}(T_p^*M)$$

is linear so the statement follows again from the chain rule. □

Moreover, the integral with respect to time also commutes with the differential operators, as the following Lemma shows.

Lemma 2.11. *Let α_t be a time-dependent family of differential forms defined on some interval I and X a vector field on a manifold M. Given $a, b \in I$ the following holds:*

(i) $d\int_a^b \alpha_t dt = \int_a^b d\alpha_t dt$

(ii) $\mathcal{L}_X\int_a^b \alpha_t dt = \int_a^b \mathcal{L}_X\alpha_t dt$

(iii) $\iota_X\int_a^b \alpha_t dt = \int_a^b \iota_X\alpha_t dt$

Proof. For (i) choose local coordinates $x_1, \dots, x_n$ and let $\alpha^{i_1,\dots,i_r}(t,x)$ be a component function of α_t with respect to these coordinates. Then

$$\frac{\partial}{\partial x_i}\int_a^b \alpha^{i_1,\dots,i_r}(t,x)dt = \int_a^b \frac{\partial}{\partial x_i}\alpha^{i_1,\dots,i_r}(t,x)dt$$

since $[a,b]$ is compact. The statement follows. Again (ii) follows from (i) and (iii) using Cartan's magic formula. (iii) follows from linearity of the integral (see Lemma 11.8) by again considering the linear map ι_{X_p} for $p \in M$. □

Finally, we explain how to deal with time dependent vector fields in the differential operators.

Lemma 2.12. *Let X_t be a time-dependent vector field on a manifold M. Then the following holds:*

(i) $\frac{d}{dt}\mathcal{L}_{X_t} = \mathcal{L}_{\frac{d}{dt}X_t}$

(ii) $\frac{d}{dt}\iota_{X_t} = \iota_{\frac{d}{dt}X_t}$

Proof. (i) follows from (ii) as well as Lemma 2.10 (i) using Cartan's magic formula. For (ii) let $\alpha \in \Omega^r(M)$ and consider the linear map

$$A\colon T_pM \to \Lambda^{r-1}(T_p^*M),\ \ v \mapsto \alpha_p(v,-)$$

for $p \in M$. Then

$$\left(\frac{d}{dt}\iota_{X_t}\alpha\right)_p = \frac{d}{dt}A(X_t(p))$$

and so the statement follows again from the chain rule. □

Remark 2.13. Of course analogous statements hold for $\mathbb{R}^n$-valued time parameters $t \in \mathbb{R}^n$.

Moser's trick

In many cases one has to differentiate a time-dependent family of differential forms, which is given by an expression where the time variable appears at multiple instances. To understand how a formula for this is obtained, we begin by discussing an important example, namely Moser's trick, which is used to proof the Darboux Theorem about symplectic/contact manfiolds. Following that, we present a straightforward cooking recipe on how to compute the derivative of such an expression in general, illustrated by another example.

Let M be a manifold (without boundary). Given a smooth family α_t of differential forms on M indexed over $t \in \mathbb{R}$ we want to find a family of diffeomorphisms $\varphi_t\colon M \to M$ (i.e. an isotopy), such that $\varphi_0 = id_M$ and

$$\varphi_t^*\alpha_t = \alpha_0 \ \text{ for all } t \in [0,1],$$

or equivalently:

$$\frac{d}{dt}\varphi_t^*\alpha_t = 0. \tag{3}$$

Directly constructing such an isotopy is difficult but we can make use of the following trick: The isotopy φ_t corresponds to the level preserving diffeomorphism

$$\Phi\colon M \times [0,1] \to M \times [0,1], (p,t) \mapsto (\varphi_t(p), t).$$

This diffeomorphism then defines a vector field $\Phi_*\frac{\partial}{\partial t}$ on $M \times [0,1]$, whose integral curves uniquely determine φ_t. This suggests that in order to construct an isotopy φ_t we can construct a vector field Y on $M \times [0,1]$, of the form

$$Y_{(p,t)} = ((X_t)_p, \frac{\partial}{\partial t})$$

for a time dependent vector field X_t on M (this trick is also explained in [Mil65] Thm. 5.6).

Then (3) can be reformulated as a condition on X_t, which is shown in the following Lemma.

Lemma 2.14. *In the situation above the following holds:*

$$\frac{d}{dt}\varphi_t^*\alpha_t = \varphi_t^*(L_{X_t}\alpha_t + \dot{\alpha}_t)$$

Proof. The idea is to "decouple" the time variables and consider the two parameter family

$$g\colon [0,1]\times\mathbb{R}\to T_p^*M,\ (t,s)\mapsto(\varphi_t^*\alpha_s)_p$$

The partial derivatives are now much easier to compute. By Proposition 2.8 we have:

$$\left.\frac{\partial}{\partial t}\right|_{t=t_0}\varphi_t^*\alpha_s=\varphi_{t_0}^*(\mathcal{L}_{X_{t_0}}\alpha_s)$$

Moreover, by Lemma 2.9

$$\left.\frac{\partial}{\partial s}\right|_{s=t_0}\varphi_t^*\alpha_s=\varphi_t^*\left(\left.\frac{\partial}{\partial s}\right|_{s=t_0}\alpha_s\right)=\varphi_t^*\dot\alpha_{t_0}.$$

In order to compute the derivative of

$$f\colon[0,1]\to T_p^*M,\ t\mapsto(\varphi_t^*\alpha_t)_p$$

consider the diagonal map

$$\Delta\colon[0,1]\to[0,1]\times\mathbb{R},\ t\mapsto(t,t).$$

Then

$$f=g\circ\Delta,$$

hence

$$\begin{aligned}\left.\frac{d}{dt}\right|_{t=t_0}f(t)&=D_{t_0}f(1)=D_{(t_0,t_0)}g\circ D_{t_0}\Delta(1)\\&=D_{(t_0,t_0)}(1,1)=D_{(t_0,t_0)}(1,0)+D_{(t_0,t_0)}(0,1)\\&=\frac{\partial g}{\partial t}(t_0,t_0)+\frac{\partial g}{\partial s}(t_0,t_0)\end{aligned}$$

The Lemma follows. □

So (3) can be reformulated to the condition:

$$0=L_{X_t}\alpha_t+\dot\alpha_t=d\iota_{X_t}\alpha_t+\iota_{X_t}d\alpha_t+\dot\alpha_t,$$

where we used Theorem 2.1 (iii) in the second equation. Hence it suffices to construct a time dependent vector field X_t satisfying the above conditon and whose integral curves exist for time $t=1$.

A cooking recipe

The proof of Lemma 2.14 now gives the announced cooking recipe for computing derivatives of smooth families of differential forms indexed over some time t, where t appears at multiple instances. Namely, fix t for all but one instance and differentiate. Then take the sum of all of the results. We illustrate this by an example.

Example 2.15. Let X_t be some time dependent vector field on a manifold M with flow ϕ_t. Furthermore, let α_t be a smooth family of differential forms and f_t a smooth family of functions on M. We want to compute:

$$\frac{d}{dt}\phi_t^*(e^{f_t}\alpha_t).$$

Begin by fixing all but the first occurence of t and differentiate, i.e. compute

$$\begin{aligned}\frac{\partial}{\partial t}\phi_t^*(e^{f_s}\alpha_s)=\phi_t^*(\mathcal{L}_{X_t}(e^{f_s}\alpha_s))&=\phi_t^*((X_t\cdot e^{f_s})\alpha_s+e^{f_s}\mathcal{L}_{X_t}\alpha_s)\\&=\phi_t^*((X_t\cdot f_s)e^{f_s}\alpha_s+e^{f_s}\mathcal{L}_{X_t}\alpha_s)\end{aligned}$$

Here we used Proposition 2.8 and the Leibniz rule for the Lie derivative. Next fix all but the second occurence of t and using Lemma 2.9 compute:

$$\frac{\partial}{\partial t}\phi_s^*(e^{f_t}\alpha_s)=\phi_s^*(\frac{\partial}{\partial t}e^{f_t}\alpha_s)=\phi_s^*(\dot f_te^{f_t}\alpha_s)$$

And lastly
$$\frac{\partial}{\partial t}\phi_s^*(e^{f_s}\alpha_t) = \phi_s^*(e^{f_s}\frac{\partial}{\partial t}\alpha_t) = \phi_s^*(e^{f_s}\dot{\alpha}_t)$$
Finally plug in $s = t$ and sum the results:
$$\begin{aligned}\frac{d}{dt}\phi_t^*(e^{f_t}\alpha_t) &= \phi_t^*((X_t \cdot f_t)e^{f_t}\alpha_t + e^{f_t}\mathcal{L}_{X_t}\alpha_t) + \phi_t^*(\dot{f}_t e^{f_t}\alpha_t) + \phi_t^*(e^{f_t}\dot{\alpha}_t)\\ &= \phi_t^*(e^{f_t}((X_t \cdot f_t)\alpha_t + \mathcal{L}_{X_t}\alpha_t + \dot{f}_t\alpha_t + \dot{\alpha}_t))\end{aligned}$$

2.3 Formulas for the Hodge star operator and codifferential

In this section we briefly discuss the definition of the Hodge star operator and codifferential as well as their basic properties. This is supposed to serve as a canonical reference for the correct signs in the formulas involved. After the basic properties we proof a few more formulas that might be interesting.

Let $(V, \langle\cdot,\cdot\rangle)$ be an oriented n-dimensional euclidian vector space. Then the Hodge star operator is the unique linear map
$$\star\colon \Lambda^k V \to \Lambda^{n-k}V$$
satisfying
$$\alpha \wedge \star\beta = \langle\alpha,\beta\rangle \mathrm{vol},$$
for all $\alpha,\beta \in \Lambda^k V$, where $\mathrm{vol} \in \Lambda^n V$ is the canonical volume form on V. If $e_1,\dots,e_n \in V$ is an oriented orthonormal basis we see that for any ordered tuple $(i_1,\dots,i_k)$
$$\star(e_{i_1} \wedge \ldots \wedge e_{i_k}) = e_{j_1} \wedge \ldots \wedge e_{j_{n-k}},$$
where $(j_1,\dots,j_{n-k})$ is the complementary ordered tuple. Since tuples of this form define an orthonormal basis it follows that $\star$ is an isometry. Using this fact we compute
$$\begin{aligned}\langle\star\star\beta,\alpha\rangle\mathrm{vol} &= \star\star\beta \wedge \star\alpha = (-1)^{k(n-k)} \star\alpha \wedge \star\star\beta\\ &= (-1)^{k(n-k)}\langle\star\alpha,\star\beta\rangle\mathrm{vol} = (-1)^{k(n-k)}\langle\alpha,\beta\rangle\mathrm{vol}\end{aligned}$$
from which we conclude that
$$\star\star = (-1)^{k(n-k)}\mathrm{id}_{\Lambda^k V}.$$
The Hodge star operator can also be used to relate the interior and exterior product. For $v \in V$ and $\omega \in \Lambda^k V^*$ we denote the interior product by $v \lrcorner \omega \in \Lambda^{k-1}V^*$, that is
$$(v \lrcorner \omega)(v_1,\dots,v_{k-1}) := \omega(v, v_1,\dots,v_{k-1}).$$
(Some authors define this with the opposite sign, i.e. $v \lrcorner \omega := -\omega(v,-)$). Moreover the notation $v \wedge \omega \in \Lambda^{k+1}V^*$ is to be understood as first dualizing v and then taking the usual wedge product, that is
$$v \wedge \omega := \langle v, -\rangle \wedge \omega.$$
We then have the following Lemma.

Lemma 2.16. *For $v \in V$, $\omega \in \Lambda^k V$ and $\eta \in \Lambda^{k-1}V$ the following holds*

(i) $\langle\eta, v \lrcorner \omega\rangle = \langle v \wedge \eta, \omega\rangle$

(ii) $v \lrcorner \omega = (-1)^{n(k+1)} \star (v \wedge \star\omega)$

Proof. For (i) note that the expression on both sides is linear in v, ω and η, such that it suffices to show equality on basis vectors. So let $e_1,\dots,e_n \in V$ be an oriented orthonormal basis and $\varepsilon^1,\dots,\varepsilon^n \in V^*$ its dual basis. We have to compute (i) for
$$v = e_m, \;\; \omega = \varepsilon^{i_1} \wedge \ldots \wedge \varepsilon^{i_k}, \;\; \eta = \varepsilon^{j_1} \wedge \ldots \wedge \varepsilon^{j_{k-1}}.$$
First note that we may assume $m = i_p$ and
$$(i_1,\dots,\widehat{i_p},\dots,i_k) = (j_1,\dots,j_{k-1}).$$

for some $1 \leq p \leq k$ since otherwise both sides vanish. But then

$$v \lrcorner \omega = (-1)^{p-1}\eta$$

and

$$v \wedge \eta = (-1)^{p-1}\omega$$

from which (i) follows. Using this we can show (ii):

$$\begin{aligned}\langle \eta, v \lrcorner \omega\rangle \mathrm{vol} &= \langle v \wedge \eta, \omega\rangle \mathrm{vol} \\ &= v \wedge \eta \wedge \star\omega \\ &= (-1)^{k-1}\eta \wedge v \wedge \star\omega \\ &= (-1)^{k-1+(n-k+1)(k-1)}\eta \wedge \star\star(v \wedge \star\omega) \\ &= (-1)^{(n-k+2)(k-1)}\langle \eta, \star(v \wedge \star\omega)\rangle \mathrm{vol}\end{aligned}$$

Moreover we compute modulo 2:

$$(n-k+2)(k-1) = (n-k)(k-1) = n(k-1) - k(k-1) = n(k+1)$$

Since η was arbitrary the statement follows. □

Remark 2.17. Part (i) of Lemma 2.16 states that the interior product is the adjoint of the exterior product.

Next we show a useful formula for the exterior derivative and codifferential in terms of the Levi-Civita connection. Let (M, g) be an oriented n-dimensional Riemannian manfiold. First let's fix a sign convention for the codifferential δ. On k-forms we define δ by

$$\begin{array}{ccc}\Omega^k(M) & \xrightarrow{(-1)^k\delta} & \Omega^{k-1}(M) \\ \downarrow\star & & \downarrow\star \\ \Omega^{n-k}(M) & \xrightarrow{d} & \Omega^{n-k+1}(M)\end{array}$$

or in formulas

$$\delta := (-1)^k \star^{-1} d\star = (-1)^{n(k-1)+1} \star d\star \colon \Omega^k(M) \to \Omega^{k-1}(M).$$

With this sign convention we have for $\omega \in \Omega^k(M)$ and $\eta \in \Omega^{k+1}(M)$

$$d(\omega \wedge \star\eta) = d\omega \wedge \star\eta - \omega \wedge \star\delta\eta. \tag{4}$$

If $\alpha, \beta \in \Omega^k(M)$, such that the intersection of their supports is compact, we denote by

$$\langle\langle \alpha, \beta\rangle\rangle := \int_M \langle \alpha, \beta\rangle \mathrm{vol}$$

their L^2-scalar product. Using Stoke's Theorem it then follows from (4):

Lemma 2.18. *Let $\omega \in \Omega^k(M)$ and $\eta \in \Omega^{k+1}(M)$, such that the intersection of their supports is compact and contained in $M \setminus \partial M$. Then*

$$\langle\langle d\omega, \eta\rangle\rangle = \langle\langle \omega, \delta\eta\rangle\rangle.$$

With this sign conventions in mind we can now show:

Lemma 2.19. *Let M be a Riemannian manifold and $\omega \in \Omega^*(M)$. Let $e_1, \dots, e_n$ be an orthonormal local frame of TM, then*

$$d\omega = \sum_{i=1}^n e_i \wedge \nabla_i\omega,$$

where $\nabla_i\omega := \nabla_{e_i}\omega$ denotes the connection on the bundle Λ^*T^*M induced by the Levi-Civita connection on M. Moreover, if M is oriented we have

$$\delta\omega = -\sum_{i=1}^n e_i \lrcorner \nabla_i\omega.$$

Proof. We first show the formula for the exterior derivative for 1-forms $\alpha \in \Omega^1(M)$. Using Corollary 2.7 and the symmetry of the Levi-Civita connection we get

$$\begin{aligned} d\alpha(X,Y) &= X \cdot \alpha(Y) - Y \cdot \alpha(X) - \alpha([X,Y]) \\ &= \nabla_X(\alpha(Y)) - \nabla_Y(\alpha(X)) - \alpha(\nabla_X Y - \nabla_Y X) \\ &= (\nabla_X \alpha)(Y) - (\nabla_Y \alpha)(X) \end{aligned}$$

for any smooth vector fields $X, Y \in \mathscr{X}(M)$. Let $\varepsilon^1, \dots, \varepsilon^n$ be the dual frame of $e_1, \dots, e_n$. It then follows that

$$\begin{aligned} d\alpha &= \sum_{i,j=1}^n d\alpha(e_i, e_j)\varepsilon^i \otimes \varepsilon^j \\ &= \sum_{i=1}^n \varepsilon^i \otimes \left(\sum_{j=1}^n (\nabla_i \alpha)(e_j)\varepsilon^j \right) - \sum_{j=1}^n \left(\sum_{i=1}^n (\nabla_j \alpha)(e_i)\varepsilon^i \right) \otimes \varepsilon^j \\ &= \sum_{i=1}^n \varepsilon^i \otimes \nabla_i \alpha - \sum_{j=1}^n \nabla_j \alpha \otimes \varepsilon^j \\ &= \sum_{i=1}^n \varepsilon^i \wedge \nabla_i \alpha \\ &= \sum_{i=1}^n e_i \wedge \nabla_i \alpha \end{aligned}$$

where we have used $\varepsilon^i = \langle e_i, - \rangle$ in the last equality (since $e_1, \dots, e_n$ is orthonormal). Now define a differential operator D by

$$D\omega := \sum_{i=1}^n e_i \wedge \nabla_i \omega.$$

Clearly D agrees with the exterior derivative on 0-forms and we have shown above that they furthermore agree on 1-forms. Moreover

$$\begin{aligned} D(\omega \wedge \eta) &= \sum_{i=}^n e_i \wedge \nabla_i \omega \wedge \eta + e_i \wedge \omega \wedge \nabla_i \eta \\ &= (D\omega) \wedge \eta + (-1)^{|\omega|} \omega \wedge D\eta. \end{aligned}$$

So D and d also satisfy the same Leibniz rule. It follows that $D = d$. From this we can deduce the formula for the codifferential by using Lemma 2.16. So let $\omega \in \Omega^k(M)$.

$$\begin{aligned} \delta\omega &= (-1)^{n(k-1)+1} \star d \star \omega \\ &= (-1)^{n(k-1)+1} \sum_{i=1}^n \star(e_i \wedge \nabla_i \star \omega) \\ &= (-1)^{n(k-1)+1} \sum_{i=1}^n \star(e_i \wedge \star \nabla_i \omega) \\ &= (-1)^{n(k-1)+1+n(k+1)} \sum_{i=1}^n e_i \lrcorner \nabla_i \omega \\ &= -\sum_{i=1}^n e_i \lrcorner \nabla_i \omega. \end{aligned}$$

In the third equality we used the fact that covariant differentiation and the Hodge star operator commute, as shown in the Lemma 2.20 below. ☐

Lemma 2.20. *Let M be a an oriented Riemannian manifold, $\omega \in \Omega^*(M)$ and $X \in \mathscr{X}(M)$. Then*

$$\nabla_X \star \omega = \star \nabla_X \omega.$$

Proof. Let $\omega \in \Omega^k(M)$ and $\eta \in \Omega^{n-k}(M)$. Recall that the volume form $\mathrm{vol} \in \Omega^n(M)$ is parallel, i.e. $\nabla_X \mathrm{vol} = 0$. Hence

$$\nabla_X(\langle \eta, \star\omega\rangle \mathrm{vol}) = (\langle \nabla_X \eta, \star\omega\rangle + \langle \eta, \nabla_X \star \omega\rangle)\mathrm{vol}$$

On the other hand

$$\begin{aligned}
\nabla_X(\langle \eta, \star\omega\rangle \mathrm{vol}) &= \nabla_X(\eta \wedge \star \star \omega) \\
&= (-1)^{k(n-k)}(\nabla_X \eta \wedge \omega + \eta \wedge \nabla_X \omega) \\
&= \nabla_X \eta \wedge \star \star \omega + \eta \wedge \star \star \nabla_X \omega \\
&= (\langle \nabla_X \eta, \star\omega\rangle + \langle \eta, \star\nabla_X \omega\rangle)\mathrm{vol}.
\end{aligned}$$

Thus

$$\langle \eta, \nabla_X \star \omega\rangle = \langle \eta, \star\nabla_X \omega\rangle.$$

Since η was arbitrary the statement follows. $\square$

3 Riemannian Geometry

3.1 Product Manifolds

In the following let (M_1, g_1) and (M_2, g_2) be Riemannian manifolds. Then we can consider the product manifold

$$M := M_1 \times M_2$$

equipped with the product metric $g := g_1 \times g_2$ defined by

$$g(X_1 + X_2, Y_1 + Y_2) = g_1(X_1, Y_1) + g_2(X_2, Y_2), \tag{5}$$

where $X_i, Y_i \in \mathscr{X}(M_i)$. In this section we compute the Levi-Civita connection, curvature tensors, covariant derivative along curves and geodesics of (M, g) and see how they relate to the corresponding objects on (M_1, g_1) and (M_2, g_2).

Let us establish some notation used in the discussion that follows. For a vector field $Z \in \mathscr{X}(M)$ we write $\tilde{Z} \in \mathscr{X}(M)$ for the horizontal lift of Z, that ist

$$\tilde{Z}|_{(p_1,p_2)} := Z|_{p_1}$$

for $(p_1, p_2) \in M$. Moreover for a smooth function $f \in C^\infty(M_1)$ we define

$$\tilde{f} := f \circ \pi_1 \in C^\infty(M),$$

where $\pi_1 \colon M \to M_1$ denotes the projection. The same notation is used for vector fields and smooth functions on M_2. So to be precise, equation (5) should read

$$g(\tilde{X}_1 + \tilde{X}_2, \tilde{Y}_1 + \tilde{Y}_2) = \widetilde{g_1(X_1, Y_1)} + \widetilde{g_2(X_2, Y_2)}.$$

For the sake of notational simplicity we will often suppress these subtleties, if there is no risk of confusion and only write $X_1 + X_2$ instead of $\tilde{X}_1 + \tilde{X}_2$ or f instead of $\tilde{f}$. It should always be clear from the context what is meant. Let us moreover denote by $\nabla^1, D_t^1, \mathrm{Rm}_1, \mathrm{Ric}_1, \mathrm{scal}_1$ the Levi-Civita connection, covariant derivative along curves, Riemannian curvature tensor, Ricci tensor and scalar curvature of (M_1, g_1) respectively. Similiarly we denote the same objects on (M_2, g_2) by $\nabla^2, D_t^2, \mathrm{Rm}_2, \mathrm{Ric}_2, \mathrm{scal}_2$ and on (M, g) simply by $\nabla, D_t, \mathrm{Rm}, \mathrm{Ric}, \mathrm{scal}$. Finally we define $m_1 := \dim(M_1)$ and $m_2 := \dim(M_2)$.

Of course our first step will be to compute ∇ in terms of ∇^1 and ∇^2. For this we begin with a more general statement on constructing a canonical connection on product vector bundles, where we use the same notational convention for sections in the product bundle as for vector fields introduced above.

Proposition 3.1. *Let $E_1 \to M_1$ and $E_2 \to M_2$ be smooth vector bundles and let ∇^1, ∇^2 be connections on E_1, E_2 respectively. Then there exists a unique connection ∇ on the product bundle*

$$E_1 \times E_2 \to M_1 \times M_2,$$

such that for sections $s \in \Gamma(E_1 \times E_2)$ of the form $s = s_1 + s_2$, where $s_i \in \Gamma(E_i)$, and vector fields $X \in \mathscr{X}(M_1 \times M_2)$ of the form $X = X_1 + X_2$, where $X_i \in \mathscr{X}(M_i)$, the following holds:

$$\nabla_X s = \nabla^1_{X_1} s_1 + \nabla^2_{X_2} s_2.$$

We call ∇ the product connection of ∇^1 and ∇^2 and write $\nabla = \nabla^1 \times \nabla^2$.

Proof. We first show uniqueness. Let $X \in \mathscr{X}(M_1 \times M_2)$ be an arbitary vector field and $s \in \Gamma(E_1 \times E_2)$ an arbitary section. For $(p_1, p_2) \in M_1 \times M_2$ we can find open neighbourhoods $U_i \subseteq M_i$ of p_i, such that there exist local frames

$$\sigma^i_1, \ldots, \sigma^i_{n_i} \in \Gamma(E_i|_{U_i}),$$

where n_i is the rank of E_i and there exist local frames

$$E^i_1, \ldots, E^i_{m_i} \in \Gamma(TM_i|_{U_i}),$$

On $U := U_1 \times U_2$ we can write

$$s = s_1^{j_1}\sigma^1_{j_1} + s_2^{j_2}\sigma^2_{j_2}$$

and

$$X = X_1^{k_1}E^1_{k_1} + X_2^{k_2}E^2_{k_2}$$

for smooth functions $s_1^{j_1}, s_2^{j_2}, X_1^{k_1}, X_2^{k_2} \in C^\infty(U)$. Then

$$\begin{aligned}
\nabla_X s &= X_1^{k_1}\left(\nabla_{E^1_{k_1}}(s_1^{j_1}\sigma^1_{j_1}) + \nabla_{E^1_{k_1}}(s_2^{j_2}\sigma^2_{j_2})\right) + X_2^{k_2}\left(\nabla_{E^2_{k_2}}(s_1^{j_1}\sigma^1_{j_1}) + \nabla_{E^2_{k_2}}(s_2^{j_2}\sigma^2_{j_2})\right) \\
&= X_1^{k_1}\left((E^1_{k_1}\cdot s_1^{j_1})\sigma^1_{j_1} + s_1^{j_1}\nabla^1_{E^1_{k_1}}\sigma^1_{j_1} + (E^1_{k_1}\cdot s_2^{j_2})\sigma^2_{j_2}\right) \\
&\quad + X_2^{k_2}\left((E^2_{k_2}\cdot s_1^{j_1})\sigma^1_{j_1} + (E^2_{k_2}\cdot s_2^{j_2})\sigma^2_{j_2} + s_2^{j_2}\nabla^2_{E^2_{k_2}}\sigma^2_{j_2}\right) \qquad (6)
\end{aligned}$$

Since the last expression does not depend on ∇ this shows uniqueness. To show existence define a connection ∇^U on U by formula (6). It is easy to see that this does in fact define a connection on $E_1 \times E_2|_U$, which is compatible with the restricted connections $\nabla^1|_{U_1}$ and $\nabla^2|_{U_2}$ in the sense of the statement of the proposition. So by the uniqueness result this connection does not depend on the chosen frames, thus we get a well-defined connection ∇ on $E_1 \times E_2$ by

$$(\nabla_X s)|_U := \nabla^U_{X|_U} s|_U.$$

□

Proposition 3.2. *The Levi-Civita connection ∇ on (M, g) is given by*

$$\nabla = \nabla^1 \times \nabla^2.$$

Proof. To ensure clarity, we will be more precise with the notation in this proof, as explained at the beginning of the section. Let $X, Y \in \mathscr{X}(M)$ be of the form $X = \tilde{X}_1 + \tilde{X}_2, Y = \tilde{Y}_1 + \tilde{Y}_2$, where $X_i, Y_i \in \mathscr{X}(M_i)$. By Proposition 3.1 it suffices to show that the Levi Civita connection satisfies

$$\nabla_X Y = \widetilde{\nabla^1_{X_1}Y_1} + \widetilde{\nabla^2_{X_2}Y_2}.$$

We would like to use the Koszul formula to compute $\nabla_X Y$ and begin by simplifying some of the terms involved in this formula. For $Z \in \mathscr{X}(M_1), W \in \mathscr{X}(M_2)$ we claim that

$$[\tilde{Z}, \tilde{W}] = 0.$$

To see this choose coordinates $x^1, \ldots, x^{m_1}$ of M_1 and $y^1, \ldots, y^{m_2}$ of M_2. Write

$$Z = f^i\frac{\partial}{\partial x^i},\ W = g^j\frac{\partial}{\partial y^j}$$

for smooth functions f^i and g^j defined on the chosen coordinate neighbourhoods of M_1 and M_2 respectively. Then $x^1, \ldots, x^{m_1}, y^1, \ldots, y^{m_2}$ define coordinates of M and

$$\tilde{Z} = \tilde{f}^i\frac{\partial}{\partial x^i},\ \tilde{W} = \tilde{g}^j\frac{\partial}{\partial y^j}.$$

We now compute

$$\begin{aligned}
[\tilde{Z}, \tilde{W}] &= \tilde{f}^i\frac{\partial \tilde{g}^j}{\partial x^i}\frac{\partial}{\partial y^j} + \tilde{g}^j\left[\tilde{f}^i\frac{\partial}{\partial x^i}, \frac{\partial}{\partial y^j}\right] \\
&= \tilde{f}^i\frac{\partial \tilde{g}^j}{\partial x^i}\frac{\partial}{\partial y^j} - \tilde{g}^j\frac{\partial \tilde{f}^i}{\partial y^j}\frac{\partial}{\partial x^i} - \tilde{g}^j\tilde{f}^i\left[\frac{\partial}{\partial y^j}, \frac{\partial}{\partial x^i}\right].
\end{aligned}$$

Since $\tilde{g}^j$ does not depend on x^i we get

$$\frac{\partial \tilde{g}^j}{\partial x^i} = 0.$$

Similiarly

$$\frac{\partial \tilde{f}^i}{\partial y^j} = 0,$$

such that $[\tilde{Z},\tilde{W}]=0$ follows. For Z as above the Koszul formula now reads

$$
\begin{aligned}
g(\nabla_X Y,\tilde{Z}) &= \frac{1}{2}\Big(Xg(Y,\tilde{Z})+Yg(\tilde{Z},X)-\tilde{Z}g(X,Y)\\
&\quad -g(Y,[X,\tilde{Z}])-g(\tilde{Z},[Y,X])+g(X,[\tilde{Z},Y])\Big)\\
&= \frac{1}{2}\Big(\tilde{X}_1 g(\tilde{Y}_1,\tilde{Z})+\tilde{Y}_1 g(\tilde{Z},\tilde{X}_1)-\tilde{Z}g(\tilde{X}_1,\tilde{Y}_1)\\
&\quad -g(\tilde{Y}_1,[\tilde{X}_1,\tilde{Z}])-g(\tilde{Z},[\tilde{Y}_1,\tilde{X}_1])+g(\tilde{X}_1,[\tilde{Z},\tilde{Y}_1])\Big)
\end{aligned}
\tag{7}
$$

Besides the claim about the Lie bracket shown above we have further used, that $g(Y,\tilde{Z})=g(\tilde{Y}_1,\tilde{Z})$ is constant in the M_2-direction, such that $\tilde{X}_2\, g(\tilde{Y}_1,\tilde{Z})=0$, hence

$$Xg(Y,\tilde{Z})=\tilde{X}_2\, g(\tilde{Y}_1,\tilde{Z}).$$

An analogous statement holds for the second and third term. Moreover $[\tilde{X}_1,\tilde{Z}]=\widetilde{[X,Z]}$, such that $g(\tilde{Y}_2,[\tilde{X}_1,\tilde{Z}])=0$, hence

$$g(Y,[X,\tilde{Z}])=g(\tilde{Y}_1,[\tilde{x}_1,\tilde{Z}]).$$

Again analogous statements hold for the last two terms. Applying the Koszul formula to $\nabla^1_{X_1}Y_1$ we get

$$
\begin{aligned}
g_1(\nabla^1_{X_1}Y_1,Z) &= \frac{1}{2}\Big(X_1 g_1(Y_1,Z)+Y_1 g_1(Z,X_1)-Zg_1(X_1,Y_1)\\
&\quad -g_1(Y_1,[X_1,Z])-g_1(Z,[Y_1,X_1])+g_1(X_1,[Z,Y_1])\Big)
\end{aligned}
\tag{8}
$$

To compare this expression to (7) we note the following identities for $U,V\in\mathscr{X}(M_1)$ and $f\in C^\infty(M_1)$:

$$g(\tilde{U},\tilde{V})=\widetilde{g_1(U,V)}$$

just by definition of g and

$$\tilde{V}\tilde{f}=\widetilde{Vf}.$$

So (7) and (8) now yield

$$g(\nabla_X Y,\tilde{Z})=\widetilde{g_1(\nabla^1_{X_1}Y_1,Z)}=g(\widetilde{\nabla^1_{X_1}Y_1},\tilde{Z}).$$

Similiarly we get for $W\in\mathscr{X}(M_2)$

$$g(\nabla_X Y,\tilde{Z})=g(\widetilde{\nabla^2_{X_2}Y_2},\tilde{W}).$$

Together this implies

$$g(\nabla_X Y,\tilde{Z}+\tilde{W})=g(\widetilde{\nabla^1_{X_1}Y_1}+\widetilde{\nabla^2_{X_2}Y_2},\tilde{Z}+\tilde{W}).$$

From which the assertion follows.

□

Knowing the Levi-Civita connection on M, it is now easy to compute the curvature, which behaves just as expected.

Corollary 3.3. *Let $X=X_1+X_2$, $Y=Y_1+Y_2$, $Z=Z_1+Z_2$, $W=W_1+W_2$ be smooth vector fields on M, where $X_i,Y_i,Z_i,W_i\in\mathscr{X}$. Then the following holds*

(i) $\mathrm{Rm}(X,Y,Z,W)=\mathrm{Rm}_1(X_1,Y_1,Z_1,W_1)+\mathrm{Rm}_2(X_2,Y_2,Z_2,W_2)$

(ii) $\mathrm{Ric}(X,Y)=\mathrm{Ric}_1(X_1,Y_1)+\mathrm{Ric}_2(X_2,Y_2)$

(iii) $\mathrm{scal}=\mathrm{scal}_1+\mathrm{scal}_2$

Proof. By Proposition 3.2

$$\nabla_X \nabla_Y Z = \nabla_X (\nabla^1_{Y_1} Z_1 + \nabla^2_{Y_2} Z_2) = \nabla^1_{X_1}\nabla^1_{Y_1} Z_1 + \nabla^2_{X_2}\nabla^2_{Y_2} Z_2.$$

An analogous equation holds for $\nabla_Y \nabla_X Z$. Moreover

$$[X, Y] = [X_1, Y_1] + [X_2, Y_2]$$

as we have seen in the proof of Proposition 3.2, such that

$$\nabla_{[X,Y]} Z = \nabla^1_{[X_1,Y_1]} Z_1 + \nabla^2_{[X_2,Y_2]} Z_2.$$

Then (i) follows from the definition of Rm and the product metric g. For (ii) write

$$\mathrm{Ric}(X, Y) = \sum_{i=1}^{m_1} \mathrm{Rm}(X, E^1_i, E^1_i, Y) + \sum_{j=1}^{m_2} \mathrm{Rm}(X, E^2_j, E^2_j, Y)$$

for local orthonormal frames $E^1_1, \dots, E^1_{m_1}$ of TM_1 and $E^2_2, \dots, E^2_{m_2}$ of TM_2. By (i)

$$\begin{aligned}\mathrm{Rm}(X, E^1_i, E^1_i, Y) &= \mathrm{Rm}_1(X_1, E^1_i, E^1_i, Y_1) + \mathrm{Rm}_2(X_2, 0, 0, Y_2)\\ &= \mathrm{Rm}_1(X_1, E^1_i, E^1_i, Y_1).\end{aligned}$$

An analogous statement holds for the second term, such that

$$\begin{aligned}\mathrm{Ric}(X, Y) &= \sum_{i=1}^{m_1} \mathrm{Rm}_1(X_1, E^1_i, E^1_i, Y_1) + \sum_{j=1}^{m_2} \mathrm{Rm}_2(X_2, E^2_j, E^2_j, Y_2)\\ &= \mathrm{Ric}_1(X_1, Y_1) + \mathrm{Ric}_2(X_2, Y_2).\end{aligned}$$

Finally (iii) follows from (ii) in the same way. □

Finally we compute the covariant derivative along curves and geodesics of M, which again behave in exactly the way one would expect.

Corollary 3.4. *Let $\gamma = (\gamma_1, \gamma_2)\colon I \to M$ be a smooth curve and $V\colon I \to \gamma^*TM \cong \gamma_1^*TM_1 \oplus \gamma_2^*TM_2$ a smooth vector field along γ. Write $V = V_1 + V_2$ for $V_i\colon I \to \gamma_i^*TM$. Then*

$$D_t V = D^1_t V_1 + D^2_t V_2.$$

Proof. As in the proof of Proposition 3.2, we will take some care with the notation in this proof. To show the statement we compute both sides locally. To this end let $E^1_1, \dots, E^1_{m_1}$ be a local frame of TM_1 and $E^2_2, \dots, E^2_{m_2}$ a local frame of TM_2. Write

$$V_1(t) = V^i_1(t) E^1_i|_{\gamma_1(t)}$$

and

$$V_2(t) = V^j_2(t) E^2_j|_{\gamma_2(t)}.$$

Then

$$V(t) = V^i_1(t)\tilde{E}^1_i|_{\gamma(t)} + V^j_2(t)\tilde{E}^2_j|_{\gamma(t)}$$

and hence

$$\begin{aligned}D_t V = {}&\dot{V}^i_1(t)\tilde{E}^1_i|_{\gamma(t)} + V^i_1(t)\nabla_{\dot\gamma(t)}\tilde{E}^1_i\\ &+ \dot{V}^j_2(t)\tilde{E}^2_j|_{\gamma(t)} + V^j_2(t)\nabla_{\dot\gamma(t)}\tilde{E}^2_j.\end{aligned}$$

Moreover

$$D^1_t V_1 = \dot{V}^i_1(t) E^1_i|_{\gamma_1(t)} + V^i_1(t)\nabla^1_{\dot\gamma_1(t)} E^1_i$$

and similiarly for $D^2_t V_2$. By Proposition 3.2

$$\nabla_{\dot\gamma(t)}\tilde{E}^1_i = \widetilde{\nabla^1_{\dot\gamma_1(t)} E^1_i} \quad \text{and} \quad \nabla_{\dot\gamma(t)}\tilde{E}^2_j = \widetilde{\nabla^2_{\dot\gamma_2(t)} E^2_j}$$

and thus

$$D_t V = \widetilde{D^1_t V_1} + \widetilde{D^2_t V_2}.$$

□

Corollary 3.5. *Let $\gamma = (\gamma_1, \gamma_2)\colon I \to M$ be a smooth curve. Then γ is a geodesic if and only if γ_1 and γ_2 are geodesics.*

Proof. This now follows immediately form the formula from Corollary 3.4. □

3.2 Formulas for connections

Proposition 3.6. *Let $E_1 \to M$ and $E_2 \to M$ be smooth vector bundles and let ∇^1, ∇^2 be connections on E_1, E_2 respectively. Then there exists a unique connection ∇ on the tensor product bundle*

$$E_1 \otimes E_2 \to M,$$

such that for any vector field $X \in \mathscr{X}(M)$ and sections $s_1 \in \Gamma(E_1)$ and $s_2 \in \Gamma(E_2)$ the following holds:

$$\nabla_X(s_1 \otimes s_2) = \nabla^1_X s_1 \otimes s_2 + s_1 \otimes \nabla^2_X s_2.$$

Proof. The proof is analogous to the proof of Proposition 3.1, so we only summarize the idea brielfy. One notes that any section $s \in \Gamma(E_1 \otimes E_2)$ can be written locally as

$$s = \sum_{i,j} s_1^i \otimes s_2^j,$$

where s_1^i and s_2^j are local sections of E_1 and E_2 respectively. This then shows uniqueness and moreover gives us a way to define ∇ locally, which can be pasted to together to define ∇ globally just as in Proposition 3.1. □

Lemma 3.7. *Let M be a Riemannian manifold, $\omega, \eta \in \Omega^*(M)$ differential forms and $X \in \mathscr{X}(M)$ a vector field. Then*

$$\nabla_X(\omega \wedge \eta) = \nabla_X \omega \wedge \eta + \omega \wedge \nabla_X \eta,$$

where ∇_X denotes the connection on the bundle $\Lambda^* T^* M$ induced by the Levi-Civita connection on M.

Proof. By definition of the wedge product it suffices to show that ∇_X commutes with the anti-symmetrization map

$$\mathrm{Alt}\colon (T^*M)^{\otimes k} \to (T^*M)^{\otimes k}.$$

To see this let $\alpha_1, \ldots, \alpha_k \in \Omega^1(M)$. Then

$$\begin{aligned}\nabla_X(\mathrm{Alt}(\alpha_1 \otimes \ldots \otimes \alpha_k)) &= \nabla_X \left(\frac{1}{k!} \sum_\sigma \mathrm{sgn}(\sigma) \alpha_{\sigma(1)} \otimes \ldots \otimes \alpha_{\sigma(k)} \right) \\ &= \frac{1}{k!} \sum_\sigma \sum_i \mathrm{sgn}(\sigma) \alpha_{\sigma(1)} \otimes \ldots \otimes \nabla_X \alpha_{\sigma(i)} \otimes \ldots \otimes \alpha_{\sigma(k)}\end{aligned}$$

and

$$\begin{aligned}\mathrm{Alt}(\nabla_X(\alpha_1 \otimes \ldots \otimes \alpha_k)) &= \mathrm{Alt} \left(\sum_i \alpha_1 \otimes \ldots \otimes \nabla_X \alpha_i \otimes \ldots \otimes \alpha_k \right) \\ &= \sum_i \sum_\sigma \frac{1}{k!} \mathrm{sgn}(\sigma) \alpha_{\sigma(1)} \otimes \ldots \otimes \nabla_X \alpha_{\sigma(i)} \otimes \ldots \otimes \alpha_{\sigma(k)}.\end{aligned}$$

Since locally any section $\omega \in \Gamma((T^*M)^{\otimes k})$ can be written as a sum of tensor products of 1-forms the statement follows. □

Lemma 3.8. *Let $E \to M$ be a smooth vector bundle equipped with a metric and compatible connection. Then for any $X \in \mathscr{X}(M)$ the following diagram commutes*

$$\begin{array}{ccc} \Gamma(E) & \xrightarrow{\nabla_X} & \Gamma(E) \\ \big\downarrow \cong & & \big\downarrow \cong \\ \Gamma(E^*) & \xrightarrow{\nabla_X} & \Gamma(E^*) \end{array}$$

where the vertical isomorphisms are given by dualizing and the connection on E^* is the unique connection defined by

$$X \cdot (\alpha(s)) = (\nabla_X \alpha)(s) + \alpha(\nabla_X s)$$

for $s \in \Gamma(E), \alpha \in \Gamma(E^*)$.

Proof. This follows immediately from the definition of the connection on E^* and the compatibility of the metric. □

4 Algebraic Topology

4.1 General Statements about Homology

In the following let $(H_\bullet, \partial_\bullet)$ be any homology theory.

Lemma 4.1. *Let $A \subseteq X$ be a retract. Then $H_\bullet(X) \cong H_\bullet(X, A) \oplus H_\bullet(A)$.*

Proof. Let $r\colon X \to A$ be a retraction and $i\colon A \to X$ the inclusion. Then $H_\bullet(r) \circ H_\bullet(i) = \mathrm{id}_{H_\bullet(A)}$, in particular $H_\bullet(i)$ is injective. Hence the long exact sequence for the pair (X, A) breaks up into short split exact sequences

$$\begin{tikzcd} 0 \arrow[r] & H_n(A) \arrow[r] & H_n(X) \arrow[l, bend right, "H_n(r)"'] \arrow[r] & H_n(X,A) \arrow[r] & 0 \end{tikzcd}$$

and the splitting lemma yields the desired isomorphism. □

Lemma 4.2. *Let $f\colon (X, A) \to (Y, B)$ be a continuous map of pairs, such that $f\colon X \to Y$ and $f\colon A \to B$ are homotopy equivalences. Then f induces an isomorphism*

$$H_\bullet(f)\colon H_\bullet(X, A) \to H_\bullet(Y, B).$$

Remark 4.3. Note that $f\colon (X, A) \to (Y, B)$ is not necessarily a homotopy equivalence of *pairs*. See chapter 3 in [Ste25] for a counterexample.

Proof. Consider the map between the long exact sequences of the pairs (X, A) and (Y, B) induced by f:

$$\begin{tikzcd} \cdots \arrow[r] & H_n(A) \arrow[r] \arrow[d, "H_n(f)", "\cong"'] & H_n(X) \arrow[r] \arrow[d, "H_n(f)", "\cong"'] & H_n(X,A) \arrow[r] \arrow[d, "H_n(f)"] & H_{n-1}(A) \arrow[r] \arrow[d, "H_{n-1}(f)", "\cong"'] & H_{n-1}(X) \arrow[r] \arrow[d, "H_{n-1}(f)", "\cong"'] & \cdots \\ \cdots \arrow[r] & H_n(B) \arrow[r] & H_n(Y) \arrow[r] & H_n(Y,B) \arrow[r] & H_{n-1}(B) \arrow[r] & H_{n-1}(Y) \arrow[r] & \cdots \end{tikzcd}$$

The outer maps are isomorphisms by homotopy invariance and so the five lemma implies that the middle map is an isomorphism as well. □

Lemma 4.4. *Let $A \subseteq X$ be a closed neighbourhood-deformation retract. Then the projection $X \to X/A$ induces a natural isomorphism*

$$H_\bullet(X, A) \to H_\bullet(X/A, *)$$

.

Proof. Applying excision in the push-out

$$\begin{tikzcd} A \arrow[r] \arrow[d] & X \arrow[d] \\ * \arrow[r] & X/A \end{tikzcd}$$

shows that the projection induces an isomorphism. Furthermore, any map $(X, A) \to (Y, B)$ descends to a map $X/A \to Y/B$ and naturality then just follows from functoriality.

□

4.2 CW-Complexes

Proposition 4.5. *Let (X, A) be a relative CW-complex. If A is Hausdorff (resp. normal) then X is Hausdorff (resp. normal) as well. In particular every absolute CW-complex is Hausdorff and normal.*

Proof. We show the statement for the case that A is normal. The Hausdorff case follows completely analogously by replacing the sets to be seperated by point sets. So let $B, C \subseteq X$ be closed disjoint sets. We will show the following claim inductively:

Claim. There exist open disjoint sets $U_n, V_n \subseteq X_n$, such that $B_n := B \cap X_n \subseteq U_n$ and $C_n := C \cap X_n \subseteq V_n$ as well as $U_{n-1} \subseteq U_n, V_{n-1} \subseteq V_n$.

The case $n = -1$ follows from the assumption on A. For the induction step choose a push-out

$$\begin{array}{ccc} \coprod_{i\in I} S^{n-1} & \xrightarrow{(q_i)} & X_{n-1} \\ \downarrow & & \downarrow \\ \coprod_{i\in I} D^n & \xrightarrow{(Q_i)} & X_n \end{array}$$

and set $U'_{n-1} := (q_i)^{-1}_{i\in I}(U_{n-1})$, $V'_{n-1} := (q_i)^{-1}_{i\in I}(V_{n-1})$. By Lemma 4.6 shown below we can find open neighbourhoods $U'_n, V'_n \subseteq \coprod_I D^n$ with the following properties:

(i) $(Q_i)_I^{-1}(B_n) \subseteq U'_n$ and $(Q_i)_I^{-1}(C_n) \subseteq V'_n$

(ii) $U'_n \cap V'_n = \emptyset$

(iii) $U'_{n-1} = U'_n \cap \coprod_I S^{n-1}$ and $V'_{n-1} = V'_n \cap \coprod_I S^{n-1}$

Then define

$$U_n := U_{n-1} \cup (Q_i)_I(U'_n), \quad V_n := V_{n-1} \cup (Q_i)_I(V'_n)$$

By (iii) $U_n \cap X_{n-1} = U_{n-1}$ and $(Q_i)_I^{-1}(U_n) = U'_n$, so $U_n \subseteq X_n$ is open. The same holds for V_n. Furthermore, $B_n \subseteq U_n$, $C_n \subseteq V_n$ by (i) and $U_n \cap V_n = \emptyset$ by (ii). This shows the claim. Now consider

$$U := \bigcup_n U_n, \quad V := \bigcup_n V_n.$$

By construction $U \cap X_n \subseteq$ is open for all $n \geq -1$ and hence $U \subseteq X$ is open by definition of the colimit topology. The same holds for V. Moreover, $B \subseteq U, C \subseteq V$ and $U \cap V = \emptyset$. □

It remains to proof the lemma mentioned in the proof of the above proposition.

Lemma 4.6. *Let $B, C \subseteq D^n$ be closed disjoint sets and $U', V' \subseteq S^{n-1}$ open disjoint neighbourhoods of $B \cap S^{n-1}, C \cap S^{n-1}$, respectively. Then there exist open sets $U, V \subseteq D^n$ with the following properties:*

(i) $B \subseteq U, C \subseteq V$

(ii) $U \cap V = \emptyset$

(iii) $U' = U \cap S^{n-1}, V' = V \cap S^{n-1}$

Proof. Consider the open disjoint sets

$$U'' := \{rx \mid x \in U',\, 0 < r \leq 1\}$$

and

$$V'' := \{rx \mid x \in V',\, 0 < r \leq 1\}.$$

Moreover, since B and $\overline{V}'$ are disjoint and closed we can find an open neighbourhood W_B of B, such that

$$\overline{W}_B \cap \overline{V}' = \emptyset.$$

Similialry we can find an open neighbourhood W_C of C with

$$\overline{W}_C \cap \overline{U}' = \emptyset.$$

After potentially shrinking W_B and W_C we may also assume, that they are disjoint. Now set

$$U := (U'' - \overline{W}_C) \cup (W_B - S^{n-1})$$

and

$$V := (V'' - \overline{W}_B) \cup (W_C - S^{n-1}).$$

It is then easy to verify, that these sets satisfy the statement. $\square$

4.3 Remarks about Urs' lecture on Algebraic Topology

In this section we present an alternative proof on how to extend the fact that de-Rham cohomology and singular cohomology of open subsets of $\mathbb{R}^n$ are isomorphic to manifolds with or without boundary, while at the same time formulating the statement in more category theoretical terms.

Lemma 4.7. *The map*

$$\Gamma\colon \Omega^k(M) \to C_k^\infty(M)^*$$
$$\omega \mapsto (\sigma \mapsto \int_\sigma \omega := \int_{\Delta^k} \sigma^*\omega)$$

defines a natural map of cochain complexes

$$(\Omega^\bullet(M), d) \to (C_k^\infty(M)^*, \partial^*),$$

i.e. a natural transformation of functors from the category of smooth manifolds to the category of cochain complexes over $\mathbb{R}$.

Proof. For smooth manifolds M, N and a smooth map $f\colon M \to N$ we have to show the commutativity of the following diagrams:

$$\begin{tikzcd} \Omega^k(M) \arrow[r, "\Gamma"] \arrow[d, "d"] & C_k^\infty(M)^* \arrow[d, "\partial^*"] \\ \Omega^{k+1}(M) \arrow[r, "\Gamma"] & C_{k+1}^\infty(M)^* \end{tikzcd} \qquad \begin{tikzcd} \Omega^k(N) \arrow[r, "\Gamma"] \arrow[d, "f^*"] & C_k^\infty(N)^* \arrow[d, "f^*"] \\ \Omega^k(M) \arrow[r, "\Gamma"] & C_k^\infty(M)^* \end{tikzcd}$$

For the left diagram we compute:

$$(\Gamma \circ d)(\omega) = \left(\sigma \mapsto \int_\sigma d\omega\right) = \left(\sigma \mapsto \int_{\partial\sigma} \omega\right) = (\partial^* \circ \Gamma)(\omega)$$

where we used Stokes' Theorem for singular chains in the second equality. Moreover, we compute for the right diagram:

$$(\Gamma \circ f^*)(\omega) = \left(\sigma \mapsto \int_\sigma f^*\omega\right)$$
$$(f^* \circ \Omega)(\omega) = \left(\sigma \mapsto \int_{f\circ\sigma} \omega\right)$$

But by definition:

$$\int_\sigma f^*\omega = \int_{\Delta^k} \sigma^* f^*\omega = \int_{\Delta^k} (f\circ\sigma)^*\omega = \int_{f\circ\sigma} \omega$$

This completes the proof. $\square$

Lemma 4.8. *The inclusion $C_k^\infty(M) \to C_k(M)$ induces a natural map of cochain complexes*

$$(C_k^\infty(M)^*, \partial^*) \to (C_k(M)^*, \partial^*)$$

Proof. Obvious. □

In the lecture we have shown that the cochain maps from the above two lemmas induce isomorphisms in homology at least for open subsets of $\mathbb{R}^n$. A short proof of how this generalizes to all manifolds with or without boundary is given in the following theorem (which notably uses facts that we have not established in the lecture).

Theorem 4.9. *The cochain maps from Lemma 4.7 and 4.8 induce isomorphisms in homology.*

Proof. We first show the statement if $\partial M = \emptyset$. Begin by choosing an embedding

$$\iota\colon M \hookrightarrow U$$

into a tubular neighbourhood $U \subseteq \mathbb{R}^n$. In particular, U is open and ι a homotopy equivalence. Then by Lemma 4.7 the following diagram commutes:

$$\begin{tikzcd} H^k_{dR}(U) \arrow[r, "\Gamma"] \arrow[d, "\iota^*"] & H^k_\infty(U) \arrow[d, "\iota^*"] \\ H^k_{dR}(M) \arrow[r, "\Gamma"] & H^k_\infty(M) \end{tikzcd}$$

The vertical maps are isomorphisms due to homotopy invariance and in the lecture we have shown that the top horizontal map is an isomorphism as well. The claim follows.

If $\partial M \neq \emptyset$ we can reduce this to the case $\partial M = \emptyset$ by considering the homotopy equivalence

$$\mathrm{int}(M) \hookrightarrow M$$

and using a completely analogous argument to above. The same argument shows the statement for the map from Lemma 4.8. □

Composing the two maps we get a natural isomorphism

$$H^k_{dR}(M) \to H^k(M).$$

Under the identification $H^k(M) \cong H^*_k(M)$ (which is also natural, see Lemma 4.10 below) this isomorphism is given by

$$[\omega] \mapsto \left([\eta] \mapsto \int_{\widetilde{\eta}} \omega\right),$$

where $\widetilde{\eta}$ is a smooth representative of the class $[\eta]$.

Lemma 4.10. *Let $(C_\bullet, d_\bullet)$ be a chain complex over $\mathbb{R}$ (or any other field). Then*

$$\begin{aligned} D\colon H^k(C^*_\bullet, d^*_\bullet) &\to H^*_k(C_\bullet, d_\bullet) \\ [\phi] &\mapsto ([c] \mapsto \phi(c)) \end{aligned}$$

defines an isomorphism, which is natural for maps between chain complexes over $\mathbb{R}$.

Proof. First we show well-definedness, so let $\phi' = \phi + d^*\psi$ for some $\psi \in C^*_{k-1}$. But then

$$\phi'(c) - \phi(c) = d^*\psi(c) = \psi(dc) = 0$$

since c is closed. Moreover, if $c' = c + da$ for some $a \in C_{k+1}$ we get

$$\phi(c' - c) = \phi(da) = d^*\phi(a) = 0$$

since ϕ is closed. Next we show injectivity, i.e. for closed $\phi \in C^*_k$ with $\phi(c) = 0$ for all closed $c \in C_k$ we have to construct $\psi \in C^*_{k-1}$ with $d^*\psi = \phi$. For $a = dc \in C_{k-1}$ we define ψ by

$$\psi(a) := \phi(c)$$

which is well defined since for $a = dc'$ we have $d(c - c') = 0$ such that

$$\phi(c - c') = 0.$$

Now extend ψ arbitrarily to a linear map $C_{k-1} \to \mathbb{R}$ (which is possible since C_{k-1} is a vector space.) To show surjectivity let $\tau \in H_k^*(C_\bullet, d_\bullet)$. We have to construct a closed $\phi \in C_k^*$ with

$$\phi(c) = \psi([c])$$

for all closed $c \in C_k$. To this end write

$$C_k = \ker d_k \oplus V$$

for some subspace $V \subseteq C_k$ and define ϕ by

$$\begin{aligned} \phi(c) &:= \psi([c]),\ c \in \ker d_k \\ \phi|_V &:= 0 \end{aligned}$$

By construction $d^*\phi = 0$ and $D([\phi]) = \psi$. Finally, we will show naturality. Let

$$f_\bullet \colon (C_\bullet, d_\bullet) \to (B_\bullet, b_\bullet)$$

be a chain map. We have to show commutativity of the following diagram:

$$\begin{tikzcd} H^k(C^*_\bullet, d^*_\bullet) \arrow[r, "D"] & H_k^*(C_\bullet, d_\bullet) \\ H^k(B^*_\bullet, b^*_\bullet) \arrow[r, "D"] \arrow[u, "H^k(f_\bullet)"] & H_k^*(B_\bullet, b_\bullet) \arrow[u, "H_k^*(f_\bullet)"'] \end{tikzcd}$$

For $\phi \in H^k(B^*_\bullet, b^*_\bullet)$ we compute:

$$\begin{aligned} (H_k^*(f_\bullet) \circ D)([\phi]) &= (H_k(f_\bullet))^*([b] \mapsto \phi(b)) \\ &= ([c] \mapsto \phi(f_k c)) \end{aligned}$$

as well as

$$\begin{aligned} (D \circ H^k(f_\bullet))([\phi]) &= D([\phi \circ f_k]) \\ &= ([c] \mapsto \phi(f_k c)). \end{aligned}$$

□

4.4 Local Computation of the Degree

In this section we proof the following theorem.

Theorem 4.11. *Let $f \colon S^n \to S^n$ be smooth around a point $x \in S^n$, $y = f(x)$ and $T_x f \colon T_x S^n \to T_y S^n$ an isomorphism. Then for the local degree of f at x we have*

$$\deg_x(f) = +1$$

if $T_x f$ is orientation preserving and

$$\deg_x(f) = -1$$

if $T_x f$ is orientation reversing.

Proof. First assume $T_x f$ is orientation preserving. We have to show that for an open neighbourhood $U \subseteq S^n$ of x the map

$$H_n(S^n) \xrightarrow{\cong} H_n(S^n, S^n - y_1) \xleftarrow{\cong} H_n(U, U - x) \xrightarrow[H_n(f|_U)]{\cong} H_n(S^n, S^n - y_1) \xleftarrow{\cong} H_n(S^n)$$

is the identity (where all other maps are induced by inlcusions). This follows immediately from the following claim.

Claim. For orientation preserving embeddings $g_i\colon U \to S^n$ with $g_i(x) = y_i$, $i = 1, 2$ the map

$$H_n(S^n) \xrightarrow{\cong} H_n(S^n, S^n - y_1) \xleftarrow[H_n(g_1)]{\cong} H_n(U, U - x) \xrightarrow[H_n(g_2)]{\cong} H_n(S^n, S^n - y_1) \xleftarrow{\cong} H_n(S^n) \qquad (\star)$$

is the identity

Firstly we may assume, that $y_1 = y_2$, otherwise replace g_1 with $R \circ g_1$, where $R\colon S^n \to S^n$ is a rotation with $R(y_1) = y_2$. Then the map $(\star)$ is unchanged as seen by the commutativity of the following diagramm

$$\begin{tikzcd}
H_n(S^n) \arrow[r, "\cong"] \arrow[dd, "H_n(R) = id"'] & H_n(S^n, S^n - y_1) \arrow[dd, "H_n(R)"'] & \\
& & H_n(U, U - x) \arrow[ul, "H_n(g_1)"'] \arrow[dl, "H_n(R \circ g_1)"] \\
H_n(S^n) \arrow[r, "\cong"] & H_n(S^n, S^n - y_2) &
\end{tikzcd}$$

Note that $R \simeq id_{S^n}$ since $\mathrm{SO}(n)$ is path-connected. Moreover we may assume $U \cong \mathbb{R}^n$, otherwise shrink U to some $U' \subseteq U$ and excision yields $H_n(U', U' - x) \xrightarrow{\cong} H_n(U, U - x)$. So if we consider g_1 and g_2 in suitable charts we get orientation preserving embeddings

$$\tilde{g}_1, \tilde{g}_2\colon \mathbb{R}^n \to \mathbb{R}^n$$

with $\tilde{g}_1(0) = 0 = \tilde{g}_2(0)$. So by Lemma 7.1 $\tilde{g}_1$ and $\tilde{g}_2$ are isotopic $\operatorname{rel}\{0\}$. In particular we get a homotopy

$$H\colon U \times I \to S^n$$

from g_1 to g_2 with $H((U - x) \times I) \subseteq S^n - y_1$, i.e. a homotopy of *pairs*

$$g_1, g_2\colon (U, U - x) \to (S^n, S^n - y_1)$$

and thus

$$H_n(g_1) = H_n(g_2)\colon H_n(U, U - x) \to H_n(S^n, S^n - y_1)$$

which proofs the claim. It remains to consider the case where $T_x f$ is orientation reversing. For this consider the following commutative diagramm

$$\begin{tikzcd}
H_n(S^n) \arrow[r, "\cong"] & H_n(S^n, S^n - x) & H_n(U, U - x) \arrow[l, "\cong"'] \arrow[r, "\cong", "H_n(f|_U)"'] \arrow[dr, bend right, "\cong", "H_n(r \circ f|_U)"'] & H_n(S^n, S^n - y) \arrow[d, "\cong"', "H_n(r)"] & H_n(S^n) \arrow[l, "\cong"'] \arrow[d, "\cong"', "H_n(r)"] \\
& & & H_n(S^n, S^n - r(y)) & H_n(S^n) \arrow[l, "\cong"]
\end{tikzcd}$$

where $r\colon S^n \to S^n$ is the reflection along the equator. Then $T_x(r \circ f)$ is orientation preserving and the statement follows from what has been shown already and the fact that $\deg(r) = -1$. $\square$

5 Point Set Topology

Lemma 5.1. *Let $\pi\colon X \to Y$ be a quotient map and $A \subseteq X$ with $\pi^{-1}(\pi(A)) = A$. If $A \subseteq X$ is either closed or open then the restriction*

$$\pi|_A\colon A \to \pi(A)$$

is also a quotient map.

Proof. Let $p := \pi|_A\colon A \to \pi(A)$ be the restriction and A be open. Given $V \subseteq \pi(A)$ such that $U := p^{-1}(V) \subseteq A$ is open we get from our assumptions on A that U is also open in X and that $U = \pi^{-1}(V)$, hence V is also open in Y and thus V is open in $\pi(A)$. The same argument shows the statement if A is closed, since we can equivalently describe the quotient topology in terms of closed sets. □

6 Fiber Bundles

6.1 Construction of fiber bundles

For convencience, let us state the definition of a fiber bundle.

Definition 6.1 (Fiber Bundle)**.** Let G be a topological group acting effectively on a space F. A fiber bundle E over B with fiber F and structure group G consists of a surjective continuous map $p\colon E \to B$ and a collection of maps

$$(\phi_\alpha\colon U_\alpha \times F \to p^{-1}(U_\alpha))_\alpha,$$

called a trivialising atlas[3], for open sets $U_\alpha \subseteq B$, such that

(i) The maps ϕ_α are homeomorphisms and the following diagrams commute:

$$\begin{tikzcd} U_\alpha \times F \arrow[rr, "\phi_\alpha"] \arrow[dr, "\pi"'] & & p^{-1}(U_\alpha) \arrow[dl, "p"] \\ & U_\alpha & \end{tikzcd}$$

where π is the usual projection onto the first factor.

(ii) For any two maps ϕ_α, ϕ_β there exists a continuous map

$$\tau_{\alpha,\beta}\colon U_\alpha \cap U_\beta \to G,$$

such that

$$\phi_\alpha^{-1} \circ \phi_\beta\colon (U_\alpha \cap U_\beta) \times F \to (U_\alpha \cap U_\beta) \times F$$

is given by $(b, f) \mapsto (b, \tau_{\alpha,\beta} \cdot f)$.

(iii) $(U_\alpha)_\alpha$ is an open cover of B.

(iv) The collection (ϕ_α) is maximal with respect to the first 3 conditions.

The maps ϕ_α are called local trivialisations, The sets U_α are called trivialising neighbourhoods and the map $\tau_{\alpha,\beta}$ is called the transition function for ϕ_α, ϕ_β. Furthermore, we call E the total space, B the base space and F the fiber. We write

$$F \hookrightarrow E \xrightarrow{p} B$$

to denote the fiber bundle.

Remark 6.2. We require surjectivity of p to exclude the case $F = \emptyset$. The requirement, that G acts effectively guarantees for the transition maps to be unique. However, this is not a strong requirement to impose, since if G does not act effectively, i.e.

$$\Phi\colon G \to \mathrm{Aut}(F)$$

is not injective we get an effective group action by taking a quotient of G:

$$G/\ker\Phi \to \mathrm{Aut}(F).$$

It follows from the universal property of the quotient topology, that $G/\ker\Phi$ still acts continuously on F, i.e. the induced map $G/\ker\Phi \times F \to F$ is still continuous (if F is locally compact we can define the exponential topology on $\mathrm{Aut}(F)$ and the statement is equivalent to $G/\ker\Phi \to \mathrm{Aut}(F)$ being continuous).

[3]Wikipedia calls this a *G-Atlas*

Remark 6.3. The maximality condition for the trivialising atlas is well-defined: Given any trivialising atlas $\mathcal{F} = (\phi_\alpha)_\alpha$ satisfying conditions (i) - (iii) of Definition 6.1 we can define

$$\begin{aligned}\hat{\mathcal{F}} := \{\phi\colon U \times F \to p^{-1}(U) \mid U \subseteq B \text{ open }, \phi \text{ homeomorphism, for any } \phi_\alpha \in \mathcal{F} \\ \text{exists a map } \tau_{\alpha,\phi}\colon U \cap U_\alpha \to G, \text{ with } (\phi_\alpha^{-1} \circ \phi)(b,f) = (b, \tau_{\alpha,\phi}(b) \cdot f)\}.\end{aligned}$$

This is the maximal trivialising atlas induced by $\mathcal{F}$ satisfying (i) - (iii). Note however, that there is no unique trivialising atlas, since beginning with a different $\mathcal{F}$ may yield a different $\hat{\mathcal{F}}$ (as is the case for smooth atlases of smooth manifolds).

Proof of above remark. Given $\phi\colon U \times F \to p^{-1}(U), \phi'\colon U' \times F \to p^{-1}(U') \in \hat{\mathcal{F}}$, we claim that there exists $\tau\colon U \cap U' \to G$, such that

$$(\phi^{-1} \circ \phi')(b,f) = (b, \tau(b) \cdot f).$$

For $b_0 \in B$ choose $(\phi_\alpha\colon U_\alpha \times F \to p^{-1}(U_\alpha)) \in \mathcal{F}$ with $b_0 \in U_\alpha$. Then there exists

$$\tau_{\alpha,\phi}\colon U \cap U_\alpha \to G,\ \tau_{\alpha,\phi'}\colon U' \cap U_\alpha \to G$$

with

$$(\phi_\alpha^{-1} \circ \phi)(b,f) = (b, \tau_{\alpha,\phi}(b) \cdot f),\ (\phi_\alpha^{-1} \circ \phi')(b,f) = (b, \tau_{\alpha,\phi'}(b) \cdot f).$$

Then

$$(\phi^{-1} \circ \phi_\alpha) \circ (\phi_\alpha^{-1} \circ \phi')\colon (U_\alpha \cap U \cap U') \times F \to (U_\alpha \cap U \cap U') \times F$$

is given by

$$(b,f) \mapsto (b, \tau_{\alpha,\phi}(b)^{-1}\tau_{\alpha,\phi'}(b) \cdot f).$$

We conclude that around $\{b_0\} \times F$ we have

$$(\phi^{-1} \circ \phi')(b,f) = (b, \tau(b) \cdot f)$$

for $\tau(b) := \tau_{\alpha,\phi}(b)^{-1}\tau_{\alpha,\phi'}(b)$. Since group inversion and multiplication are continuous, τ is continuous. As b_0 was chosen arbitrary, such a map τ can be defined on all of $U \cap U'$. Furthermore, the group action of G is effective, hence τ is unique and the above argument shows that it is continuous. □

For the transition functions we have the following Lemma, where again the effectiveness of the group action is crucial.

Lemma 6.4. *Let $F \hookrightarrow E \xrightarrow{p} B$ be a fiber bundle with structure group G. Then (using the notation in Definition 6.1):*

(i) $\tau_{\alpha,\alpha}(b) = e$

(ii) $\tau_{\alpha,\beta}^{-1}(b) = \tau_{\beta,\alpha}(b)$

(iii) $\tau_{\gamma,\alpha}(b)\tau_{\alpha,\beta}(b) = \tau_{\gamma,\beta}(b)$

Proof. Let us first proof (iii), then (i) and (ii) follow directly from the group axioms. We compute:

$$\begin{aligned}(b, \tau_{\gamma,\beta}(b) \cdot f) &= (\phi_\gamma^{-1} \circ \phi_\beta)(b,f) \\ &= (\phi_\gamma^{-1} \circ \phi_\alpha \circ \phi_\alpha^{-1} \circ \phi_\beta)(b,f) = (b, \tau_{\gamma,\alpha}(b)\tau_{\alpha,\beta}(b) \cdot f).\end{aligned}$$

Since the transition functions are uniquely determined, the statement follows. □

Interestingly, the transition functions already uniquely determine the fiber bundle, in a sense made precise by the following theorem.

Theorem 6.5 (Fiber bundle construction A)**.** *Let B, F be (non-empty) topological spaces and G a topological group acting effectively on F. Suppose we are given the following data:*

(i) An open cover $\mathcal{U} := (U_\alpha)_\alpha$ of B (where α ranges over some index set A)

(ii) A collection $(\tau_{\alpha,\beta})$ of continuous maps

$$\tau_{\alpha,\beta} \colon U_\alpha \cap U_\beta \to G$$

for each pair $U_\alpha, U_\beta \in \mathcal{U}$

(iii) $\tau_{\gamma,\alpha}(b)\tau_{\alpha,\beta}(b) = \tau_{\gamma,\beta}(b)$ for all $b \in U_\alpha \cap U_\beta \cap U_\gamma$

Then there exists a unique (up to isomorphism) fiber bundle $F \hookrightarrow E \xrightarrow{p} B$ with structure group G, such that the maximal trivialising atlas is induced by a trivialising atlas of the form

$$(\phi_\alpha \colon U_\alpha \times F \to p^{-1}(U_\alpha))_\alpha,$$

such that $(\phi_\alpha^{-1} \circ \phi_\beta)(b, f) = (b, \tau_{\alpha,\beta}(b) \cdot f)$.

Proof. Consider the space

$$E := \left(\coprod_{\alpha \in A} U_\alpha \times F \right) \Big/ \sim$$

where the equivalence relation $\sim$ is defined by

$$(b, f, \beta) \sim (b, \tau_{\alpha,\beta}(b) \cdot f, \alpha)$$

and let $p \colon E \to B$, $[(b, f, \alpha)] \mapsto b$. Furthermore, let $\pi \colon \coprod_\alpha U_\alpha \times F \to E$ be the projection map.

Step 1. *$\sim$ is an equivalence relation.*

Transitivity follows directly from condition (iii). Reflexivity and symmetry then follow from the group axioms.

Step 2. *p is well-defined, continuous and surjective.*

Clearly p is well-defined and surjective as $F \neq \emptyset$. Continuity then follows directly from the universal property of the quotient topology.

Step 3. *Defining the trivialising atlas*

Consider

$$\phi_\alpha \colon U_\alpha \times F \to p^{-1}(U_\alpha),\ (b, f) \mapsto [(b, f, \alpha)].$$

Clearly ϕ_α is well-defined ($[(b, f, \alpha)] \in p^{-1}(U_\alpha)$) and continuous (the projection map π is continuous). The inverse is given by

$$\phi_\alpha^{-1}([(b, f, \beta)]) = (b, \tau_{\alpha,\beta}(b) \cdot f).$$

To show that ϕ_α^{-1} is continuous, we show that ϕ_α is an open map. To this end, let $U \times V \subseteq U_\alpha \times F$ be open and consider

$$\theta_{\alpha,\beta} \colon (U_\alpha \cap U_\beta) \times F \to (U_\alpha \cap U_\beta) \times F,\ (b, f) \mapsto (b, \tau_{\alpha,\beta}(b) \cdot f).$$

Clearly $\theta_{\alpha,\beta}$ is continuous (in fact it is a homeomorphism), hence

$$V_\beta := \theta_{\alpha,\beta}^{-1}((U \cap U_\beta) \times V) \subseteq (U_\alpha \cap U_\beta) \times F$$

is open and thus

$$\pi^{-1}(\phi_\alpha(U \times V)) = \coprod_{\beta \in A} V_\beta \subseteq \coprod_{\beta \in A} U_\beta \times F$$

is open. Thus $\phi_\alpha(U \times V)$ is open in E and since $p^{-1}(U_\alpha)$ is also open, we conclude that ϕ_α is an open map. Furthermore, the diagram

$$\begin{tikzcd} U_\alpha \times F \arrow[rr, "\phi_\alpha"] \arrow[dr] & & p^{-1}(U_\alpha) \arrow[dl, "p"] \\ & U_\alpha & \end{tikzcd}$$

commutes, such that ϕ_α is a local trivialisation. Moreover,

$$(\phi_\alpha^{-1} \circ \phi_\beta)(b, f) = (b, \tau_{\alpha,\beta}(b) \cdot f).$$

So the collection of maps $(\phi_\alpha)_\alpha$ is a trivialising atlas with the desired transition functions, making $F \hookrightarrow E \xrightarrow{p} B$ into a fiber bundle with structure group G.

Step 4. *Uniqueness*

Given two fiber bundles

$$\delta := (F \hookrightarrow E \xrightarrow{p} B)$$
$$\varepsilon := (F \hookrightarrow \tilde{E} \xrightarrow{\tilde{p}} B)$$

with structure group G and trivialising atlases

$$(\phi_\alpha \colon U_\alpha \times F \to p^{-1}(U_\alpha))_{\alpha \in A}$$
$$(\tilde{\phi}_\alpha \colon U_\alpha \times F \to \tilde{p}^{-1}(U_\alpha))_{\alpha \in A}$$

respectively, such that

$$(\phi_\alpha^{-1} \circ \phi_\beta)(b, f) = (b, \tau_{\alpha,\beta}(b) \cdot f) = (\tilde{\phi}_\alpha^{-1} \circ \tilde{\phi}_\beta)(b, f).$$

We can then construct an isomorphism $\Gamma \colon \delta \to \varepsilon$ as follows: for $x \in E$ choose $U_\alpha \in \mathcal{U}$ with $p(x) \in U_\alpha$. Then define

$$\Gamma(x) := (\tilde{\phi}_\alpha \circ \phi_\alpha^{-1})(x).$$

Well-definedness of Γ follows from the commutativity of the following diagram:

$$\begin{tikzcd}
 & (U_\alpha \cap U_\beta) \times F \arrow[dl, "\phi_\beta"'] \arrow[dd, "\tau"] \arrow[dr, "\tilde{\phi}_\beta"] & \\
p^{-1}(U_\alpha \cap U_\beta) & & \tilde{p}^{-1}(U_\alpha \cap U_\beta) \\
 & (U_\alpha \cap U_\beta) \times F \arrow[ul, "\phi_\alpha"] \arrow[ur, "\tilde{\phi}_\alpha"'] &
\end{tikzcd}$$

where $\tau(b, f) := (b, \tau_{\alpha,\beta}(b) \cdot f)$. Locally Γ is given by a continuous map, so it is continuous. Analogously we define the inverse map. Furthermore,

$$\begin{tikzcd}
E \arrow[rr, "\Gamma"] \arrow[dr, "p"'] & & \tilde{E} \arrow[dl, "\tilde{p}"] \\
 & B &
\end{tikzcd}$$

commutes, so Γ is a bundle isomorphism. This completes the proof. $\square$

We may also formulate Theorem 6.5 slightly different, if we are given the fibers $F_x = p^{-1}(x)$ as sets. This is usually more practical, as the description of the bundle is more explicit:

Theorem 6.6 (Fiber bundle construction B)**.** *Let B, F be (non-empty) topological spaces and G a topological group acting effectively on F. Given sets F_x for each $x \in B$ we define*

$$E := \coprod_{x \in B} F_x$$

and $p \colon E \to B$ by sending each element in F_x to x. Suppose we are given the following data:

(i) An open cover $\mathcal{U} := (U_\alpha)_\alpha$ of B (where α ranges over some index set A)

(ii) A collection of bijections $\phi_\alpha \colon U_\alpha \times F \to p^{-1}(U_\alpha)$, that restrict to bijections $\{x\} \times F \to F_x$

(iii) A collection $(\tau_{\alpha,\beta})$ of continuous maps

$$\tau_{\alpha,\beta} \colon U_\alpha \cap U_\beta \to G$$

for each pair $U_\alpha, U_\beta \in \mathcal{U}$, such that $(\phi_\alpha^{-1} \circ \phi_\beta)(b, f) = (b, \tau_{\alpha,\beta}(b) \cdot f)$

Then E has a unique topology making $F \hookrightarrow E \xrightarrow{p} B$ into a fiber bundle with structure group G, where the maps ϕ_α are local trivialisations.

Proof. Define the topology on E as the final topology with respect to the maps ϕ_α, i.e.

$$V \subseteq E \text{ is open } \Leftrightarrow \phi_\alpha^{-1}(V) \subseteq U_\alpha \times F \text{ is open for each } \alpha$$

We show that the maps ϕ_α are open with this topology. To this end, let $W \subseteq U_\alpha \times F$ be open. Then

$$\phi_\beta^{-1}(\phi_\alpha(W)) = (\phi_\beta^{-1} \circ \phi_\alpha)(W \cap ((U_\alpha \cap U_\beta) \times F)) \subseteq U_\beta \times F$$

is open, since $\phi_\beta \circ \phi_\alpha$ is a homeomorphism by condition (iii). Thus ϕ_α is a homeomorphism. In particular, for any open set $U \subseteq U_\alpha$, we get that

$$p^{-1}(U) = \phi_\alpha(U \times F)$$

is open, since by (ii) the following diagram commutes:

$$\begin{tikzcd} U_\alpha \times F \arrow[rr, "\phi_\alpha"] \arrow[dr] & & p^{-1}(U_\alpha) \arrow[dl, "p"] \\ & U_\alpha & \end{tikzcd}$$

So p is also continuous. Moreover, as $F \neq \emptyset$, we get that p is surjective. It remains to show uniqueness. So let E be equipped with any topology, such that p is continuous and ϕ_α are homeomorphisms. Let $V \subseteq E$, such that

$$W_\alpha := \phi_\alpha^{-1}(V) \subseteq U_\alpha \times F$$

is open for all α. Then

$$V = \bigcup_\alpha W_\alpha,$$

since the U_α cover B (and hence $E = \bigcup_\alpha \mathrm{im}\phi_\alpha$). As the ϕ_α are homeomorphisms and $p^{-1}(U_\alpha)$ is open, we conclude that V is open. Furthermore, the topology can not be finer than the final topology defined above, such that the two must coincide. $\square$

Example 6.7. Suppose we are given a (toplogical) rank n vector bundle $E \xrightarrow{\pi} B$. We can use Theorem 6.6 to define the **frame bundle** $\mathrm{GL}(E)$ of E as follows. Let $\mathrm{GL}(E_x)$ denote the set of all ordered bases of the vector space $E_x := \pi^{-1}(x)$ and set

$$\mathrm{GL}(E) := \coprod_{x \in B} \mathrm{GL}(E_x).$$

Now choose an open cover (U_α) of B over which E is trivial, i.e. there exist local frames $X_1^\alpha, \dots, X_n^\alpha$ over U_α. We define the local trivialisations for the bundle by

$$\begin{aligned} \phi_\alpha \colon U_\alpha \times \mathrm{GL}(n) &\to p^{-1}(U_\alpha) \\ (x, A) &\mapsto \left(\sum_{i=1}^n a_{i1} X_i^\alpha(x), \dots, \sum_{i=1}^n a_{in} X_i^\alpha(x) \right), \end{aligned}$$

where $A = (a_{ij})$, i.e. $\phi_\alpha(x, A)$ is the basis of E_x, whose coordinates in the frame (X_i^α) are precisely the columns of the matrix A. Clearly ϕ_α is a bijection. Moreover for $x \in U_\alpha \cap U_\beta$ we have

$$(\phi_\alpha^{-1} \circ \phi_\beta)(x, A) = (x, \tau_{\alpha,\beta}(x)A),$$

where $\tau_{\alpha,\beta}(x) \in \mathrm{GL}(n)$ is the change of basis matrix from $(X_i^\beta(x))$ to $(X_i^\alpha(x))$, which depends continuously on x. So applying Theorem 6.6 we can see $\mathrm{GL}(E)$ as principal $\mathrm{GL}(n)$-bundle over B. Using Theorem 6.12 the same construction works in the smooth category. It is interesting to note that E is trivial if and only if $\mathrm{GL}(E)$ admits a global section.

If E carries a fiber metric we may similiarly construct the **orthonormal frame bundle** $O(M)$ as a principal $\mathrm{O}(n)$-bundle and if E is oriented we can construct the **oriented frame bundle** $\mathrm{GL}^+(M)$ as a principal $\mathrm{GL}^+(n)$-bundle. Of course we can also combine the two to get the **oriented orthonormal frame bundle** $\mathrm{SO}(M)$.

Given a fiber bundle, Theorem 6.5 allows to change the fiber:

Corollary 6.8 (Changing the fiber). *Let $F \hookrightarrow E \xrightarrow{p} B$ be a fiber bundle with structure group G. Given a space F' upon which G acts effectively, there exists a unique fiber bundle*

$$F' \hookrightarrow E' \xrightarrow{p'} B$$

which has the same trivialising neighbourhoods (U_α) and the same transition functions $(\tau_{\alpha,\beta})$ as $F \hookrightarrow E \xrightarrow{p} B$.

Theorem 6.5 tells us, that the new bundle still retains all of the information of the old bundle, as it is uniquely determined by the transition functions, i.e. we can simply change the fiber again to get the original bundle back.

Example 6.9. Given any fiber bundle $F \hookrightarrow E \xrightarrow{p} B$ with structure group G we can apply the above corollary with $F' := G$ upon which G acts effectively by left-translations to get a principal G-bundle over B. This bundle is called the **underlying principal bundle** of $E \to B$.

Example 6.10. Using the above corollary we can also construct the frame bundle from Example 6.7 more abstractly by defining it as the underlying principal bundle of $E \to F$. Note that the uniqueness results in Theorem 6.5 shows, that the bundle constructed in this way is in fact isomorphic to $\mathrm{GL}(E)$. The construction in Example 6.7 of course has the benefit of being more concrete.

6.2 Smooth bundles

If we are in the smooth category, we can analogously define smooth fiber bundles by requiring the spaces E, B and F to be smooth manifolds (where either F or B has no boundary), the group G to be a Lie group and all maps to be smooth. In particular, the local trivialisation are diffeomorphisms. All of the above results also hold in the smooth category.

Theorem 6.11 (Fiber bundle construction A′). *The conclusions of Theorem 6.5 remain true in the smooth category.*

Proof. The only difficulty is in defining the smooth structure on

$$E := \left(\coprod_{\alpha \in A} U_\alpha \times F \right) \Big/ \sim$$

This smooth structure is defined via the local trivialisations (ϕ_α) as follows: Choose and open cover (V_α) of B, such that

- $V_\alpha \subseteq U_\alpha$
- there exist smooth charts $h_\alpha \colon W_\alpha \to V_\alpha$

Furthermore, choose a smooth atlas

$$(k_i \colon W_i^F \to V_i^F)_i$$

of F. Then

$$(\phi_\alpha \circ (h_\alpha \times k_i) \colon W_\alpha \times W_i^F \to E)_{i,\alpha}$$

defines a smooth atlas for E, since the maps $\phi_\alpha^{-1} \circ \phi_\beta$ are diffeomorphisms. It is then easy to verify, that this atlas satisifies the second countability and Hausdorff axioms. Furthermore, this smooth structure is uniquely determined by requiring the local trivialisations to be diffeomorphisms. □

Theorem 6.12 (Fiber bundle construction B′). *The conclusions of Theorem 6.6 remain true in the smooth category.*

Proof. As in the above theorem, the local trivialisations determine a unique smooth structure on E. □

6.3 Vector bundles

Whitney Sums

Given vector bundles E_1, E_2 over the same base B, we can define their **Whitney Sum** $E_1 \oplus E_2$ using Theorem 6.6 as the vector bundle whose fibers are given by $E_1|_p \oplus E_2|_p$ with the obvious choice as local trivialisations. The following Lemma characterizes the Whitney sum and at the same time provides the technical grounds for an intuition.

Lemma 6.13 (Characterisation of the Whitney sum)**.** *Let E_1 and E_2 be vector bundles over the same base space B of rank k_1 and k_2 respectively. A rank $k_1 + k_2$ vector bundle E over B is isomorphic to $E_1 \oplus E_2$ if and only if E_1 and E_2 can be embedded into E as subbundles and $E_1|_p + E_2|_p = E|_p$ for all $p \in B$.*

Let us first clarify what we mean by embedding. A vector bundle homomorphism $F\colon E \to E'$ is called and **embedding**, if it is injective. Then $\mathrm{im}(F)$ is a subbundle of E' and $F\colon E \to \mathrm{im}(F)$ is a vector bundle isomorphism. This follows from [Lee11] Theorem 10.34 and Prop. 10.26.

Proof of Lemma 6.13. First let $E = E_1 \oplus E_2$. Then we can define embeddings by

$$F_1\colon E_1 \to E,\ v \mapsto (v, 0), F_2\colon E_2 \to E\ w \mapsto (0, w).$$

Now let $E_1, E_2 \subseteq E$ be subbundles. Let $(U_\alpha)_\alpha$ be an open cover of B, such that E_1 and E_2 are trivial over U_α. Choose local frames

$$\sigma^i_{\alpha,1}, \dots, \sigma^i_{\alpha,k_i}\colon U_\alpha \to E_i$$

and define local trivialisations

$$\phi^i_\alpha\colon U_\alpha \times \mathbb{R}^{k_i} \to E_i,\ (p, v) \mapsto \sum_{j=1}^{k_i} v_j \sigma^i_{\alpha,j}(p).$$

Let $\tau^i_{\alpha,\beta}\colon U_\alpha \cap U_\beta \to \mathrm{GL}(k_i)$ be the transition functions defined by

$$((\phi^i_\alpha)^{-1} \circ \phi^i_\beta)(p, v) = (p, \tau^i_{\alpha,\beta}(p) \cdot v).$$

Since $E|_p$ is the direct sum of $E_1|_p$ and $E_2|_p$, we get that

$$\sigma^1_{\alpha,1}, \dots, \sigma^1_{\alpha,k_1}, \sigma^2_{\alpha,1}, \dots, \sigma^2_{\alpha,k_2}$$

is a local frame for E, thus defines a local trivialisation by

$$\psi_\alpha\colon U_\alpha \times \mathbb{R}^{k_1} \times \mathbb{R}^{k_2} \to E,\ (p, v, w) \mapsto \phi^1_\alpha(p, v) + \phi^2_\alpha(p, w).$$

Then

$$(\psi_\alpha^{-1} \circ \psi_\beta)(p, v, w) = (p, \tau^1_{\alpha,\beta}(p) \cdot v, \tau^2_{\alpha,\beta}(p) \cdot w).$$

Note that $E_1 \oplus E_2$ is also trivial over U_α via

$$\varphi_\alpha\colon U_\alpha \times \mathbb{R}^{k_1} \times \mathbb{R}^{k_2} \to E_1 \oplus E_2,\ (p, v, w) \mapsto (\phi^1_\alpha(p, v), \phi^2_\alpha(p, w))$$

and $\varphi_\alpha^{-1} \circ \varphi_\beta = \psi_\alpha^{-1} \circ \psi_\beta$, thus

$$E \cong E_1 \oplus E_2$$

by the uniqueness result in Theorem 6.5. □

This characterisation gives an intuitive reasoning behind the fact, that the Whitney sum of two Moebius bundles is trivial:

Lemma 6.14. *Let E, E_1 and E_2 be vector bundles over a manifold M and $\sigma_i\colon M \to E_i$ smooth sections for $i = 1, 2$. Then the following maps are smooth:*

(i) $(\sigma_1, \sigma_2)\colon N \to E_1 \oplus E_2,\ p \mapsto (\sigma_1(p), \sigma_2(p))$

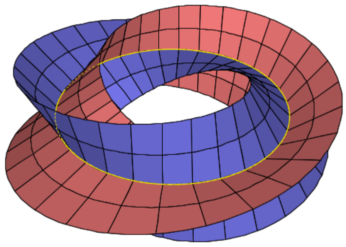

Figure 2: Illustration of the Whitney sum of two Moebius bundles [source: `https://math.stackexchange.com/questions/1009126/global-trivialization-of-m-oplus-m`]

(ii) The addition map $E \oplus E \to E$, $(v, w) \mapsto v + w$

(iii) The scalar multiplication map $\mathbb{R} \times E \to E$, $(\lambda, v) \mapsto \lambda v$

Proof. All statements follow easily by writing the maps in local trivialisations. □

The Normal Bundle

We can generalize Lemma 10.35 from [Lee11] to Riemannian manifolds:

Lemma 6.15 (Orthogonal complement bundles)**.** *Let* M *be a Riemannian manifold,* $S \subseteq M$ *a submanifold and* $E \subseteq TM|_S$ *a subbundle. Then*

$$E^\perp := \{(p, v) \mid p \in S, v \in E_p^\perp\} \subseteq TM|_S$$

is a subbundle and there exists a smooth orthonormal local frame for $E^\perp$ *around each point in* S*.*

Proof. Choose any local frame
$$\sigma_1, \dots, \sigma_k \colon U \to E.$$
After possibly shrinking U, we can extend this frame to a frame

$$\sigma_1, \dots, \sigma_n \colon U \to TM|_S$$

of $TM|_S$. Write $\sigma_i(p) = (p, X_p^i)$. Applying Gramm-Schmidt to the vectors X_p^i yields an orthonormal basis $Y_p^1, \dots, Y_p^n$ of T_pM for each $p \in U$:

$$Y_p^i := \frac{X_p^i - \sum_{j=1}^{i-1} \langle X_p^i, Y_p^j \rangle X_p^i}{\| X_p^i - \sum_{j=1}^{i-1} \langle X_p^i, Y_p^j \rangle X_p^i \|}$$

Since this formula is smooth, we can define the smooth sections

$$\tilde{\sigma}_i(p) := (p, Y_p^i).$$

Notice that $\tilde{\sigma}_i(p) \in E^\perp$ for $i \geq k+1$ and $E_p^\perp = \operatorname{span}\{Y_p^{k+1}, \dots, Y_p^n\}$. This shows the statement. □

Given a Riemannian manifold M and a submanifold $S \subseteq M$, we can define the **normal bundle** of S:
$$NS := \{(p, v) \in TM \mid p \in S, v \in T_pS^\perp\}.$$

Corollary 6.16. *Let* M *be a Riemannian manifold,* $S \subseteq M$ *a submanifold. Then* NS *is a subbundle of* $TM|_S$*.*

Proof. We only need to show, that TS is a smooth subbundle of $TM|_S$ and then apply Lemma 6.15. To this end, choose a local frame (σ_i) of TS. Then $\iota \circ \sigma_i$ are sections of $TM|_S$, where $\iota \colon TS \hookrightarrow TM|_S$ is the inclusion. So we only have to verify, that ι is smooth. But this can easily be seen in local trivialisations of TS and $TM|_S$. □

6.4 Pullback Bundles

Given a fiber bundle $F \hookrightarrow E \xrightarrow{p} B$ and a map $g\colon B' \to B$ we can construct the pullback bundle g^*E using Theorem 6.6 as the fiber bundle over B', where the fiber over $b' \in B'$ is given by the fiber of E over $g(b')$, i.e.

$$g^*E = \coprod_{b' \in B'} p^{-1}(g(b')).$$

Given a local trivialisation $\phi\colon U \times F \to E$ we define a local trivialisation of g^*E by

$$g^{-1}(U) \times F \to g^*E,\ (b', f) \mapsto (b', \phi(g(b'), f)).$$

The same construction works in the smooth category using Theorem 6.12 as long as either F has no boundary or B and B' have no boundary. By projecting onto the second factor we get a bundle homomorphism $G\colon g^*E \to E$, making the following diagram commute:

$$\begin{tikzcd} g^*E \arrow[r, "G"] \arrow[d] & E \arrow[d, "p"] \\ B' \arrow[r, "g"] & B \end{tikzcd}$$

Lemma 6.17. *In the above situation the following holds:*

(i) If g is an immersion, so is G

(ii) If g is an embedding, so is G

Proof. In local trivialisations we can write G as

$$g^{-1}(U) \times F \to U \times F,\ (b, f) \mapsto (g(b'), f),$$

such that (i) follows. Now let g be an embedding. We have $\mathrm{im}G = p^{-1}(g(B'))$, so we can define

$$H\colon \mathrm{im}G \to f^*E,\ x \mapsto (g^{-1}(p(x)), x).$$

Clearly H is continuous and $G \circ H = \mathrm{id}_{\mathrm{im}G}$ as well as $H \circ G = \mathrm{id}_{g^*E}$, so G is in fact an embedding. $\square$

Given a fiber bundle $F \hookrightarrow E \xrightarrow{p} B$ and a submanifold $S \subseteq B$ we can then define the restriction of E to S as the pullback of E under the inclusion $\iota\colon S \hookrightarrow B$:

$$E|_S := \iota^*E$$

The above Lemma shows that this is indeed a submanfold of E. In fact, in the situation of the above Lemma we have

$$\mathrm{im}G = E|_{g(B')}.$$

One usecase for pullback bundles is to define vector fields along curves $\gamma\colon I \to M$ for a smooth manifold M as sections in the pullback bundle γ^*TM. The follwoing Lemma says, that we can equivalently define them as maps

$$V\colon I \to TM$$

with $V(t) \in T_{\gamma(t)}M$ (as is done in [Lee18]).

Lemma 6.18. *Let $F \hookrightarrow E \to M$ be a smooth fiber bundle and $g\colon N \to M$ a smooth map. Then a section $N \xrightarrow{\sigma} g^*E$ is smooth if and only if the composition*

$$N \xrightarrow{\sigma} g^*E \xrightarrow{G} E$$

is smooth, where G is the induced bundle homomorphism $g^*E \to E$ discussed above.

Proof. This follows directly from the definition of the smooth structure of the pullback bundle given above. $\square$

7 Construction of Diffeomorphisms

7.1 Existence of Smooth Isotopies

In this section we proof two statements on the existence of smooth isotopies. Both follow [Mil65] quite closely. The first Lemma is a slight modification of Lemma 5.7 in [Mil65].

Lemma 7.1 (Isotopy Lemma)**.** *Let $h\colon \mathbb{R}^n \to \mathbb{R}^n$ be an orientation preserving embedding with $h(0) = 0$. Then h is isotopic to the identity* rel $\{0\}$.

Proof. The proof is basically the same as in [Mil65]. Since $h(0) = 0$ we can write

$$h(x) = x_1h_1(x) + \ldots + x_nh_n(x)$$

for smooth functions $h_i\colon \mathbb{R}^n \to \mathbb{R}^n$ and then define an isotopy

$$\frac{1}{t}h(tx) = x_1h_1(tx) + \ldots + x_nh_n(tx), \quad 0 \leq t \leq 1$$

from the linear map

$$A\colon \mathbb{R}^n \to \mathbb{R}^n, x \mapsto x_1h_1(0) + \ldots + x_nh_n(0)$$

to h. Since $\det(A) > 0$ and the space of all $n \times n$ matrices with positive determinant is path connected we moreover get an isotopy from A to id. ☐

Next we proof Theorem 5.8 in [Mil65] and give a slightly stronger formulation of it. The main idea behind the proof is beautifully explained in [Mil65].

Theorem 7.2 (Isotopy Extension Theorem)**.** *Let $M \subseteq N$ be a submanifold with or without boundary of the boundaryless manifold N and let $h_t\colon M \to N, 0 \leq t \leq 1$ be an isotopy of the inclusion $M \hookrightarrow N$. Given any compact subset $K \subseteq M$ and an open set $U \subseteq N$, such that $h_t(K) \subseteq U$ for all $t \in [0, 1]$, there exists an isotopy*

$$\overline{h}_t\colon N \to N, 0 \leq t \leq 1$$

with the following properties

i) $\overline{h}_0 = \mathrm{id}_N$

ii) $\overline{h}_t(x) = h_t(x)$ *for all* $x \in K$ *and* $t \in [0, 1]$

iii) $\overline{h}_t(x) = x$ *for all* $x \in N \setminus U$ *and* $t \in [0, 1]$

Proof. We formulate the proof only for the case $\partial M = \emptyset$, see Remark 7.3. Let $V \subseteq M$ be a relatively compact neighbourhood of K. Then

$$\begin{aligned} H\colon V \times [0, 1] &\to N \times [0, 1], \\ (x, t) &\mapsto (h_t(x), t) \end{aligned}$$

is an embedding, since the map $M \times [0, 1] \to N \times [0, 1]$ defined by the same formula is an injective immersion. So H definies a submanifold

$$S := \mathrm{im}(H) \subseteq N \times [0, 1]$$

as well as the smooth vector field $X := H_*(\frac{\partial}{\partial t})$ on S, i.e. for $y = h_t(x)$ we have

$$X_{(y,t)} := D_{(x,t)}H\left(\frac{\partial}{\partial t}\right) = \left(\frac{\partial h_t}{\partial t}(x), \frac{\partial}{\partial t}\right).$$

We can then extend X to some open neighbourhood $W \subseteq U \times [0, 1] \subseteq N \times [0, 1]$ of S at first arbitrarily and then set the coordinate of the part pointing in the $\frac{\partial}{\partial t}$ direction equal to 1. Thus we obtain a smooth vector field Y on W of the form

$$Y_{(y,t)} = (\tilde{X}_{(y,t)}, \frac{\partial}{\partial t})$$

which agrees with X on S. Choose a cut-off function

$$\psi\colon N \times [0,1] \to \mathbb{R}$$

which equals 1 on $H(K \times [0,1])$ and has its support contained in W. Then we consider the smooth vector field Z on $N \times [0,1]$ defined by

$$Z := \psi Y + (1-\psi)\frac{\partial}{\partial t}.$$

The idea now is to define the required isotopy $\overline{h}_t$ using the flow of Z. For this we want to assume that U is relatively compact, otherwise we may shrink U. The reason for this will become clear shortly, but first let us record the crucial properties of Z:

i) Z is of the form
$$Z_{(y,t)} = (\tilde{Z}_{(y,t)}, \frac{\partial}{\partial t}),$$

ii) Z agrees with X on $H(K \times [0,1])$,

iii) $Z = \frac{\partial}{\partial t}$ outside of the compact set $\overline{U} \times [0,1]$.

We claim that for any $y \in N$ the unique maximal integral curve ψ_y of Z starting at $(y,0) \in N \times [0,1]$ is defined for all $t \in [0,1]$. If $y \notin \overline{U}$ this is clear, since the ψ_y is explicitly given by

$$\psi_y(t) = (y,t), \quad 0 \leq t \leq 1.$$

For $y \in \overline{U}$ this follows from the fact, that ψ_y has to be contained in the compact set $\overline{U} \times [0,1]$ for all time, where it is defined (due to uniqueness of integral curves and the case $y \notin \overline{U}$ above). Then using property (i) of Z we see that $\psi_y(t)$ is in fact defined for all $t \in [0,1]$.

So after extending Z to $N \times \mathbb{R}$ (to avoid dealing with flows on manifolds with boundary) we can consider the flowout of $N \times \{0\}$, which is defined on $N \times [0,1]$ by the preceeding observations and hence defines a level-preserving diffeomorphism

$$\begin{aligned} \overline{H}\colon N \times [0,1] &\to N \times [0,1], \\ (y,t) &\mapsto \psi_y(t). \end{aligned}$$

Finally we define the required isotopy $\overline{h}_t$ by

$$\overline{H}(y,t) = (\overline{h}_t(y), t).$$

It remains to show that $\overline{h}_t$ agrees with h_t on K. But this follows from property (ii) of Z: for $x \in K$ let $\phi_x(t) := (x,t),\ 0 \leq t \leq 1$ be the integral curve of the vector field $\frac{\partial}{\partial t}$ on $M \times [0,1]$. Then

$$H(x,t) = H(\phi_x(t)) = \psi_x(t) = \overline{H}(x,t).$$

The theorem follows. $\square$

Remark 7.3. If M is a manifold with boundary we can define a smooth isotopy as a map

$$h\colon M \times [0,1] \to N,$$

which extends to a smooth isotopy

$$\overline{h}\colon M \times \mathbb{R} \to N.$$

Then exactly the same proof can be carried out if M is allowed to have boundary by constructing the required vector fields on $N \times \mathbb{R}$ and then restricting everything back to $N \times [0,1]$. Note that this definition of smooth isotpies does not change the isotopy classes of maps in the case $\partial M = \emptyset$ (cf. Lemma 11.6). Of course using the notion of a manifold with edges we can define smooth isotopies in the same way and the proof of Theorem 7.2 goes through unchanged.

Together the Isotopy Lemma and the Isotopy Extension Theorem can be used to extend locally defined diffeomorphisms to the whole manifold as follows: Begin with an orientation preserving embedding $\phi\colon U \to M$ of an open subset $U \subseteq M$. Then - after possibly restricting ϕ to a smaller domain - use Lemma 7.1 to construct an isotopy from ϕ to the inclusion $U \hookrightarrow M$. Lastly - after once again restricting the isotopy to a smaller domain - we can use Theorem 7.2 to extend this isotopy to an an isotopy $\overline{h}_t\colon N \to N$ starting at id_N. Then $h_1\colon N \to N$ is a diffeomorphism extending ϕ.

Next we state a few interesting corollaries of Theorem 7.2.

Corollary 7.4. *Let M be a manifold without boundary, $p, q \in M$ and $\gamma\colon [0,1] \to M$ a smooth path from p to q. Then there exists a diffeomorphism $\phi\colon M \to M$ with $\phi(p) = q$, which is smoothly isotopic to id_M. Moreover this isotopy can be chosen to be stationary outside a given open neighbourhood $U \subseteq M$ of the image of γ.*

Proof. The smooth path γ can be seen as an isotpy of the inclusion $\{p\} \hookrightarrow M$. Then the statement follows directly from Theorem 7.2. □

Remark 7.5. If $\mathrm{Aut}_{id}(M) \subset \mathrm{Aut}(M)$ denotes the subgroup of diffeomorphisms which are smoothly isotopic to id_M, the above corollary in particular implies that $\mathrm{Aut}_{id}(M)$ acts transitively on M.

We use Corollary 7.4 to show a corollary from [GP74].

Corollary 7.6. *Let M be a connected manifold without boundary of dimension at least 2. Given distinct points $x_1, \dots, x_k, y_1, \dots, y_k \in M$, there exists a diffeomorphism*

$$\psi\colon M \to M$$

with $\psi(x_i) = y_i$, which is smoothly isotopic to the identity. Moreover, we can choose this isotopy to be compactly supported.

Proof. We proof the statement inductively. For $k = 0$ there is nothing to show. So suppose we already have a diffeomorphism $\psi'\colon M \to M$ smoothly isotopic to the identity, where the isotopy is compactly supported and $\psi'(x_i) = y_i$ for $1 \leq i \leq k-1$. Note that $M - \{y_1, \dots, y_{k-1}\}$ is connected, so Corollary 7.4 implies the existence of a diffeomorphism $\phi\colon M \to M$ with $\phi(\psi'(x_k)) = y_k$ and $\phi(y_i) = y_i$ for $1 \leq i \leq k-1$, such that ϕ is smoothly isotopy to the identity via a compactly supported isotopy. Now set $\psi := \phi \circ \psi'$ and compose the isotopies to get a compactly supported isotopy from ψ to the identity. □

This next corollary essentially says, that there are many embeddings of small dimensional manifolds into large dimensional manifolds.

Corollary 7.7. *Let M be a connected manifold without boundary and S a manifold with or without boundary, such that there exists an embedding $S \hookrightarrow \mathbb{R}^{\dim(M)}$. Moreover we fix two distinct points $s_0, s_1 \in S$. Then for any two points $p, q \in M$ there exists an embedding $\iota\colon S \hookrightarrow M$, such that $\iota(s_0) = p$ and $\iota(s_1) = q$.*

Proof. By assumption we can find an embedding $\iota'\colon S \hookrightarrow U \subseteq M$ into any coordinate neighbourhood U around p, such that $\iota'(s_0) = p$. We may assume $p \neq q$, so we can find a smooth path from $\iota'(s_1)$ to q, that does not hit p (if M is 1-dimensional we may arrange the embedding ι' in such a way, that $\iota'(s_1)$ lies in the same connected component of the manifold $M - \{p\}$ as q). Then Corollary 7.4 implies the existence of a diffeomorphism $\phi\colon M \to M$ with $\phi(\iota'(s_1)) = q$ and $\phi(p) = p$. Now set $\iota := \phi \circ \iota'$. □

The above corollary should also be true if M is allowed to have boundary, but the proof fails, since Corollary 7.4 is clearly false if M has non-empty boundary. An interesting consequence of Corollary 7.7 is the existence of **embedded paths**: For any connected manifold M (without boundary) and any $p, q \in M$ there exists an embedding $\gamma\colon [0,1] \to M$ with $\gamma(0) = p, \gamma(1) = q$.

8 Some Linear Algebra

8.1 General Statements

Lemma 8.1. *Let $T\colon V \to W$ be a linear map of finite dimensional vector spaces and let A be the matrix representation of T with respect to some basis. Then the matrix representation of the dual map $T^*\colon W^* \to V^*$ with respect to the corresponding dual basis is given by A^T.*

Proof. Let $X\colon \mathbb{R}^n \to V$ and $Y\colon \mathbb{R}^m \to W$ be linear isomorphisms given by the chosen basis, i.e.

$$\begin{tikzcd} V \arrow[r, "T"] & W \\ \mathbb{R}^n \arrow[u, "X"'] \arrow[r, "A"] & \mathbb{R}^m \arrow[u, "Y"'] \end{tikzcd}$$

commutes. Moreover, let $\phi\colon \mathbb{R}^n \to (\mathbb{R}^n)^*$, $\psi\colon \mathbb{R}^m \to (\mathbb{R}^m)^*$ be the canonical isomorphisms induced by the standard basis $e_1, \ldots, e_n$ and $d_1, \ldots, d_m$ on $\mathbb{R}^n$ and $\mathbb{R}^m$, respectively. We then get the following commutative diagram

$$\begin{tikzcd} V^* & W^* \arrow[l, "T^*"'] \arrow[d, "Y^*"] \\ (\mathbb{R}^n)^* \arrow[from=u, "X^*"] & (\mathbb{R}^m)^* \arrow[l, "A^*"'] \\ \mathbb{R}^n \arrow[u, "\phi"'] & \mathbb{R}^m \arrow[l, "B"'] \arrow[u, "\psi"'] \end{tikzcd}$$

where B is the desired matrix representation of T^*. Write $B = (b_{ij})$. Then b_{ij} is given as the i-th component of the vector Bd_j. By the above diagram this is precisely

$$b_{ij} = (d_j^* \circ A)(e_i) = a_{ji},$$

i.e. $B = A^T$. ☐

Lemma 8.2. *Let $V, V_1, \ldots, V_k$ be finite dimensional vector spaces. There exist natural isomorphisms*

$$V_1^* \otimes \ldots \otimes V_k^* \xrightarrow{\cong} (V_1 \otimes \ldots \otimes V_k)^*$$
$$\Lambda^k V^* \xrightarrow{\cong} (\Lambda^k V)^*$$
$$\Sigma^k V^* \xrightarrow{\cong} (\Sigma^k V)^*$$

where $\Lambda^k V$ denotes the k-th exterior power and $\Sigma^k V$ the k-th symmetric power (see section 8.4).

Proof. Define the first isomorphism $V_1^* \otimes \ldots \otimes V_k^* \to (V_1 \otimes \ldots \otimes V_k)^*$ by

$$\phi_1 \otimes \ldots \otimes \phi_k \mapsto \left(v_1 \otimes \ldots \otimes v_k \mapsto \prod_{i=1}^{k} \phi_i(v_i) \right).$$

using the universal property of the tensor product. This map is clearly injective, hence indeed an isomorphism. Moreover this map induces the second isomorphism $\Lambda^k V^* \to (\Lambda^k V)^*$ by

$$\begin{tikzcd} (V^*)^{\otimes k} \arrow[r, "\cong"] & (V^{\otimes k})^* \arrow[d] \\ \Lambda^k V^* \arrow[u] \arrow[r] & (\Lambda^k V)^* \end{tikzcd}$$

where the vertical maps are (induced by) the inclusions $\Lambda^k V^* \to (V^*)^{\otimes k}$ and $\Lambda^k V \to V^{\otimes k}$. To see that this does in fact define an isomorphism note that an inverse is explicitly given by

$$\begin{tikzcd} (V^*)^{\otimes k} \arrow[r, "\cong"] \arrow[d, "\mathrm{Alt}"'] & (V^{\otimes k})^* \\ \Lambda^k V^* & (\Lambda^k V)^* \arrow[l] \arrow[u, "\mathrm{Alt}^*"'] \end{tikzcd}$$

where Alt denotes the antisymmetrization map (again see section 8.4). The third isomorphism $\Sigma^k V^* \to (\Sigma^k V)^*$ is defined analogously using the symmetrization map instead. Naturality of the defined isomorphisms follows directly from the definiton by a straightforward computation. $\square$

8.2 Multilinear Algebra

Here are two statements about multilinear maps, which are generalizations of the corresponding statements for linear maps.

Lemma 8.3. *Let $V_1, \dots, V_k$ and W be normed vector spaces and*

$$A \colon V_1 \times \dots \times V_k \to Z$$

be a multilinear map, then A is continous (with respect to the product norm on $V_1 \times \dots \times V_k$) if and only if it is bounded, i.e. there exists $C > 0$ with

$$\|A(v_1, \dots, v_k)\| \leq C \prod_{i=1}^{k} \|v_i\|$$

for all $v_i \in V_i$.

Proof. The proof is analogous to the linear case. First assume that A is continous, then in particular there exists $\delta > 0$, such that

$$\sum_{i=1}^{k} \|v_i\| \leq k\delta \quad \Rightarrow \quad \|A(v_1, \dots, v_k)\| \leq 1.$$

So for arbitrary $v_i \in V_i$, we have

$$\|A(v_1, \dots, v_k)\| = \frac{1}{\delta^k} \prod_{i=1}^{k} \|v_i\| \, \|A(\frac{\delta v_1}{\|v_1\|}, \dots, \frac{\delta v_k}{\|v_k\|})\| \leq \frac{1}{\delta^k} \prod_{i=1}^{k} \|v_i\|$$

Now assume, that A is bounded. Given any $v_i, w_i \in V_i$ we compute:

$$\begin{aligned} \|A(v_1, \dots, v_k) - A(w_1, \dots, w_k)\| &= \|\sum_{i=1}^{k} A(w_1, \dots, w_{i-1}, v_i - w_i, v_{i+1}, \dots, v_k)\| \\ &\leq C \sum_{i=1}^{k} \|w_1\| \cdots \|w_{i-1}\| \, \|v_i - w_i\| \, \|v_{i+1}\| \cdots \|v_k\| \end{aligned}$$

Since the left hand side vanishes for $v_i \to w_i$ the claim follows. $\square$

Lemma 8.4. *Multilinear maps on finite dimensional vector spaces are continous (with respect to any norm).*

Proof. For ease of notation we only proof the Lemma for bilinear maps, but the proof is completely analogous for the general case. So let $A \colon V \times W \to Z$ be a bilinear map for finite dimensional vetcor spaces V, W and let $e_1, \dots e_k$ and $d_1, \dots, d_m$ be a basis for V and W, respectively. Given $v \in V$ and $w \in W$ write

$$v = \sum_{i=1}^{k} v^i e_i, \; w = \sum_{j=1}^{m} w^j d_j.$$

Define $C := \max_{i,j} \|A(e_i, d_j)\|$, then

$$\|A(v, w)\| \leq \sum_{i,j} |v^i||w^j| \, \|A(e_i, d_j)\| \leq C \sum_{i,j} |v^i||w^j| = C \left(\sum_i |v^i| \right) \left(\sum_j |w^j| \right).$$

Note that

$$\|v\| := \sum_i |v^i|$$

defines a norm, so A is bounded with respect to these norms on V, W, hence continous by Lemma 8.3. Since all norms on finite dimensional vector spaces are equivalent, the statement follows. $\square$

The following lemma extends the universal property of the Tensor Product to multilinear maps.

Lemma 8.5. *For $i = 1, \dots n$ let $V_1^i, \dots V_{k_i}^i$ be vector spaces and*

$$\pi_i \colon V_1^i \times \dots \times V_{k_i}^i \to V_1^i \otimes \dots \otimes V_{k_i}^i$$

the canonical product maps. Then for every multilinear map $L \colon \prod_{i=1}^n V_1^i \times \dots \times V_{k_i}^i \to Z$ to a vector space Z, there exists a unique multilinear map $A \colon \prod_{i=1}^n V_1^i \otimes \dots \otimes V_{k_i}^i \to Z$ such that the following diagram commutes

$$\begin{tikzcd}
\prod_{i=1}^n V_1^i \times \dots \times V_{k_i}^i \arrow[r, "L"] \arrow[d, "\prod_i \pi_i"'] & Z \\
\prod_{i=1}^n V_1^i \otimes \dots \otimes V_{k_i}^i \arrow[ur, dashed, "A"'] &
\end{tikzcd}$$

Proof. Consider the following commutative diagram

$$\begin{tikzcd}
\prod_{i=1}^n V_1^i \times \dots \times V_{k_i}^i \arrow[d, "\pi"'] \arrow[dr, "L"] \arrow[dd, bend right=60, "\prod_i \pi_i"'] & \\
\bigotimes_{i=1}^n V_1^i \otimes \dots \otimes V_{k_i}^i \arrow[r, dashed, "B"] & Z \\
\prod_{i=1}^n V_1^i \otimes \dots \otimes V_{k_i}^i \arrow[u, "p"] \arrow[ur, "A"] &
\end{tikzcd}$$

where p and π are the canonical product maps, the map B is induced by L by the universal property of the tensor product and A is defined as the composition of B and p. Clearly A is the unique map satisfying the statement of the lemma. $\square$

8.3 Symplectic Vector Spaces

Lemma 8.6. *Let (V, ω) be a symplectitc vector space and $W \subseteq V$ an isotropic subspace. Then any basis of W can be extended to a symplectic basis of V.*

Proof. We proof this by induction over $\dim(V)$. If $\dim(V) = 2$ the statement follows directly from non-degeneracy of ω. Now assume the statement holds for $\dim(V) = 2(n-1)$ and let $\dim(V) = 2n$. Given a basis $u_1, \dots, u_k$ of W define

$$U := \operatorname{span}(u_2, \dots, u_k) \subsetneq W$$

Then $W^\omega \subsetneq U^\omega$, hence there exists $v_1 \in U^\omega$ with $v_1 \notin W^\omega$, so by construction we must have

$$\omega(u_1, v_1) \neq 0.$$

After normalizing we may assume $\omega(u_1, v_1) = 1$. Now consider

$$Q := \operatorname{span}(u_1, v_1).$$

Then Q is a symplectic subspace, thus

$$(Q^\omega, \sigma := \omega|_{Q^\omega})$$

again a symplectic vector space. Moreover $U \subseteq Q^\omega$ and

$$U \subseteq U^\sigma.$$

So by our induction hypothesis we can extend $u_2, \dots, u_k$ to a symplectic basis $u_2, \dots, u_n, v_2, \dots, v_n$ of (Q^ω, σ). Then

$$u_1, \dots, u_n, v_1, \dots, v_n$$

is a symplectic basis of (V, ω). $\square$

The following Lemma gives a useful characterisation of symplectic matrices analogous to orthogonal matrices.

Lemma 8.7. *A matrix $A \in \mathbb{R}^{2n\times 2n}$ is symplectic if and only if the column vectors of A form a symplectic basis of $(\mathbb{R}^{2n}, \omega_0)$.*

Proof. Write $A = (a_{x_1} \cdots a_{x_n} a_{y_1} \cdots a_{y_n})$ for $a_{x_i}, a_{y_i} \in \mathbb{R}^{2n}$. If A is symplectic we have

$$\omega_0(a_{x_i}, a_{y_j}) = \omega_0(Ae_{x_i}, Ae_{y_i}) = \omega_0(e_{x_i}, e_{y_i}) = \delta_{ij}$$

and similiarly $\omega_0(a_{x_i}, a_{x_j}) = 0 = \omega_0(a_{y_i}, a_{y_j})$. For the other implication it suffices to show that $\omega_0(\cdot,\cdot) = \omega_0(A\cdot, A\cdot)$ holds on basis vectors, which follows from the same calculation. $\square$

8.4 The Exterior Algebra

This section is meant to give a quick overview of the definition and most essential properties of the exterior algebra $\Lambda^* V$ of a finite dimensional vector space V (over some field $\mathbb{K}$). To define the k-fold exterior power $\Lambda^k V$ consider the permutation maps

$$P_\sigma \colon V^{\otimes k} \to V^{\otimes k}$$

for $\sigma \in S(k)$[4] given by

$$P_\sigma(v_1 \otimes \ldots \otimes v_k) := \operatorname{sgn}(\sigma) v_{\sigma(1)} \otimes \ldots \otimes v_{\sigma(k)}.$$

These are well-defined by the universal property of the tensor product. We then call a vector $\alpha \in V^{\otimes k}$ **alternating** if

$$P_\sigma(\alpha) = \alpha \quad \forall \sigma \in S(k)$$

and define $\Lambda^k V \subseteq V^{\otimes k}$ as the subspace of all alternating vectors. Given any vector in $V^{\otimes k}$ there is a canonical way to make it into an alternating one using the **antisymmetrization map**

$$\operatorname{Alt} \colon V^{\otimes k} \to \Lambda^k V, \quad \alpha \mapsto \frac{1}{k!} \sum_{\sigma \in S(k)} P_\sigma(\alpha).$$

Note that this map is surjective and $\operatorname{Alt}(\alpha) = \alpha$ if and only if α is alternating. Using the antisymmetrization map we can define the **exterior product** (or **wedge product**) as the bilinear map

$$\begin{aligned} &\Lambda^k V \times \Lambda^l V \to \Lambda^{k+l} V, \\ &(\alpha, \beta) \mapsto \alpha \wedge \beta := \frac{(k+l)!}{k!l!} \operatorname{Alt}(\alpha \otimes \beta). \end{aligned}$$

Without proof we state the facts, that this product is associative and graded commutative. Moreover if $e_1, \ldots, e_n \in V$ is a basis of V then

$$\{e_{i_1} \wedge \ldots \wedge e_{i_k}\}_{1 \leq i_1 \leq \ldots \leq i_k \leq n}$$

is a basis of $\Lambda^k V$. If we identify $\Lambda^k V$ with the space of alternating maps $V^* \times \ldots \times V^* \to \mathbb{R}$, these basis vectors satisfy

$$(e_{i_1} \wedge \ldots \wedge e_{i_k})(\varepsilon^{j_1}, \ldots, \varepsilon^{j_k}) = \begin{cases} 1 & , (i_1, \ldots, i_k) = (j_1, \ldots, j_k) \\ 0 & \text{otherwise} \end{cases}$$

where $\varepsilon^1, \ldots, \varepsilon^n \in V^*$ is the dual basis of $e_1, \ldots, e_n$ and $1 \leq j_1 < \ldots < j_k \leq n$ is any ordered tuple. This is the reason for the normalization factor in the definition of the wedge product. The exterior algebra is then defined as

$$\Lambda^* V := \bigoplus_{k=0}^{\infty} \Lambda^k V$$

with multiplication given by the wedge product (we set $V^{\otimes 0} = \Lambda^0 V := \mathbb{K}$).

We can also characterize the exterior power and the exterior algebra by means of a universal property.

[4] S(k) denotes the symmetric group on n elements

Definition 8.8 (Universal property of the exterior power)**.** The (k-fold) exterior power of V, denoted $\Lambda^k V$, is a vector space together with an alternating multilinear map

$$\pi\colon V^{\times k} \to \Lambda^k V,$$

such that for any other alternating multilinear map

$$T\colon V^{\times k} \to W$$

into a vector space W there exists a unique linear map

$$\widetilde{T}\colon \Lambda^k V \to W$$

making the following diagram commute

$$\begin{tikzcd} V^{\times k} \arrow[r, "T"] \arrow[d, "\pi"'] & W \\ \Lambda^k V \arrow[ur, dashed, "\widetilde{T}"'] & \end{tikzcd}$$

Definition 8.9 (Universal property of the exterior algebra)**.** The exterior algebra of V, denoted $\Lambda^* V$, is an algebra together with a linear map

$$\iota\colon V \to \Lambda^* V,$$

which satisfies $\iota(v)^2 = 0$ for all $v \in V$ and such that for any linear map

$$\psi\colon V \to A'$$

into an algebra A' with $\psi(v)^2 = 0$ for all $v \in V$ there exists a unique algebra homomorphism

$$L\colon \Lambda^* V \to A'$$

making the following diagram commute

$$\begin{tikzcd} V \arrow[r, "\iota"] \arrow[dr, "\psi"'] & \Lambda^* V \arrow[d, dashed, "L"] \\ & A' \end{tikzcd}$$

Remark 8.10. The universal property of the exterior power is analogous to the one of the tensor product, where we think of π as the product map $(v_1, \ldots, v_k) \mapsto v_1 \wedge \ldots \wedge v_k$. So alternating multilinear maps on $V^{\times k}$ are in $1:1$ correspondence to linear maps on $\Lambda^k V$. In Definition 8.9 we think of ι as an inlcusion and of L as extending the map ψ.

Lemma 8.11. *The construction of $\Lambda^k V \subseteq V^{\otimes k}$ at the beginning of the section together with the product map $\pi\colon V^{\times k} \to \Lambda^k V$, $(v_1, \ldots, v_k) \mapsto v_1 \wedge \ldots \wedge v_k$ satisifes the universal property in Definition 8.8.*

Proof. First of all it follows from the graded commutativity of the wedge product, that π is alternating. Uniqueness of $\widetilde{T}$ is also clear. To show existence choose a basis $e_1, \ldots, e_n \in V$ of V. As mentioned above this yields a basis of $\Lambda^k V$ and we define $\widetilde{T}$ on this basis (necessarily) by

$$\widetilde{T}(e_{i_1} \wedge \ldots \wedge e_{i_k}) := T(e_{i_1}, \ldots, e_{i_k})$$

for $1 \leq i_1 < \ldots < i_k \leq n$. To see that

$$\begin{tikzcd} V^{\times k} \arrow[r, "T"] \arrow[d, "\pi"'] & W \\ \Lambda^k V \arrow[ur, dashed, "\widetilde{T}"'] & \end{tikzcd}$$

does in fact commute it suffices to verify commutativity on basis vectors $(e_{i_1}, \ldots, e_{i_k})$ of $V^{\times k}$ (where the tuple $(i_1, \ldots, i_k)$ is not necessarily ordered!). But since T and π are both alternating we may assume after a suitable permutation of $(i_1, \ldots, i_k)$, that this tuple is ordered, such that the statement follows from the definiton of $\widetilde{T}$. $\square$

Lemma 8.12. *The construction of Λ^*V at the beginning of the section together with the canonical inclusion map $V \hookrightarrow \Lambda^*V$ satisifes the universal property in Definition 8.9.*

Proof. Again uniqueness of L is clear. For the existence of L we define it on $\Lambda^k V$ (necessarily) by

$$L(v_1 \wedge \ldots \wedge v_k) := \prod_{i=1}^{k} \psi'(v_i)$$

using the universal property of the exterior power shown in Lemma 8.11. ☐

To conclude this section let us mention another commonly used construction of the exterior power and exterior algebra. To this end let

$$S^k(V) := \operatorname{span}\{v_1 \otimes \ldots \otimes v_k \mid \exists\, 1 \leq i, j \leq k, i \neq j \text{ with } v_i = v_j\} \subseteq V^{\otimes k}$$

Then it is not hard to show that $V^{\otimes k}/S^k(V)$ together with the canonical map

$$V^{\times k} \to V^{\otimes k} \to V^{\otimes k}/S^k(V)$$

also satisifies the universal property in Definition 8.8. Hence this is also a valid and commonly used construction of the exterior power. With this definition

$$\alpha \wedge \beta = [\alpha \otimes \beta].$$

Similiarly we can also define the exterior algebra as a quotient of the tensor algebra

$$T(V) := \bigoplus_{k=0}^{\infty} V^{\otimes k},$$

by dividing out the two sided ideal $I \subseteq T(V)$ generated by elements of the form $v \otimes v \in T(V)$ for $v \in V$, i.e.

$$\Lambda^*V := T(V)/I.$$

Together with the canonical map $V \to \Lambda^*V$ this definition is also easily seen to satisfy the universal property in Definition 8.9.

9 Functional Analysis

9.1 L^p and Sobolev Spaces

The partial integration rule for C^1-functions defined on open subsets of $\mathbb{R}^n$ is shown using Gauß's Theorem, which in turn motivates the definition of weakly differentiable functions for which the partial integration rule holds by definition. Interestingly we can also go the other way around and proof Gauß's Theorem from the partial integration rule, thus establishing the theorem for weakly differentiable functions as well.

Theorem 9.1 (Gauß's Theorem). *Let $U \subseteq \mathbb{R}^n$ be open and $\Omega \subseteq U$ a compact smooth manifold with boundary of codimension 0 with outer unit normal field $\nu\colon \partial\Omega \to \mathbb{R}^n$. Given a weakly differentiable continuous vector field $\xi \in L^1_{\mathrm{loc}}(U, \mathbb{R}^n) \cap C^0(U, \mathbb{R}^n)$ with $\partial_i \xi \in L^1_{\mathrm{loc}}(U, \mathbb{R}^n)$ for $1 \leq i \leq n$, the following holds*

$$\int_\Omega \mathrm{div}(\xi)\mathrm{vol}_\Omega = \int_{\partial\Omega} \langle \xi, \nu \rangle \mathrm{vol}_{\partial\Omega}.$$

Remark 9.2. By the Sobolev embedding Theorems, the conditions on ξ are for example satisfied if $\xi \in W^{k,p}(U, \mathbb{R}^n)$ for $pk > n$.

Proof of Theorem 9.1. The idea of the proof is to construct a particular sequence of cutoff functions

$$\phi_k \colon U \to \mathbb{R},$$

which will allow us to approximate both sides of the equation and at the same time lets us relate them by the partial integration rule for ξ. We begin by extending ν arbitarily to a smooth vector field $\tilde{\nu}$ on all of U. Since $\partial\Omega$ is compact, there exists $\varepsilon > 0$, such that the integral curves of $-\tilde{\nu}$ starting at points in $\partial\Omega$ are defined on $(-\varepsilon, \varepsilon)$. Hence we can define the following bicollar neighbourhood of $\partial\Omega$ using the flow of $-\tilde{\nu}$:

$$\begin{aligned} \Psi \colon \partial\Omega \times (-\varepsilon, \varepsilon) &\to U \\ (x, t) &\mapsto \gamma_x(t), \end{aligned}$$

where $\gamma_x(t)$ is the integral curve of $-\tilde{\nu}$ with $\gamma_x(0) = x$. We will now use this bicollar neighbourhood to construct ϕ_k out of cutoff functions $\lambda_k \in C^\infty(\mathbb{R})$, with the following properties

(i) $\mathrm{supp}\lambda_k \subseteq (-\frac{1}{k}, \frac{1}{k})$

(ii) $\int_{-\infty}^{\infty} \lambda_k = 1$

(iii) $\lambda_k(t) = \lambda_k(-t)$

So λ_k is a special kind of Dirac-sequence. The reason for this definition will become clear later. Define

$$\Lambda_k \colon \partial\Omega \times (-\varepsilon, \varepsilon) \to \mathbb{R},\ (x, t) \mapsto \int_{-\infty}^{t} \lambda_k(s) ds.$$

Note that for k sufficiently large we have

$$\Lambda_k(x, t) = \begin{cases} 1, & t > \dfrac{\varepsilon}{3} \\ 0, & t < \dfrac{\varepsilon}{3} \end{cases}$$

Moreover choosing ε sufficiently small we may assume, that $\Psi(\partial\Omega \times (0, \varepsilon)) \subseteq \mathrm{int}\Omega$ and $\Psi(\partial\Omega \times (-\varepsilon, 0)) \subseteq U \setminus \Omega$ (here we need compactness of $\partial\Omega$ again). This now allows us to define the desired cutoff functions (for large k) as

$$\phi_k(y) := \begin{cases} \Lambda_k(\Psi^{-1}(y)), & y \in \mathrm{im}\Psi \\ 1, & y \in \mathrm{int}\Omega \setminus \Psi\left(\partial\Omega \times \left[0, \dfrac{\varepsilon}{2}\right]\right) \\ 0, & y \in (U \setminus \Omega) \setminus \Psi\left(\partial\Omega \times \left[-\dfrac{\varepsilon}{2}, 0\right]\right) \end{cases}$$

By construction $\phi_k \in C_c^\infty(U)$, since Ω is compact and as $k \to \infty$

$$\phi_k(y) \to \chi_\Omega(y) := \begin{cases} 1, & y \in \Omega \\ 0, & y \notin \Omega \end{cases}$$

for almost all $y \in U$ (for $y \in \partial\Omega = \Psi(\partial\Omega \times \{0\})$ we have $\phi_k(y) = \frac{1}{2}$ for all k). By definition of weak differentiablity we now get[5]

$$\int_U \operatorname{div}(\xi)\phi_k = -\sum_{i=1}^n \int_U \xi_i \, \partial_i \phi_k = -\int_U \langle \xi, \nabla \phi_k \rangle. \tag{9}$$

Since $\partial_i \xi_i \in L^1_{\mathrm{loc}}(U)$ we can use the dominated convergence theorem to conclude that the left hand side converges to

$$\int_\Omega \operatorname{div}(\xi)$$

as $k \to \infty$. To show convergence of the right hand side to

$$\int_{\partial\Omega} \langle \xi, \nu \rangle$$

we will have to analyze this term more closely, which is where our very deliberate construction of ϕ_k comes into play. We begin by computing the gradient $\nabla \phi_k$. Let

$$\beta_k \colon U \to \mathbb{R},\ y \mapsto \begin{cases} \lambda_k(t), & y = \Psi(x,t) \\ 0, & \text{otherwise} \end{cases}$$

Again for k sufficiently large this is a well defined smooth function. Moreover let

$$\pi \colon \operatorname{im}(\Psi) \to \mathbb{R},\ \Psi(x,t) \mapsto t.$$

We now claim that

$$\nabla \phi_k = \beta_k \nabla \pi$$

where the right hand side is to be understood to vanish outisde of $\operatorname{im}(\Psi)$, so clearly equality holds outside of $\operatorname{im}(\Psi)$. Note that Λ_k is the composition of the projection $\partial\Omega \times (-\varepsilon, \varepsilon) \to (-\varepsilon, \varepsilon)$ with the map

$$t \mapsto \int_{-\infty}^t \lambda_k(s) ds,$$

so for $y = \Psi(x,t)$ we get by the chain rule

$$d_y \phi_k = d_{(x,t)} \Lambda_k \circ d_y \Psi^{-1} = \lambda_k(t) d_y \pi.$$

Since $\lambda_k(t) = \beta_k(y)$ the assertion follows. Furthermore for $y \in \partial\Omega$ and $\tau \in T_y \partial\Omega$ we have

$$d_y \pi(\tau) = 0$$

and

$$d_y \pi(-\nu(y)) = d_y \pi(\dot{\gamma}_y(0)) = 1,$$

such that $\nabla \pi = -\nu$ on $\partial\Omega$. We can now compute the right hand side of (9) as

$$\int_U \langle \xi, \nabla \phi_k \rangle \mathrm{vol}_U = \int_{\operatorname{im}\Psi} \langle \xi, \nabla \pi \rangle \beta_k \mathrm{vol}_{\operatorname{im}\Psi} = \int_{\partial\Omega \times (-\varepsilon,\varepsilon)} (\langle \xi, \nabla \pi \rangle \circ \Psi)(\beta_k \circ \Psi) \Psi^* \mathrm{vol}_{\operatorname{im}\Psi} \tag{10}$$

Write

$$\Psi^* \mathrm{vol}_{\operatorname{im}\Psi} = \mu \mathrm{vol}_{\partial\Omega \times (-\varepsilon,\varepsilon)}$$

for a smooth function $\mu \in C^\infty(\partial\Omega \times (-\varepsilon, \varepsilon))$. By construction

$$d_{(y,0)} \Psi \left(\frac{\partial}{\partial t} \right) = -\nu(y)$$

and $d_{(y,0)} \Psi(\tau) = \tau$ for all $\tau \in T_y \partial\Omega$. It follows that $\mu(y,0) = 1$ (if we choose the orientation on $\partial\Omega$ correctly).

[5] For better readibility we will sometimes not write the volume forms vol_Ω and $\mathrm{vol}_{\partial\Omega}$ in the integrals

Write
$$f := (\langle \xi, \nabla\pi \rangle \circ \Psi)\mu \colon \partial\Omega \times (-\varepsilon, \varepsilon) \to \mathbb{R}.$$
Then (10) becomes
$$\int_{\partial\Omega\times(-\varepsilon,\varepsilon)} f(\beta_k \circ \Psi)\mathrm{vol}_{\partial\Omega\times(-\varepsilon,\varepsilon)} = \int_{-\varepsilon}^{\varepsilon}\int_{\partial\Omega} f(-,t)\lambda_k(t)\mathrm{vol}_{\partial\Omega}dt \tag{11}$$
Let
$$\rho(t) := \int_{\partial\Omega} f(-,t)\mathrm{vol}_{\partial\Omega}.$$
Since $\partial\Omega$ is compact and f is continuous (here our assumption $\xi \in C^0(U,\mathbb{R}^n)$ finally comes into play) it follows that ρ is continuous as well. Using this we can simplify (11):
$$\int_{-\varepsilon}^{\varepsilon} \lambda_k(t)\rho(t)dt = \int_{-\varepsilon}^{\varepsilon} \lambda_k(-t)\rho(t)dt \tag{12}$$
For large k this integral only depends on the values of ρ near 0, so using a cutoff function we may assume that ρ is a continuous bounded function defined on all of $\mathbb{R}$. Then finally we can write (12) as
$$\int_{-\infty}^{\infty} \lambda_k(-t)\rho(t)dt = (\rho * \lambda_k)(0),$$
where $\rho * \lambda_k$ denotes the convolution. Since λ_k is a Dirac-sequence it follows from [Sch12] Theorem 4.16. (ii), that
$$(\rho * \lambda_k)(0) \to \rho(0)$$
as $k \to \infty$. Hence we have shown, that the right hand side of (9) converges to
$$-\rho(0) = -\int_{\partial\Omega} f(-,0)\mathrm{vol}_{\partial\Omega}.$$
But $\mu(y,0) = 1$ and $\nabla\pi(y) = -\nu(y)$ for $y \in \partial\Omega$, so by definition of f
$$f(y,0) = -\langle \xi(y), \nu(y) \rangle,$$
from which the theorem follows.

□

On two equivalent definitions of $L^p(U,\mathbb{R}^n)$

Let $U \subseteq \mathbb{R}^n$ be open. Starting with the space $L^p(U)$ of real valued L^p functions, there are two possible defintions for the space $L^p(U,\mathbb{R}^k)$ of $\mathbb{R}^k$ valued L^p functions. The first definition requires each component function to be L^p:

Definition 9.3. $L^p(U,\mathbb{R}^k) := \{f \colon U \to \mathbb{R}^k \text{ measurable } \mid f_i \in L^p(U) \text{ for all } 1 \leq i \leq k\}$

The second definition is more similar to the definition of $L^p(U)$:

Definition 9.4. $L^p(U,\mathbb{R}^k) := \{f \colon U \to \mathbb{R}^k \text{ measurable } \mid ||f|| \in L^p(U)\}$

These definitions are however equivalent (as one would certainly hope). To see this we compute for a measurable function $f \colon U \to \mathbb{R}^k$:
$$||f_i||_p^p = \int_U |f_i|^p d\lambda \leq \int_U \sqrt{|f_1|^2 + \ldots + |f_k|^2}^{\,p} d\lambda = \int_U ||f||^p d\lambda$$
and
$$\int_U ||f||^p d\lambda \leq \int_U (|f_1| + \ldots + |f_k|)^p d\lambda \leq \frac{1}{k}\sum_{i=1}^{k}\int_U |f_i|^p d\lambda,$$
where we used the fact that
$$a^2 + b^2 \leq (a+b)^2$$
for $a, b \geq 0$ in the first inequality and made use of Jensen's inequality in the second inequality. We may also define two norms on $L^p(U,\mathbb{R}^k)$ corresponding to the two definitons:

1. $||f||_{p,1} := \sum_{i=1}^k ||f_i||_p$
2. $||f||_{p,2} := || \, ||f|| \, ||_p = \left(\int_U ||f||^p d\lambda\right)^{\frac{1}{p}}$

Again these norms turn out to be equivalent, which can be seen by a similar computation as above:

$$||f||_{p,1} \leq \sum_{i=1}^k \left(\int_U ||f||^p d\lambda\right)^{\frac{1}{p}} = k||f||_{p,2}$$

and conversely

$$\begin{aligned} ||f||_{p,2} &\leq \left(\left(\frac{1}{k}\right) \sum_{i=1}^k \int_U |f_i|^p d\lambda\right)^{\frac{1}{p}} = \left(\frac{1}{k}\right)^{\frac{1}{p}} \left(\sum_{i=1}^k ||f_i||_p^p\right)^{\frac{1}{p}} \\ &\leq \left(\frac{1}{k}\right)^{\frac{1}{p}} \left(\sum_{i=1}^k \max_{1\leq i\leq k} ||f_i||_p^p\right)^{\frac{1}{p}} = \max_{1\leq i\leq k} ||f_i||_p \leq \sum_{i=1}^k ||f_i||_p = ||f||_{p,1}. \end{aligned}$$

Note that for $p = 2$ the second norm is the one induced by the L^2-scalar product on $L^2(U, \mathbb{R}^k)$.

9.2 Complemented Subspaces

Given a vector space X and a subspace $X_0 \subseteq X$ we may want to decompose X as the direct sum of X_0 and some other subspace $X_1 \subseteq X$. More precisely this means that the canonical map

$$X_0 \oplus X_1 \to X$$

induced by the inclusions $X_i \hookrightarrow X$ should be an isomorphism. In the category of vector spaces this is of course always possible and we call X_1 and **algebraic complement** of X_0. However the situation looks quite a bit different in the category of topological vector spaces, where such X_1 may not always exist. If it exists we call X_1 a **topological complement** of X_0 and the space X_0 **complemented**. The following Lemma gives a characterization for when an algebraic complement is also a topological complement (at least in the case of Banach spaces).

Lemma 9.5. *Let X be a Banach space and $X_0, X_1 \subseteq X$ subspaces, where X_1 is an algebraic complement of X_0. Then X_1 is a topological complement of X_0 if and only if X_0 and X_1 are closed.*

Proof. Clearly $X_0, X_1 \subseteq X_0 \oplus X_1$ are closed, so if X is toplinearly isomorphic to $X_0 \oplus X_1$ they are also closed in X. On the contrary if X_0 and X_1 are closed, they are again Banach spaces and since we assume the canonical map

$$X_0 \oplus X_1 \to X$$

to be a bijection it follows from the open mapping theorem that this map is already a toplinear isomorphism. $\square$

Here are two simple sufficient conditions for the existence of topological complements. For a vector space X and a subspace $X_0 \subseteq X$ we define its **codimension** by

$$\operatorname{codim}(X_0) := \dim\left(X/X_0\right).$$

Lemma 9.6. *Let X be a Banach space and $X_0 \subseteq X$ a subspace. The following two conditions are both sufficient for the existence of a topological complement of X_0:*

(i) $\dim(X_0) < \infty$

(ii) $\operatorname{codim}(X_0) < \infty$ *and X_0 is closed*

Proof. First assume (i). Choose a basis $e_1, \ldots, e_n$ of X_0 and define linear maps $\phi_i \colon X_0 \to \mathbb{R}$ by

$$\phi_i(e_j) = \delta_{ij}.$$

Since X_0 is finite dimensional the ϕ_i are continuous. So using the Hahn-Banach theorem we can extend them to continuous linear functionals Φ_i defined on all of X. Now set

$$X_1 := \{x \in X \mid \Phi_1(x) = \ldots = \Phi_n(x) = 0\}.$$

Clearly $X_1 \subseteq X$ is closed, so by Lemma 9.5 we only have to show that X_1 is an algebraic complement of X_0. To this end let $x \in X$ be arbitary and define

$$x_0 := \sum_{i=1}^{n} \Phi_i(x) e_i \in X_0.$$

Then $x_1 := x - x_0 \in X_1$ and $x = x_0 + x_1$. Finally observe that $X_0 \cap X_1 = 0$ from which the statement follows.

For (ii) we begin by choosing any algebraic complement X_1 of X_0. Since the restriction of the projection

$$X \to X/_{X_0}$$

to X_1 is an isomorphism, our assumption tells us that X_1 is finite dimensional and so in particular closed. Thus the statement follows again from Lemma 9.5. □

9.3 The Arzelà-Ascoli Theorem

In this section we give a quite general formulation of the Arzelà-Ascoli theorem. The proof we give here is a very straightforward generalization of the one found in [Wer18]. Before stating the theorem let us recall two definitions. Given metric spaces (X, d_X) and (Y, d_Y) and a sequence $(f_n)_{n\in\mathbb{N}} \subseteq \mathrm{C}(X,Y)$ of continuous functions we say that $(f_n)_{n\in\mathbb{N}}$ is **pointwise relatively compact** if for all $x \in X$ the set

$$\{f_n(x) \mid n \in \mathbb{N}\} \subseteq Y$$

is relatively compact, i.e. has compact closure. Moreover we call $(f_n)_{n\in\mathbb{N}}$ **uniformly equicontinuous** if for all $\varepsilon > 0$ there exists $\delta > 0$, such that

$$f_n(B_\delta(x)) \subseteq B_\varepsilon(f_n(x))$$

for all $x \in X$ and $n \in \mathbb{N}$.

Theorem 9.7 (Arzelà-Ascoli)**.** *Let (X, d_X) be a compact metric space, (Y, d_Y) a complete metric space and $(f_n)_{n\in\mathbb{N}} \subseteq \mathrm{C}(X,Y)$ a sequence of continuous functions, which is pointwise relatively compact and uniformly equicontinuous. Then there exists a subsequence that converges uniformly.*

Proof. We begin by showing that any compact metric space is seperable. Since X is compact the open cover

$$\{B_{1/n}(x)\}_{x\in X}$$

of X admits a finite subcover for any $n \in \mathbb{N}$. So there exist finitely many points $x_{n,1}, \ldots, x_{n,k(n)} \in X$, such that

$$B_{1/n}(x_{n,1}), \ldots, B_{1/n}(x_{n,k(n)})$$

covers X. Then the countable set

$$\bigcup_{n\in\mathbb{N}} \{x_{n,1}, \ldots, x_{n,k(n)}\}.$$

is easily seen to be dense.

With this fact at hand we can now actually proof the theorem. Let $x_1, x_2, \ldots \in X$ be a dense sequence. Using a diagonalization argument we first construct a subseqence of $(f_n)_{n\in\mathbb{N}}$ which converges pointwise at all x_k. By assumption

$$\{f_n(x_1) \mid n \in \mathbb{N}\} \subseteq Y$$

is relatively compact, such that there exists a subseqence $(f_{k_n^1})_{n\in\mathbb{N}}$ of $(f_n)_{n\in\mathbb{N}}$ for which $(f_{k_n^1}(x_1))_{n\in\mathbb{N}}$ converges in Y. By the same argument we can find a subsequence $(f_{k_n^2})_{n\in\mathbb{N}}$ of $(f_{k_n^1})_{n\in\mathbb{N}}$ for which $(f_{k_n^2}(x_2))_{n\in\mathbb{N}}$ converges in Y and so on. Then the sequence

$$g_n := f_{k_n^n}$$

is a subsequence of $(f_n)_{n\in\mathbb{N}}$ for which $(g_n(x_k))_{n\in\mathbb{N}}$ converges in Y for any $k \in \mathbb{N}$.

We now claim that $(g_n)_{n\in\mathbb{N}}$ is already a Cauchy sequence in $C(X,Y)$ with respect to the metric of uniform convergence. Since Y is complete the same holds for $C(X,Y)$ thus proving the theorem. To show the claim let $\varepsilon > 0$ be arbitary. By uniform equicontinuity we can find $\delta > 0$, such that

$$g_n(B_\delta(x)) \subseteq B_\varepsilon(g_n(x))$$

for all $x \in X$ and $n \in \mathbb{N}$. Using the fact that $x_1, x_2, \dots$ is dense in the compact space X we can find $N \in \mathbb{N}$, such that

$$B_\delta(x_1), \dots, B_\delta(x_N)$$

cover X. Finally choose $L \in \mathbb{N}$, such that

$$d_Y(g_n(x_k), g_m(x_k)) < \varepsilon$$

for all $n, m \geq L$ and all $k = 1, \dots, N$. Then for arbitary $x \in X$ there exists some $1 \leq k \leq N$ with $x \in B_\delta(x_k)$. So for $n, m \geq L$ we compute

$$d_Y(g_n(x), g_m(x)) \leq d_Y(g_n(x), g_n(x_k)) + d_Y(g_n(x_k), g_m(x_k)) + d_Y(g_m(x_k), g_m(x)) < 3\varepsilon$$

The claim follows. □

Theorem 9.7 is usually stated differently in the case $Y = \mathbb{R}^k$ by replacing the condition of pointiwse relative compactness with the stronger condition of **uniform boundedness**, i.e. there exists some constant $C > 0$, such that

$$\max_{x\in X} |f_n(x)| < C$$

for all $n \in \mathbb{N}$. Since this is usually easier to check in practice let us note this version of the theorem down as well.

Theorem 9.8 ($\mathbb{R}^k$-valued Arzelà-Ascoli)**.** *Let (X, d_X) be a compact metric space and*

$$f_n \colon X \to \mathbb{R}^k$$

a sequence of continuous functions, which is uniformly bounded and uniformly equicontinuous. Then there exists a subsequence that converges uniformly.

As an application of Theorem 9.7 we show that the dual of a compact operator is again compact.

Lemma 9.9. *Let $K \colon X \to Y$ be a compact operator between normed spaces. Then its dual $K^* \colon Y^* \to X^*$ is compact as well.*

Proof. Let $(\psi_n)_{n\in\mathbb{N}} \subseteq Y^*$ be a sequence bounded in the operator norm by 1. We have to show that

$$\phi_n := K^*\psi_n$$

admits a convergent subsequence. Since

$$C := \overline{K(B_1(0))}$$

is compact (by definition of compact operators) we can try and apply the Arzelà-Ascoli theorem to the sequence

$$f_n := \psi_n|_C.$$

For uniform boundedness note that since C is compact it must be contained in some large ball of radius R, such that for $y \in C$

$$|f_n(y)| = R\left|\psi_n(\frac{y}{R})\right| \leq R\|\psi_n\|_{\mathrm{op}} \leq R.$$

To see uniform equicontinuity note that for $y, z \in C$

$$|f_n(y) - f_n(z)| = |\psi(y - z)| \leq \|y - z\|.$$

So by the Arzelà-Ascoli theorem we may assume that $(f_n)_{n\in\mathbb{N}}$ converges uniformly (at least after restriction to a subsequence). We now claim that $(\phi_n)_{n\in\mathbb{N}}$ is Cauchy with respect to the operator norm. For $x \in B_1(0)$ we compute

$$|\phi_n(x) - \phi_m(x)| = |f_n(Kx) - f_m(Kx)| \leq \max|f_n - f_m|.$$

Since $(f_n)_{n\in\mathbb{N}}$ is Cauchy the statement follows. □

10 Whitney Topologies

Let X, Y be smooth manifolds. In this section we discuss the concept of C^k-convergence for sequences of functions $(f_n) \subseteq C^k(X,Y)$ with respect to the Whitney topolgies. The notation and terminology used will be the same as in [GG12]. First we remark that X or Y is also allowed to have boundary. The definition of the jet bundles $J^k(X,Y)$ and the Whitney topologies works exactly the same (if both have boundary, $J^k(X,Y)$ will be a manifold with corners, so we might want to avoid that). Let us begin by understanding the C^0-topology.

Lemma 10.1. *Let X, Y be smooth manifolds with X compact. Then the Whitney C^0-topology (as described in [GG12]) and the exponential topology (as described in [Ste23b]) on $C^0(X,Y)$ agree.*

Remark 10.2. If X is not compact the C^0-topology will in general be strictly finer than the exponential topology. To see this recall that a subbase for the exponential topology on $C^0(X,Y)$ is given by sets of the form

$$F(K,V) := \{f \in C^0(X,Y) \mid f(K) \subseteq V\}$$

for compact sets $K \subseteq X$ and open sets $V \subseteq Y$. Set

$$W := X \times V \cup (X \setminus K) \times Y.$$

Then $W \subseteq X \times Y$ is open and

$$F(K,V) = M(W).$$

To see that the C^0 is finer in general consider the following example. ?????????????? To differentiate between the two topologies on $C^0(X,Y)$ we shall use the term C^0_{exp}-convergence to refer to convergence with respect to the exponential topology.

Next we understand convergence in the C^k-topology. Before stating the main proposition we shall introduce another type of convergence which is better suited for non-compact domains, since in the non-compact case usual C^k-convergence is often too strong, because

$$f_n \xrightarrow{C^k} f$$

necessarily requires for f_n and f to agree outside a compact subset (see [GG12]).

Definition 10.3. Let X, Y be smooth manifolds and $f, f_n \in C^k(X,Y)$. We say that

$$f_n \xrightarrow{C^k_{\text{loc}}} f$$

if for every compact codimension 0-submanifold $K \subseteq X$ we have

$$f_n|_K \xrightarrow{C^k} f|_K.$$

Note that for compact domains C^k_{loc}-convergence and C^k-convergence coincide, so we are actually interested in understanding C^k_{loc}-convergence. The following two propositions give a useful characterization of C^k_{loc}-convergence to work with in practice.

Proposition 10.4. *Let $D \subseteq \mathbb{R}^n$ be a compact codimension 0 submanifold, $V \subseteq \mathbb{R}^m$ open and $f, f_n \in C^k(D,V)$. Then*

$$f_n \xrightarrow{C^k} f$$

if and only if for every multiindex $\alpha \in \mathbb{N}_0^n$ with $0 \leq |\alpha| \leq k$

$$\frac{\partial^{|\alpha|} f_n}{\partial x^\alpha} \xrightarrow{\text{uniformly}} \frac{\partial^{|\alpha|} f}{\partial x^\alpha}.$$

Proposition 10.5. *Let X, Y be smooth manifolds and $f, f_n \in C^k(X,Y)$. Then the following statements are equivalent.*

(i) $f_n \xrightarrow{C^k_{\text{loc}}} f$,

(ii) $j^k f_n \xrightarrow{C^0_{\text{loc}}} j^k f$,

(iii) For all charts $\phi\colon U \to U' \subseteq X$ *and* $\psi\colon V \to V' \subseteq Y$ *with* $f(U'), f_n(U') \subseteq V'$ *(for* n *sufficiently large) we have*

$$\psi^{-1} \circ f_n \circ \phi \xrightarrow{C^k_{\text{loc}}} \psi^{-1} \circ f \circ \phi,$$

(iv) For every $p \in X$ *there exist charts* $\phi\colon U \to U' \subseteq X$ *around* p *and* $\psi\colon V \to V' \subseteq Y$ *around* $f(p)$ *with* $f(U'), f_n(U') \subseteq V'$ *(for* n *sufficiently large) such that*

$$\psi^{-1} \circ f_n \circ \phi \xrightarrow{C^k_{\text{loc}}} \psi^{-1} \circ f \circ \phi.$$

Before proving the propositions we recall a few facts about the exponential topology not mentioned in [Ste23b], which will be used in the proofs.

Lemma 10.6. *Let* X, Y, Z *be topological spaces and* $A \subseteq X$, $B \subseteq Y$ *subspaces. Then the following holds (assuming the exponential topology exists on each space)*

(i) The canonical map $C^0(X, B) \hookrightarrow C^0(X, Y)$ *is an embedding*

(ii) The restriction map $C^0(X, Y) \to C^0(A, Y)$ *is continuous*

(iii) The canonical bijection $C^0(X, Y) \times C^0(X, Z) \to C^0(X, Y \times Z)$ *is a homeomorphism*

Proof. For (i) we show that the exponential topology on $C^0(X, B)$ agrees with the topology induced by the map $C^0(X, B) \hookrightarrow C^0(X, Y)$ by showing the corresponding universal property. So let

$$\begin{tikzcd} & C^0(X,Y) \\ Z \arrow[ur, "F"] \arrow[r, "f"] & C^0(X,B) \arrow[u, hook] \end{tikzcd}$$

commute with F being continuous. We have to show that f is continuous, but by definiton of the exponential topology this is equivalent to

$$Z \times X \to B,\ (z, x) \mapsto f(z)(x)$$

being continuous and continuity of F is equivalent to continuity of

$$Z \times X \to Y,\ (z, x) \mapsto F(z)(x),$$

so (i) follows. For (ii) note that the restriction map is continuous if and only if

$$C^0(X, Y) \times A \to Y,\ (f, x) \mapsto f|_A(x)$$

is continuous. But this is just the restriction of the evaluation map which is continuous by [Ste23b]. Similiarly the map in (iii) is continuous if and only if

$$C^0(X, Y) \times C^0(X, Z) \times X \to Y \times Z,\ (f, g, x) \mapsto (f(x), g(x))$$

is. But again the continuity follows from the continuity of the evaluation map. $\square$

Proof of Proposition 10.4. In [GG12] it was shown that for compact domains C^k-convergence is equivalent to the k-jets converging uniformly with respect to any metric on the Jet-bundle compatible with its topology, that is

$$f_n \xrightarrow{C^k} f \ \Leftrightarrow\ j^k f_n \xrightarrow{\text{uniformly}} j^k f$$

Note that for compact domains uniform convergence and C^0_{exp}-convergence agree (see [Ste23b]). Now let $d(n, k) = \frac{(n+k)!}{n!k!}$ be the total number of all partial derivatives ∂^α of order $1 \leq |\alpha| \leq k$ and let

$$\tau\colon D \times V \times \mathbb{R}^{d(n,k)} \to J^k(D, V)$$

be the canonical diffeomorphism. Then

$$j^k f_n \xrightarrow{C^0_{\exp}} j^k f \;\Leftrightarrow\; j^k f_n \circ \tau \xrightarrow{C^0_{\exp}} j^k f \circ \tau.$$

But a sequence of maps with values in a product $C^0_{\exp}$-converges if and only if each component $C^0_{\exp}$-converges. Since the components of $j^k f_n \circ \tau$ and $j^k f \circ \tau$ are precisely the partial derivatives of f_n and f up to order k the statement follows (after again applying the equivalence of uniform and $C^0_{\exp}$-convergence on compact domains). $\square$

11 Miscellaneous

11.1 Smoothness on arbitrary subsets

Let M, N be manifolds and $A \subseteq M$ an arbitrary subset. In [Lee11] the author defines what it means for a map

$$f: A \to N$$

to be smooth, namely

We call f smooth if around each $x \in A$ there exists a smooth extension $\widetilde{f}: U \to N$ of f to some open neighbourhood $U \subseteq M$ of x (see [Lee11, p. 45]).

However this definition has a fatal flaw: if A happens to be a submanifold of M, a map $f: A \to N$ might be smooth in the usual sense (i.e. as a map between smooth manifolds) but not smooth in the sense of Lee‘s definition. Consider for example

Counterexample 11.1.

$$M := \mathbb{R}, N := [0,1], A := [0,1],$$
$$f: A \to N,\, t \mapsto t.$$

Then clearly f is smooth as a map between smooth manifolds, but around $0, 1 \in A$ it does not admit a smooth extension onto an open neighbourhood $U \subseteq M$, since the image of such an extension would fail to be contained within N.

We can however correct this flaw by defining:

Definition 11.2 (Extension-smooth)**.** Let M^n, N^m be manifolds with or without boundary, $A \subseteq M$ an arbitrary subset and $f: A \to N$ a map. We call f *extension-smooth* if around each $x \in A$ there exist charts

$$h: U \to M \text{ around } x$$
$$k: V \to N \text{ around } f(x),$$

such that

$$g := k^{-1} \circ f \circ h|_{h^{-1}(A)}$$

admits a smooth extension around $h^{-1}(x)$ in the sense that there exist open subsets $U_0 \subseteq \mathbb{R}^n$ and $V_0 \subseteq \mathbb{R}^m$ with $h^{-1}(x) \in U_0$ and $k^{-1}(f(x)) \in V_0$ and a smooth map

$$\widetilde{g}: U_0 \to V_0$$

with $g|_{U_0 \cap h^{-1}(A)} = \widetilde{g}\,|_{U_0 \cap h^{-1}(A)}$.

We now claim the following.

Lemma 11.3. *If $\partial N = \emptyset$ then Lee‘s definition and Definition 11.2 are equivalent. In general, if f is smooth in the sense of Lee‘s definition, then it is also extension-smooth but the converse may fail.*

Lemma 11.4. *If A is a submanifold of M, then $f: A \to N$ is smooth as a map of smooth manifolds if and only if it is extension-smooth.*

In particular the extension Lemma in [Lee11] (Lemma 2.26.) still holds and the proof of the Whitney approximation theorems in chapter 6 work the same (here the extension Lemma is used). But with our new definition Problem 6-7 in [Lee11] now is an actual counterexample. (with Lee‘s definition F would not be smooth on the closed subset A but only smooth as a map between manifolds).

Another way to properly define smoothness of $f: A \to N$ would be to change the codomain in Lee‘s definition to be the double of N. This should be equivalent to Definition 11.2 and is arguably more elegant at the cost of being less elementary.

Smooth Homotopies

This allows us to define what a smooth homotopy is:

Definition 11.5. Let M, N be manifolds with boundary. A homotopy

$$H\colon M \times [0,1] \to N,$$

is called smooth, if H is extension-smooth (see Definition 11.2), where we consider $M \times [0,1]$ as a subset of the manifold $M \times \mathbb{R}$.

In [Lee11] the author defines a smooth homotopy as a map $H\colon M \times [0,1] \to N$, which can be extended to a smooth map $\overline{H}\colon M \times \mathbb{R} \to N$. As we have seen in Counterexample 11.1 this is a stronger requirement compared to Definition 11.5, as less homotopies are considered to be smooth by Lee‘s definition. However the homotopy classes of smooth maps are the same. More precisely this is formulated in the following lemma.

Lemma 11.6. *Let M, N be manifolds with or without boundary and $f, g\colon M \to N$ smooth maps. Then f and g are smoothly homotopic in the sense of Definition 11.5 if and only if there exists a homotopy $H\colon M \times [0,1] \to N$ between f and g, which can be extended to a smooth map $\overline{H}\colon M \times \mathbb{R} \to N$.*

Proof. Let $G\colon M \times [0,1] \to N$ be a smooth homotopy from f to g in the sense of Definition 11.5. Choose a smooth function $\phi\colon \mathbb{R} \to [0,1]$ with

$$\phi(t) = 0 \text{ for } t \leq 0$$
$$\phi(t) = 1 \text{ for } t \geq 1$$

Then

$$\Phi\colon M \times [0,1] \to M \times [0,1], (x,t) \mapsto (x, \phi(t))$$

is extension-smooth (it can be smoothly extended to $M \times \mathbb{R}$ and this implies that it is smooth in the sense of Definition 11.2, as Lee‘s definition is stronger by Lemma 11.3). Furthermore, it is easily verified that the composition of extension-smooth maps on arbitray subsets is extension-smooth, such that

$$H := G \circ \Phi\colon M \times [0,1] \to N$$

is extension-smooth. Then H is a homotopy from f to g and we can extend H to the smooth map

$$\overline{H}\colon M \times \mathbb{R} \to N, (x,t) \mapsto G(x, \phi(t)).$$

The converse follows directly from Lemma 11.3.

□

11.2 Integrating vector valued functions

Let V be a finite dimensional real vector space and $(X, \mathcal{A}, \mu)$ a measure space. Given a function $f\colon X \to V$ we would like to define its integral. First of all any finite dimensional vector space carries a canonical topology (the topology induced by any norm), hence we can consider the Borel σ-algebra on V and get a well defined notion of what it means for f to be measurable. The definition of the integral can now be reduced to the case $V = \mathbb{R}^n$:

Definition 11.7. Let $f\colon (X, \mathcal{A}, \mu) \to V$ be a measurable function. Choose a linear isomorphism $A\colon V \to \mathbb{R}^n$. We call f integrable if $Af\colon X \to \mathbb{R}^n$ is integrable and in this case we set

$$\int_X f d\mu := A^{-1} \int_X Af d\mu$$

Linearity of the integral implies that this is well defined, i.e. independent of the choice of A. Moreover, this definition describes how to compute such an integral in practice: First choose any basis of V and express f in that basis. Then integrate each component function. The result gives the coordinates of $\int_X f d\mu$ in the chosen basis.

Linearity of the integral of $\mathbb{R}^n$-valued functions also generalizes immediately:

Lemma 11.8. *Let $f\colon (X, \mathcal{A}, \mu) \to V$ be an integrable function and $T\colon V \to V$ linear, then*

$$T \int_X f d\mu = \int_X T f d\mu$$

Proof. Choose an isomorphism $A \colon V \to \mathbb{R}^n$ then by definition

$$T \int_X f d\mu = TA^{-1} \int_X Af d\mu = A^{-1}(ATA^{-1}) \int_X Af d\mu = A^{-1} \int_X ATf d\mu$$

□

12 Solutions to selected exercises

Exercise 12.1 (Own exercise)**.** Let M be the Moebius line bundle. Consider the map $f\colon S^1 \to S^1$, $z \mapsto z^2$. Then the pullback bundle $E := f^*M$ is trivial.

Intuitively, we can see this as follows: The fiber over $e^{\pi i}$ of E is just the fiber over $e^{2\pi i}$ of M, which has already made a half-twist, i.e. the fiber of E over 1 can be imagined as a straight line and then begins twisting upon increasing the angle. When the angle $\theta = \pi$ is reached a half twist has been performed. But upon further increasing the angle, another half twist is performed as we "walk" around the moebius bundle a second time. So E is basically a double twisted moebius bundle, which is trivial.

Proof. Proof of Exercise 12.1 We define $M := \mathbb{R} \times \mathbb{R}/_\sim$, where the equivalence relation is given by

$$(x, t) \sim (x + k, (-1)^k t)$$

and the projection $p\colon M \to S^1$ is defined by $p([(x, t)]) := e^{2\pi i x}$. By identifiying S^1 with $[0, 1]/_{0\sim 1}$ we can write f as

$$f([\theta]) = e^{2\pi i\theta},$$

such that E is a line bundle over $[0, 1]/_\sim$. To see that E is trivial we define a global frame:

$$\sigma\colon [0, 1]/_\sim \to E,\ [\theta] \mapsto ([\theta], [(2\theta, 1)])$$

Note that $[(0, 1)] = [(2, 1)]$ in M, such that σ is a well-defined section. Furthermore, it is easily seen that this section is nowhere vanishing. To see that σ is smooth we could write σ in the local trivialisations of E. Alternatively we can argue as follows: Consider

$$s\colon [0, 1] \to M,\ \theta \mapsto [(2\theta, 1)]$$

and let $\pi\colon [0, 1] \to [0, 1]/_\sim$ be the projection map. Then $s = \sigma \circ \pi$. Clearly s is smooth and it is easily verified that π is a local diffeomorphism, such that smoothness of σ follows. This completes the proof. ☐

Exercise 12.2 (Exercise 10-14 from [Lee11])**.** Let M be a smooth manifold and $S \subseteq M$ a submanifold. Then TS is a subbundle of $TM|_S$.

Proof. Let $\sigma_1, \dots, \sigma_k$ be a local frame for TS. Then we can also consider the σ_i as local sections of $TM|_S$ via the inclusion $TS \hookrightarrow TM|_S$. If we can show that these are still smooth sections we are done as this shows that TS (as a set) is a subbundle of $TM|_S$ an the induced smoothness structure is the same as the usual one on TS (see the construction of the smoothness structure on the subbundle in Lemma 10.32 in [Lee11] to see that this indeed yields the same structures). But smoothness of σ_i as local sections of $TM|_S$ follows directly from the fact that the inclusion $TS \hookrightarrow TM|_S$ is just the restriction of the differential of the inclusion $\iota\colon S \hookrightarrow M$ to the submanifold $TM|_S \subseteq TM$ in the codomain (the fact that $TM|_S$ is a submanifold of TM follows for example from Lemma 6.17). ☐

Exercise 12.3 (Mentioned in [Bre13] in chapter 'Conventions for CW complexes')**.** Let (X, x_0) and (Y, y_0) be compact pointed topological spaces, then the smash product $X \wedge Y$ is homeomorphic to the one point compactification of $Z := (X - x_0) \times (Y - y_0)$. In particular

$$S^n \wedge S^m \cong S^{n+m}.$$

Proof sketch. Let Z^+ be the one point compactification of Z. Define $f\colon X \times Y \to Z^+$ by

$$f(x, y) := \begin{cases} (x, y) & , x \neq x_0 \text{ and } y \neq y_0 \\ \infty & , \text{otherwise} \end{cases}$$

It is then straightforward to show that f is continuous and the following diagram is a push-out:

$$\begin{tikzcd} X \vee Y \arrow[r] \arrow[d] & X \times Y \arrow[d, "f"] \\ * \arrow[r] & Z^+ \end{tikzcd}$$

So by the universal property of push-outs we get $Z^+ \cong X \wedge Y$. The statement about the smash product of spheres then just follows from $S^n - * \cong \mathbb{R}^n$ and that the one point compactification of $\mathbb{R}^n$ is S^n. $\square$

13 Alternative Proofs

This is a collection of alternative proofs for various statements from books, lecture notes etc.

Theorem 13.1 (Theorem 10.34 from [Lee11]). *Let E and E' be smooth vector bundles over a manifold M, and let $F\colon E \to E'$ be a bundle homomorphism. Then $\ker F$ and $\operatorname{im} F$ are smooth subbundles of E and E', respectively, if and only if F has constant rank.*

Proof. We present an alternative proof to show that $\ker F$ is a smooth subbundle. Let $\operatorname{rank} F = k$. Choose a local frame

$$\sigma_1, \dots, \sigma_n \colon U \to E,$$

such that $F \circ \sigma_1, \dots, F \circ \sigma_k$ are linearly independent sections, i.e. a local frame for $\operatorname{im} F$. So for $j > k$ we can write

$$F \circ \sigma_j = \sum_{i=1}^{k} \tau_{ij} F \circ \sigma_i$$

for smooth functions

$$\tau_{ij} \colon U \to \mathbb{R}$$

Define

$$\sigma'_j := \sigma_j - \sum_{i=1}^{k} \tau_{ij} \sigma_i$$

Then $F \circ \sigma'_j = 0$, thus $\operatorname{im} \sigma'_j \subseteq \ker F$. Furthermore, $\sigma'_{k+1}, \dots, \sigma'_n$ are linearly independent, since $\sigma_1, \dots, \sigma_n$ is a local frame. This shows the statement. ☐

Proposition 13.2 (Proposition 5.17 from [Lee18]). *The musical isomorphisms commute with the (total) covariant derivative: if F is any smooth tensor field with a contravariant i-th index position and $\flat$ represents the operation of lowering the i-th index, then*

$$\nabla_X(F^\flat) = (\nabla_X F)^\flat \tag{13}$$

for all vector fields X. In particular

$$\nabla(F^\flat) = (\nabla F)^\flat. \tag{14}$$

Similiarly, if G has covariant i-th position and $\sharp$ denotes raising the i-th index, then

$$\nabla_X(F^\sharp) = (\nabla_X F)^\sharp, \quad \nabla(F^\sharp) = (\nabla F)^\sharp. \tag{15}$$

In the book [Lee18], the author uses the identity $F^\flat = \operatorname{tr}(F \otimes g)$, which is however only correct in the case where F is a vector field, since then

$$F^\flat = \langle F, - \rangle = \operatorname{tr}(F \otimes g)$$

by definiton of the trace operator. In general this argument fails, as can be seen by writing each side in a local frame. Thus we present an alternative proof.

Proof. First let $F = Z$ be a vector field. Instead of using the identity mentioned above we can also argue as follows: Write $\alpha := \langle Z, - \rangle$. Then for an arbitrary vector field Y we compute:

$$(\nabla_X \alpha)(Y) = X \cdot \alpha(Y) - \alpha(\nabla_X Y) = \langle \nabla_X Z, Y \rangle + \langle Z, \nabla_X Y \rangle - \langle Z, \nabla_X Y \rangle = \langle \nabla_X Z, Y \rangle$$

hence

$$\nabla_X(Z^\flat) = \nabla_X \langle Z, - \rangle = \langle \nabla_X Z, - \rangle = (\nabla_X Z)^\flat.$$

For general F we may assume (by linearity of the $\flat$ operator)

$$F = Z_1 \otimes \ldots \otimes Z_k \otimes \varepsilon^1 \otimes \ldots \otimes \varepsilon^l$$

for some vector fields Z_j and covector fields ε^j. But then

$$F^\flat = Z_1 \otimes \ldots \otimes \langle Z_i, - \rangle \otimes \ldots \otimes Z_k \otimes \varepsilon^1 \otimes \ldots \otimes \varepsilon^l$$

so (13) follows from the product rule and what has been shown above. Then (14) follows from (13) and the definition of the $\flat$ operator as well as the total covariant derivative. For (15) we argue as in [Lee18]: note that $\sharp$ and $\flat$ are inverses to each other (since they operate on the same index position), so by substituting $F = G^\sharp$ in (13) and (14) and applying $\sharp$ to both sides, the statement follows. ☐

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